Assessment of Climate Change over the Indian Region

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Introduction to Modern Climate Change

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Part 1: The physics of climate change

Part 1: The physics of climate change

What climate is, evidence of warming, energy balance, the greenhouse effect, the carbon cycle, forcing and feedbacks, attribution

Part 2: Climate change over the Indian region

Models, temperature, monsoon and rainfall, greenhouse gases and aerosols, droughts and floods, Himalayas, oceans, impacts

Part 3: Emissions and climate scenarios

What drives emissions, energy and carbon intensity, SSPs and RCPs, projected warming, impact models by sector

Part 4: Economics and policy

Exponential growth and discounting, response options, carbon pricing, history of science and negotiations, long-term goals

Weather and climate

Weather

The state of the atmosphere at a particular time and place: temperature, humidity, wind, rain.

It matters for short-term decisions.

Climate

The statistics of weather over a few decades: the averages and the extremes.

It matters for long-term decisions, such as where to farm or build.

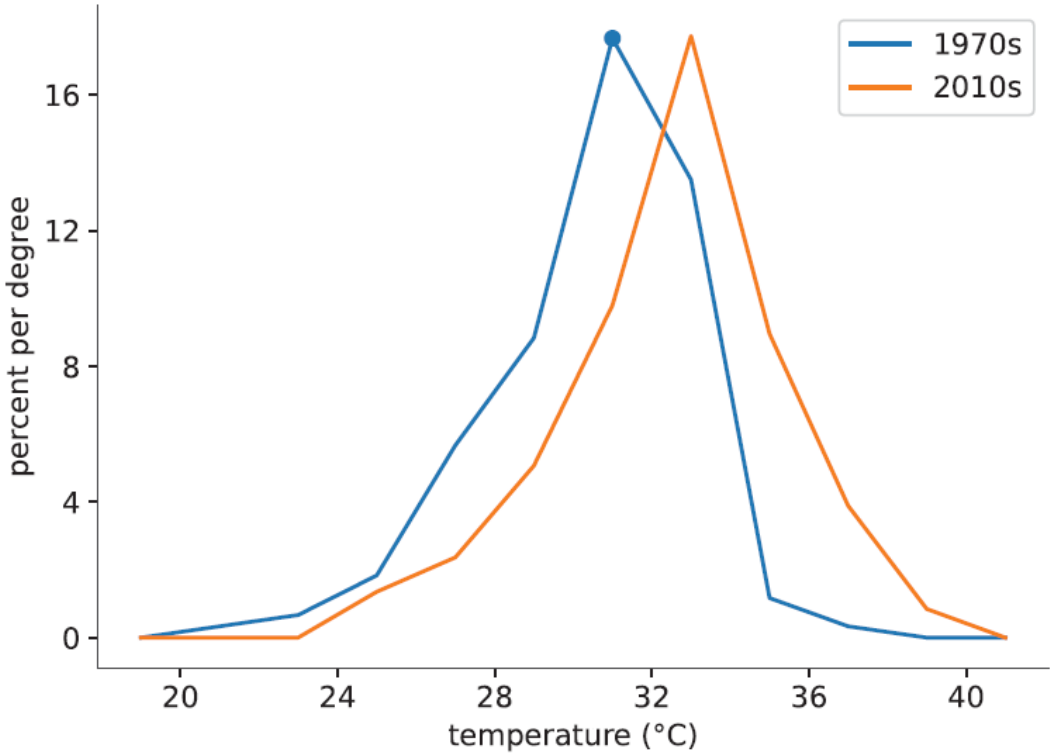
“Climate is what you expect; weather is what you get.” (Mark Twain)

- **Climate change:** any systematic change in the long-term statistics of climate elements, sustained over several decades or longer (American Meteorological Society).
- **What matters differs by person:** the farmer cares about seasonal rain, the coastal city about storm surge. Extremes often matter more than averages.

Source: Dessler (2021), Ch. 1

Climate change is a shift in the statistics

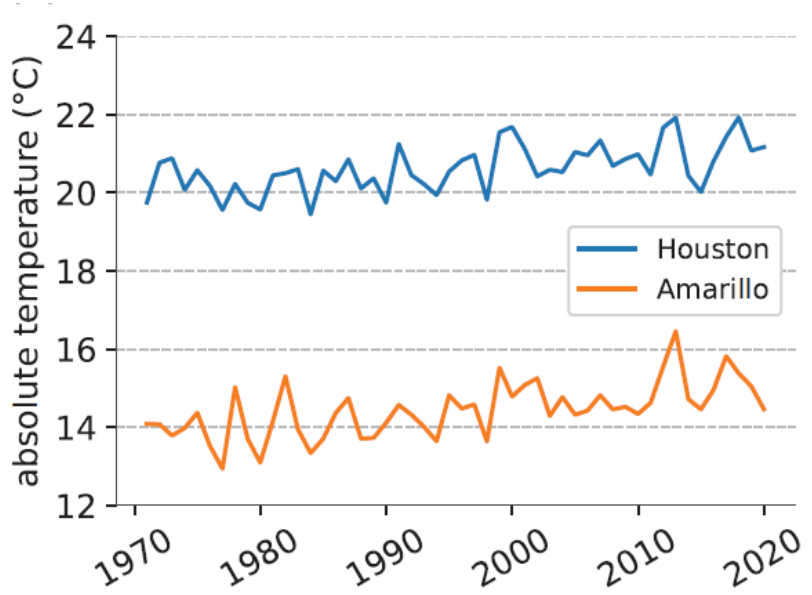
- **Example:** daily maximum temperature in September near Houston, Texas, in the 1970s and the 2010s.
- **The whole distribution moved:** the most likely daily maximum rose from 31 °C to 33 °C, a shift of about 1.5 °C.
- **Rare became common:** days above 34 °C went from 3% of days in the 1970s to 35% in the 2010s.
- **Lesson:** a small change in the average brings a large change in the frequency of extremes.

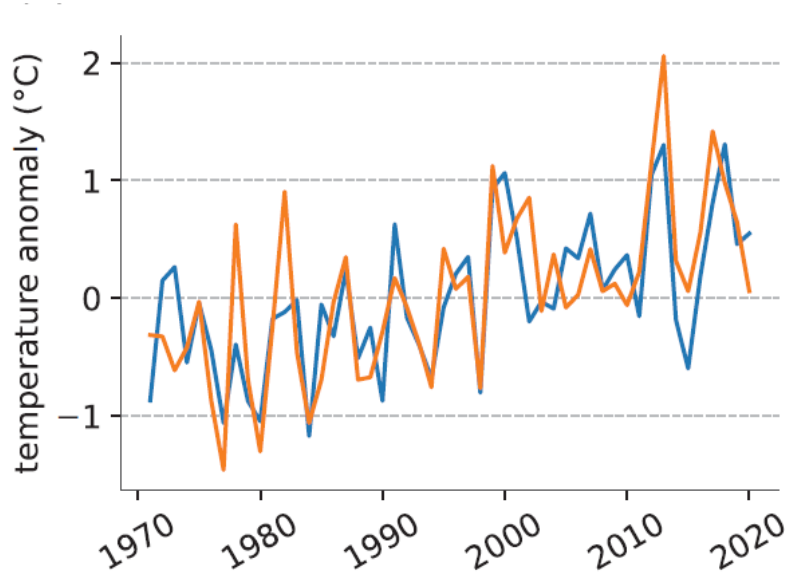


Source: Dessler (2021), Ch. 1

Measuring warming: temperature anomalies

- **Problem:** absolute temperature changes over short distances (a car park is hotter than the park beside it), so measuring it needs a very dense network.
- **Anomaly:** the temperature minus the average for a reference period at the same place.
- **Anomalies travel:** Houston and Amarillo are 1,000 km apart and differ by 6 °C, yet their anomalies rise and fall together.
- **Consequence:** about 100 well-spread thermometers are enough to measure the global average anomaly.
- **Global averages** cancel regional weather: a heat wave in one place is usually offset by a cold spell elsewhere.





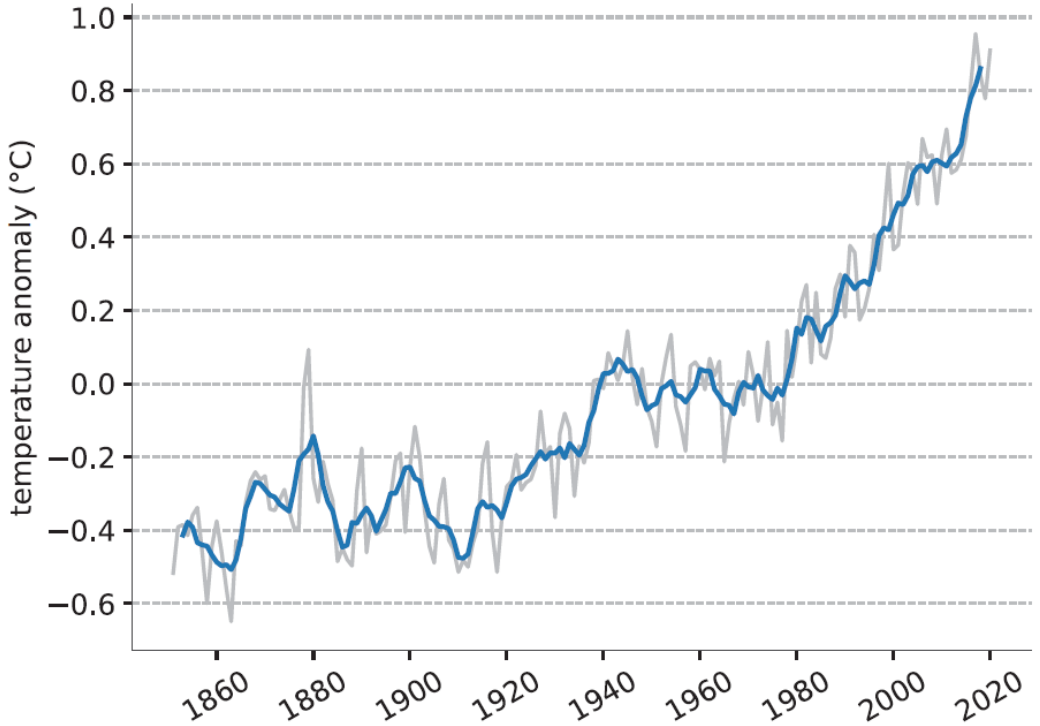
Source: Dessler (2021), Ch. 2

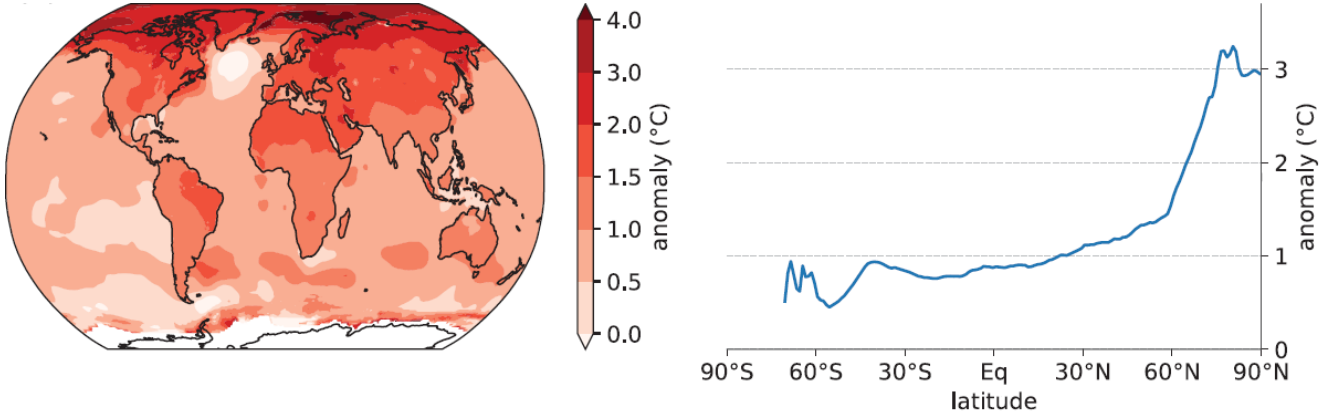
The Earth has warmed by about 1.1 °C

1.1 °C
warming from 1850–1900 to 2009–2018

0.13 °C
per decade in the satellite record since 1978

- **Not uniform:** land warmed more than ocean, and the northern hemisphere more than the tropics or the south.
- **Steady since the 1970s:** every decade warmer than the one before; the six warmest years were 2015–2020.
- **Two independent records agree:** surface thermometers and satellites.





Source: Dessler (2021), Ch. 2

Independent evidence agrees

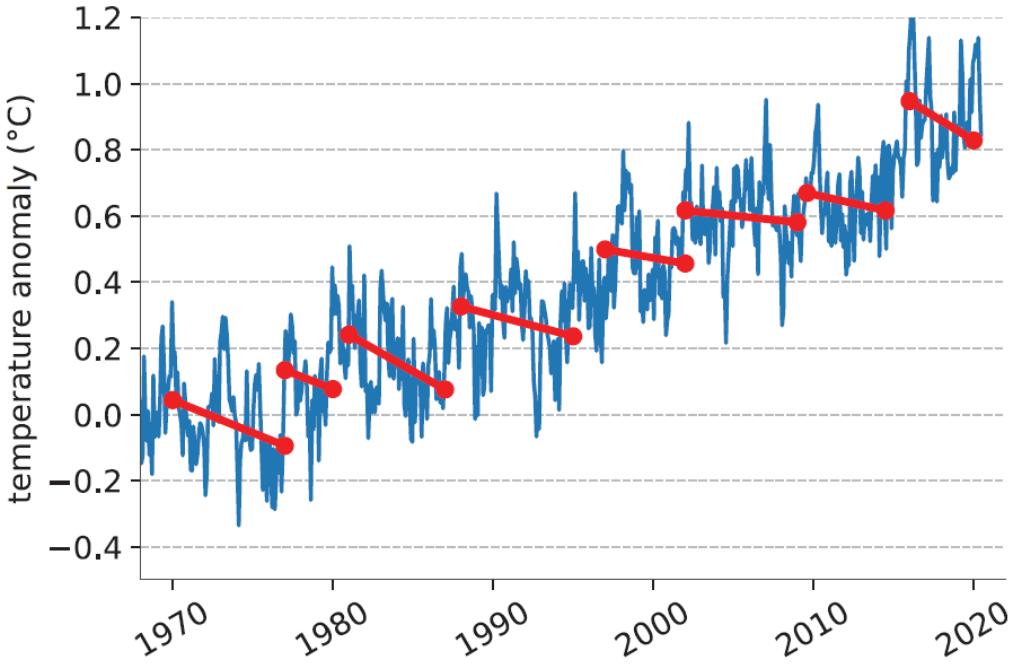
Indicator	What is observed
Glaciers	Losing mass worldwide over the last half century
Arctic sea ice	September area clearly declining, and the ice is thinner; little change around Antarctica
Ice sheets	Greenland losing about 250 billion tonnes a year, Antarctica about 180 billion
Ocean heat	The top 2 km of the ocean is warming: an enormous store of energy
Sea level	Up 0.19 m from 1900 to 2018; rate rose from 1.6 mm/yr to 3.2 mm/yr after 1993

- **Independent:** each dataset has different error sources, so no single flaw could create a false warming in all of them.
- **Consistent in detail:** more ice loss in the Arctic matches more warming there; ice melt plus thermal expansion adds up to the measured sea-level rise.
- **Conclusion:** the IPCC calls the warming “unequivocal”.

Source: Dessler (2021), Ch. 2

Reading the record: wiggles are not trends

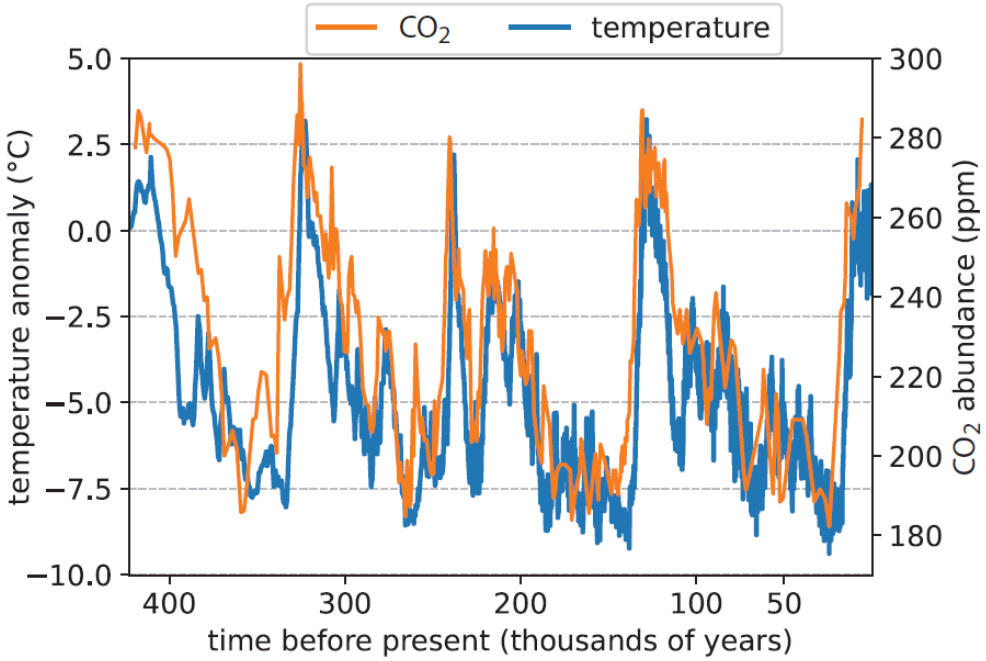
- **Short-term bumps** of a few tenths of a degree come from El Niño (warm), La Niña (cool) and volcanic eruptions (cool for a few years).
- **Cherry picking:** start in a hot El Niño year and end in a cool year, and a short “cooling trend” appears even while the planet warms at 1.7 °C per century (1970–2020).
- **Only long periods count:** over several decades the warming dominates whatever the endpoints.

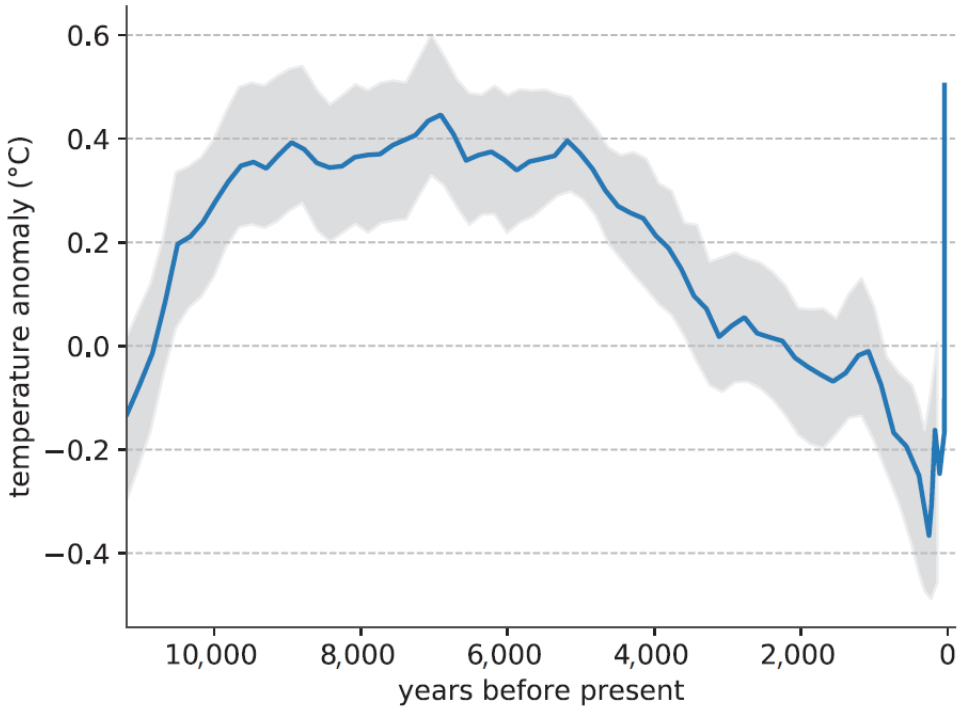


Source: Dessler (2021), Ch. 2

Climate over the Earth's history

- **Paleoproxies:** ice cores (to 2 million years), tree rings (about 1,000 years), ocean sediments (tens of millions of years).
- **Much warmer and much colder:** an ice-free "greenhouse" Earth 50 million years ago; ice ages over the last 3 million years.
- **PETM, 55 million years ago:** a massive carbon release warmed the planet by over 5 °C within a few thousand years.
- **Ice ages** were only about 6 °C colder, yet sea level was 100 m lower. CO_2 rose and fell with temperature.
- **Rate:** about 1 °C in the last century is roughly 16 times faster than the warming out of the last ice age.





Source: Dessler (2021), Ch. 2

Radiation basics

Photons and wavelength

Visible light: 0.4–0.8 µm. Infrared: 0.8–1,000 µm. The atmosphere is transparent to visible light but less so to infrared.

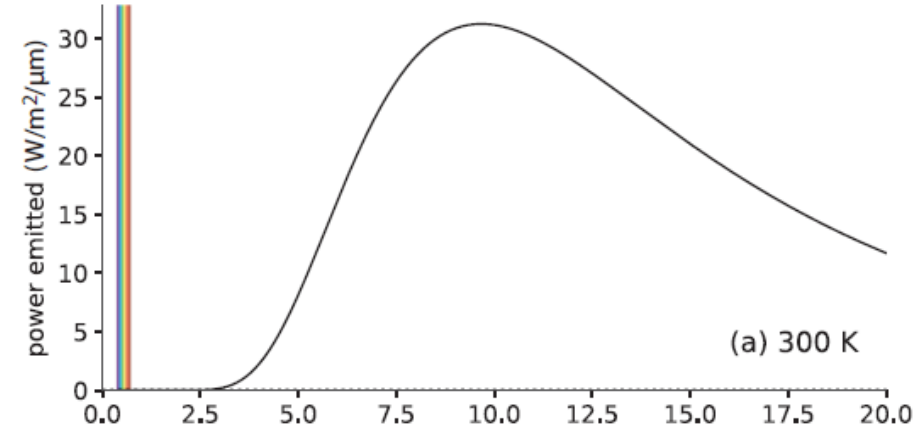
Wien's law: $\lambda max = 2{,}897 / T$

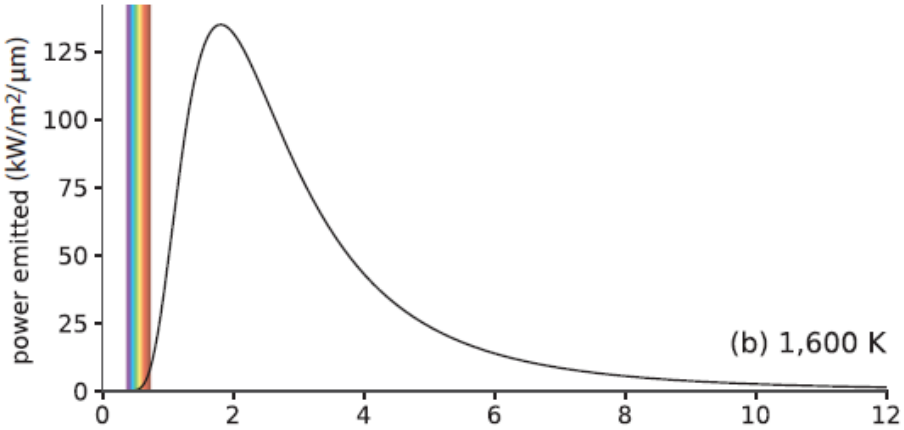
Hotter objects emit at shorter wavelengths. The Sun (6,000 K) peaks at 0.5 µm, in the visible. The Earth (about 300 K) peaks near 10 µm, in the infrared.

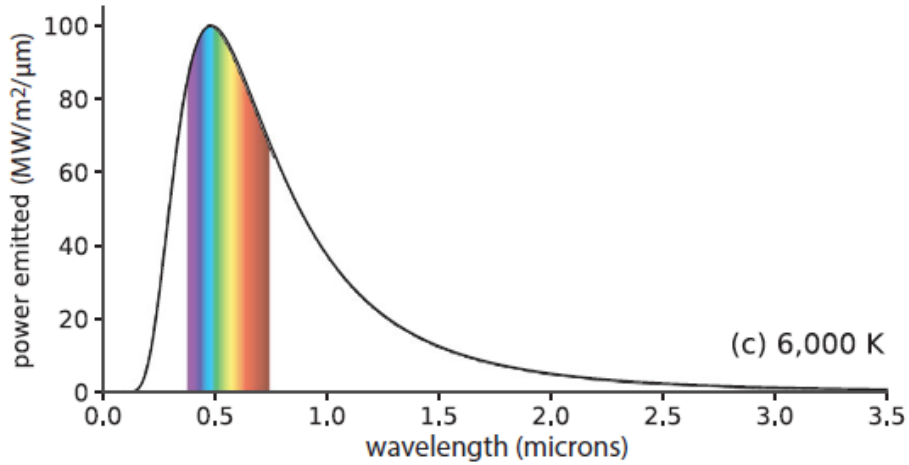
Stefan–Boltzmann: $P/a = \sigma T^4$

Hotter objects emit far more power. At 300 K a surface emits 460 W/m². $\sigma = 5.67 \times 10^{-8}$ W/m²/K⁴.

P: Power emitted, a: Surface area of the object







Source: Dessler (2021), Ch. 3

Energy balance

Condition	Internal energy	Temperature
Energy in > energy out	Increases	Rises
Energy out > energy in	Decreases	Falls
Energy in = energy out	Unchanged	Steady (equilibrium)

- **Energy is conserved:** an object that absorbs a photon warms; one that emits a photon cools.
- **Everything emits:** all objects radiate according to their temperature, mostly in the infrared at everyday temperatures.
- **For a planet:** temperature settles where energy absorbed from the Sun equals energy radiated to space. Change either side and the temperature must adjust.

Source: Dessler (2021), Ch. 3

Energy in: sunlight and albedo

1,360 W/m²

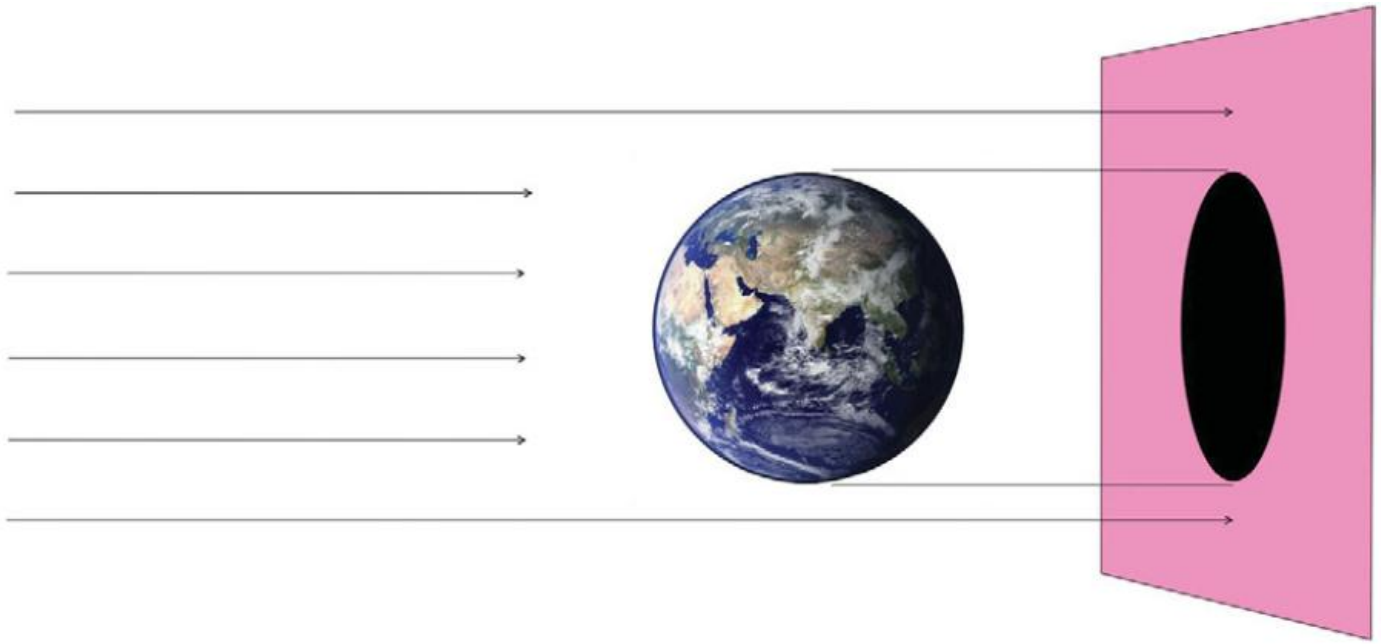
solar constant at Earth's orbit (S)

P: 3.8 x 10^{26} W (energy by sun)

a: 2.8 x 10^{23} m^2 (surface area of earth)

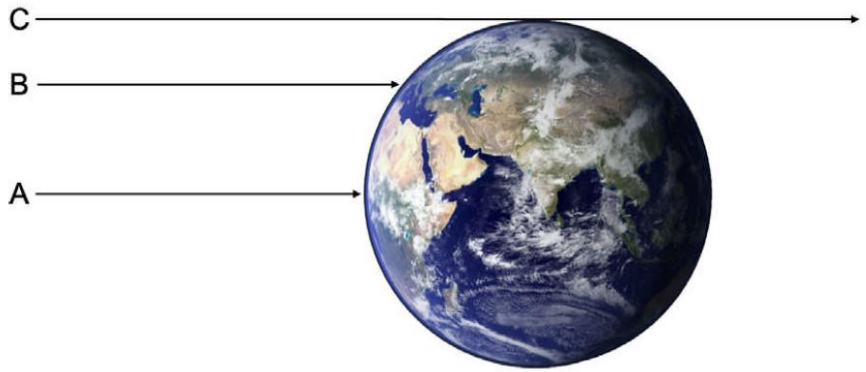
0.3

albedo (α): share of sunlight reflected



Absorbed: Ein = S(1 – α)/4 = 238 W/m²

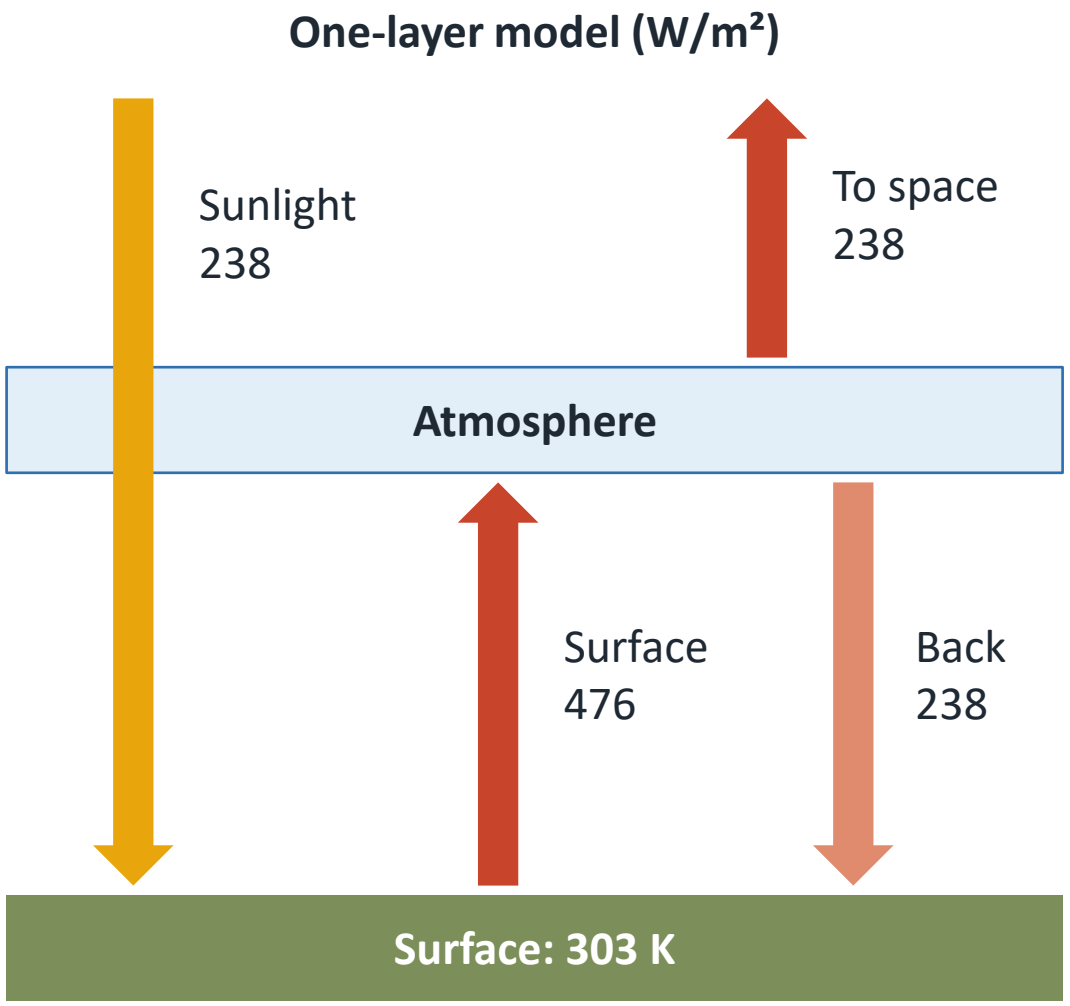
- **Why divide by 4:** the Earth intercepts sunlight over a disc (πR^2) but its surface is a sphere ($4\pi R^2$).
- **Scale:** 175,000 TW of sunlight falls on Earth; humanity uses under 20 TW.
- **Uneven:** the tropics get more sun and are dark ocean; the poles get little and are bright ice.



Source: Dessler (2021), Ch. 4

The greenhouse effect: a one-layer model

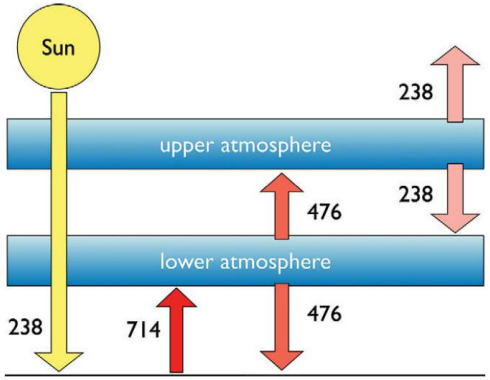
- **No atmosphere:** set $\sigma T^4 = 238$ W/m² and the Earth would be 255 K (−18 °C). The real average is 288 K (+15 °C).
- **One-layer model:** an atmosphere transparent to sunlight but opaque to infrared, radiating equally up and down.
- **Balance at the top:** the layer must send 238 to space, so it also sends 238 down.
- **Balance at the surface:** 238 from the Sun + 238 from the atmosphere = 476 emitted, which needs 303 K.
- **The greenhouse effect** is this extra heating of the surface by the atmosphere.

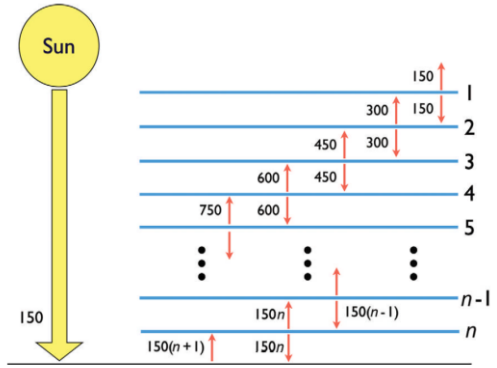


Source: Dessler (2021), Ch. 4

More greenhouse gas, more warming

$T = [(n + 1) \cdot S(1 - \alpha) / 4\sigma]^{¼}$ n = number of absorbing layers, or the amount of CO_2 in the atmosphere



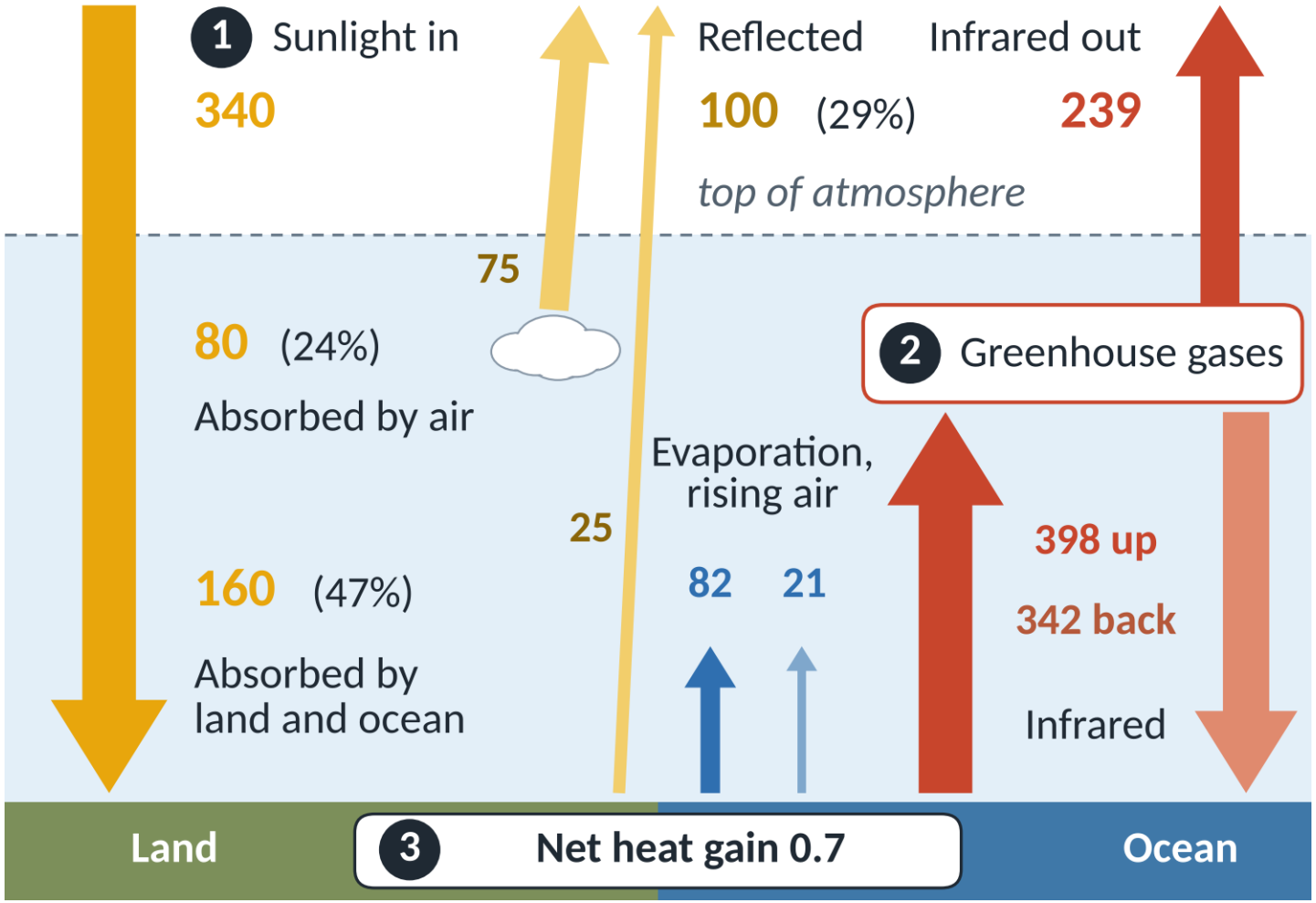


Planet	S (W/m²)	Albedo	Surface T	n
Mercury	10,000	0.1	452 K	0.05
Venus	2,650	0.7	735 K	82
Earth	1,360	0.3	289 K	0.65
Mars	580	0.15	227 K	0.22

- **Three controls:** greenhouse gas amount (n), solar constant (S) and albedo (α). Change any one and the temperature adjusts to restore balance.
- **Tested on other planets:** Venus receives less absorbed sunlight than Earth yet is 450 K hotter, because of its thick CO_2 atmosphere.
- **A simplification:** it ignores convection and heat transport, so the numbers are rough but the physics is right.

Source: Dessler (2021), Ch. 4

The greenhouse effect



All numbers in W/m², global annual average. Source: IPCC AR6 WG1 (2021), Fig. 7.2 and Box 7.2

1 Sunlight in: 340

- **100 reflected** (29%): 75 by clouds and air, 25 by the surface
- **80 absorbed by air** (24%)
- **160 absorbed by land and ocean** (47%)

2 Greenhouse effect

- **Surface sends up 398** as infrared, plus 82 by evaporation and 21 by rising air
- **Greenhouse gases send 342 back;** only 239 escapes. Surface stays near +15 °C, not −18 °C

3 Net heat gain: 0.7

- In 340, out 100 + 239 (rounded); with more CO_2 about 0.7 stays
- **Ocean 91%, land 5%, ice melt 3%, air 1%**

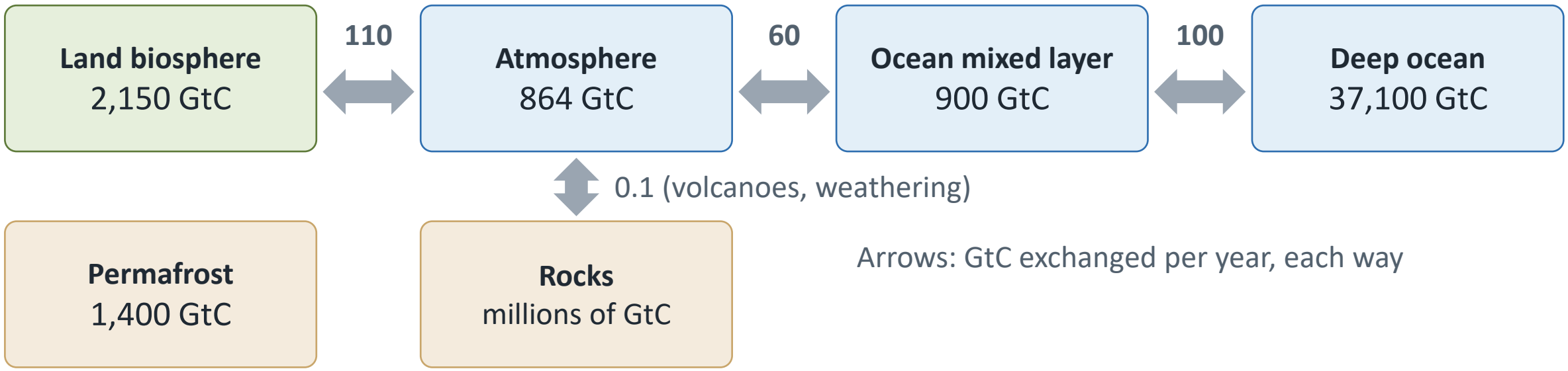
Greenhouse gases in the atmosphere

Gas	Abundance (2020)	Role
N_2, O_2, argon	over 99% of dry air	Do not absorb infrared: not greenhouse gases
Water vapour	0.2% to 4%	Most important greenhouse gas; amount is set by temperature, so it acts as a feedback
Carbon dioxide	412 ppm	Main gas humans control; rising about 2 ppm a year
Methane	1.87 ppm	Much stronger per kilogram than CO_2, but short-lived
Nitrous oxide	0.32 ppm	From nitrogen fertiliser and industry
Ozone	10–40 ppb at surface	Protective in the stratosphere; a pollutant and greenhouse gas near the ground
Halocarbons	a few ppb	Synthetic; hundreds to thousands of times stronger than CO_2 per molecule

- **Units:** ppm = molecules per million molecules of air; ppb = per billion.

Source: Dessler (2021), Ch. 5

The carbon cycle



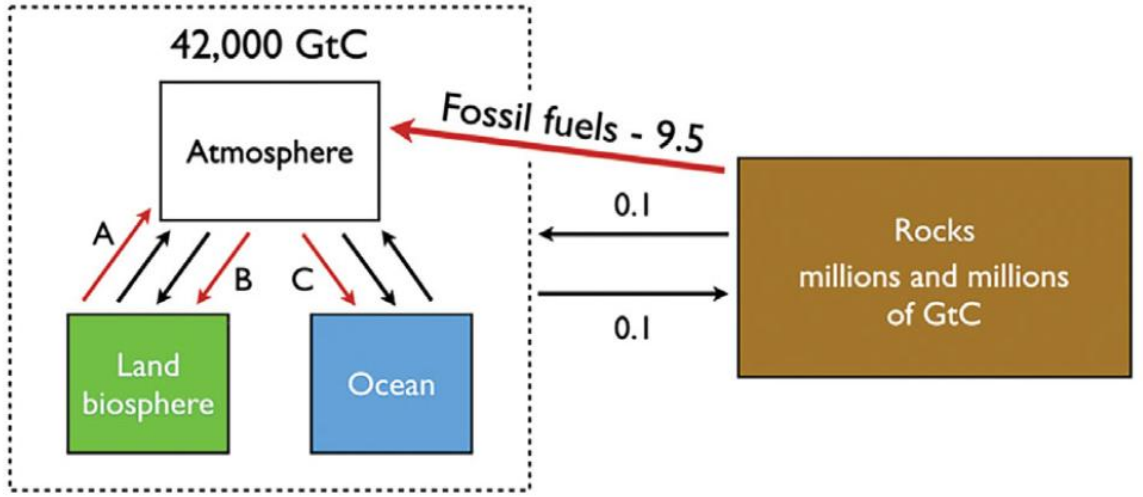
- **Fast exchange:** photosynthesis and respiration move about 110 GtC a year; air–sea exchange about 60 GtC.
- **Turnover times:** atmosphere about 5 years, land about 20 years, deep ocean about 370 years, rocks about 420,000 years.
- **Seasonal sawtooth:** northern plants draw CO_2 down by about 6 ppm each summer (the Keeling curve).

Source: Dessler (2021), Ch. 5

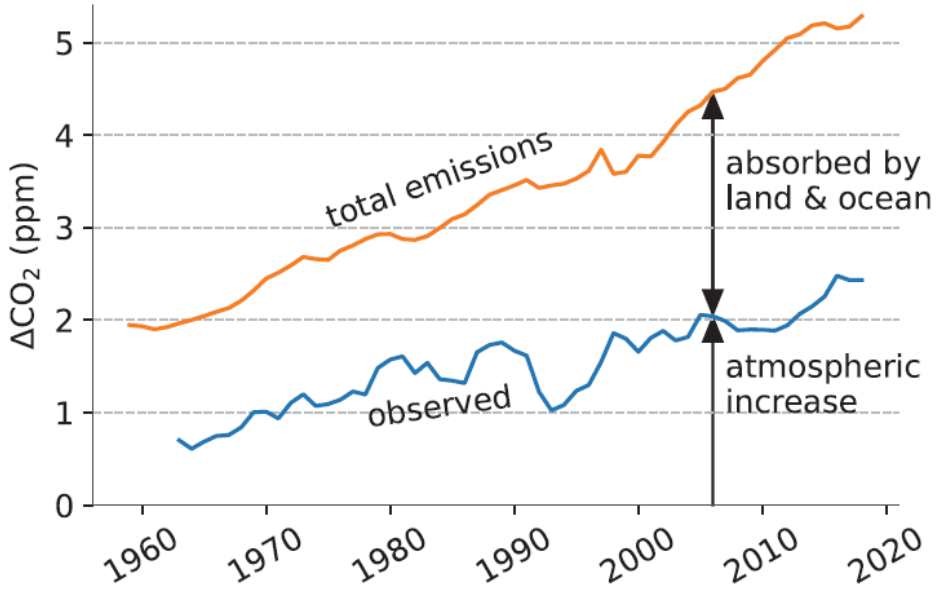
How humans perturb the carbon cycle

9.5 GtC/yr
from fossil fuels (2008–2017), about 100× volcanoes

1.5 GtC/yr
from deforestation and other land-use change



- **CO_2 rose** from 260–280 ppm (stable for 10,000 years) to 412 ppm in 2020, starting with the industrial revolution.
- **Only 44% stays in the air.** About half the rest dissolves in the ocean and half is taken up by land plants.
- **The cost:** dissolved CO_2 forms carbonic acid (ocean acidification), and the sinks may weaken as they saturate.



Source: Dessler (2021), Ch. 5

How we know the extra CO_2 is from fossil fuels

It tracks emissions

Each year's rise in atmospheric CO_2 has averaged 44% of what humans emitted that year, for half a century.

A natural source would have no reason to follow human emissions.

Carbon-13: from plants

Plants prefer carbon-12, and fossil fuels are ancient plants.

The added CO_2 is enriched in carbon-12, which rules out volcanoes and the ocean.

Carbon-14: long dead

Carbon-14 decays with a half-life of about 6,000 years, so fossil carbon has none.

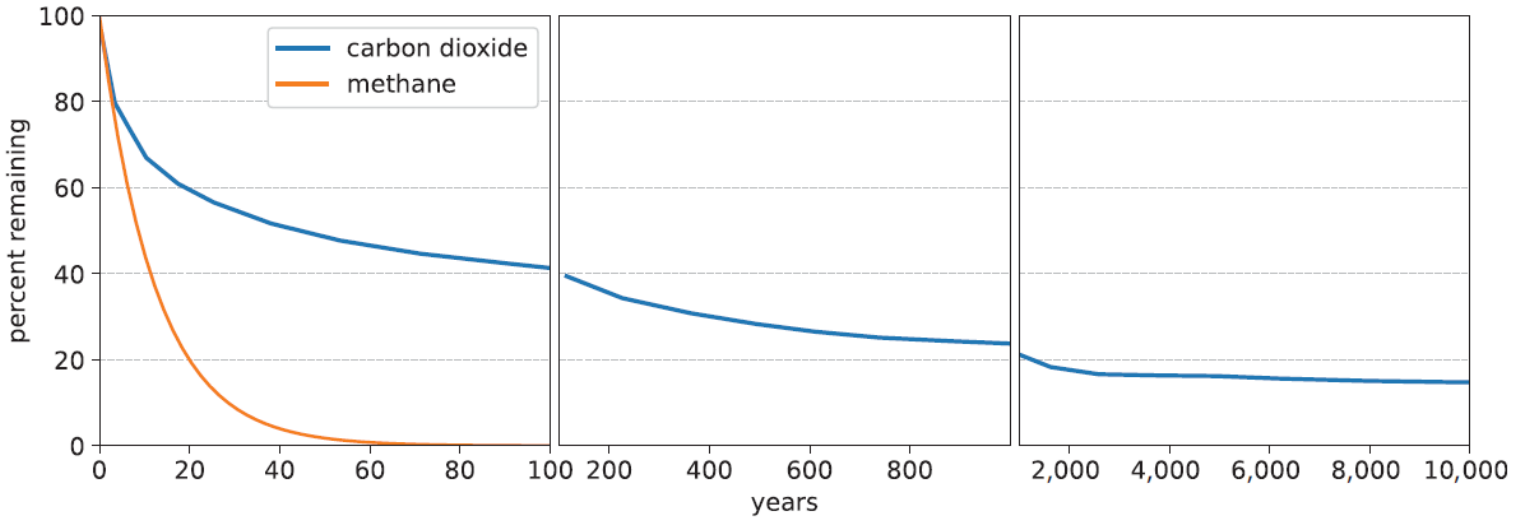
The added CO_2 is "radiocarbon dead": not from living plants.

- **But plants and animals emit ten times more?** Respiration returns carbon that plants took from the air months earlier: no net change. Fossil carbon had been locked in rock for hundreds of millions of years, so burning it is a net addition.

Source: Dessler (2021), Ch. 5

CO_2 stays for centuries to millennia

Time after emission	CO_2 still in the air	Removed by
50 years	about 50%	Land plants, upper ocean
500 years	about 28%	Deep ocean
10,000 years	about 14%	Ocean–rock reactions
100,000 years	near zero	Rock weathering



- **One number:** about 500 years (time to remove 70%).
- **Not the same as turnover:** individual molecules swap out in about 5 years, but the added amount persists.
- **Cross-check:** Archer et al. (2009) find 20–35% remains after 2–20 centuries, with a tail of thousands of years.

Source: Dessler (2021), Ch. 5; Archer et al., Annu. Rev. Earth Planet. Sci., 2009

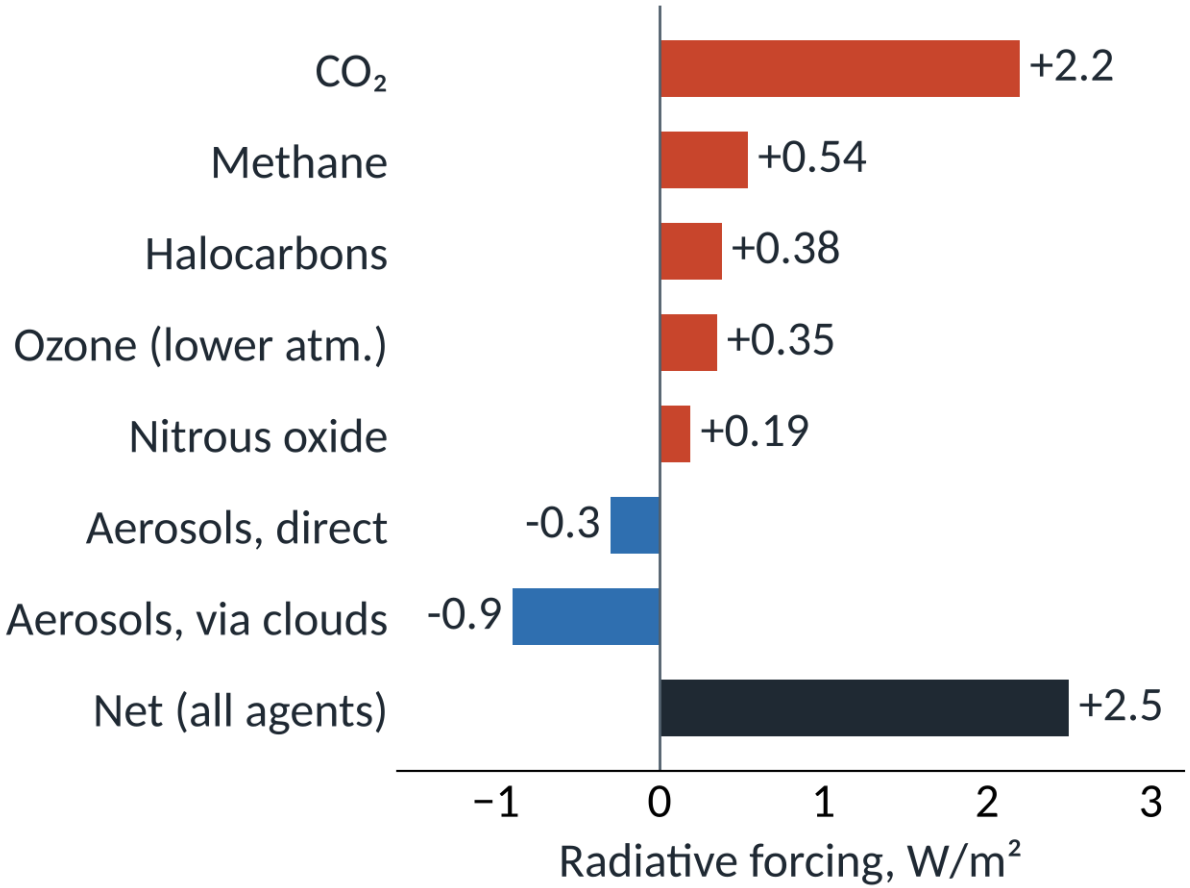
Methane and other greenhouse gases

Gas	Lifetime	Warming power per kg (CO_2 = 1)	Increase since pre-industrial
Carbon dioxide	about 500 years	1	130 ppm
Methane	12.4 years	32	1.1 ppm
Nitrous oxide	109 years	260	75 ppb
Halocarbons	years to millennia	100s to 1,000s	a few ppb

- **Methane's rise** from 0.8 to 1.9 ppm equals about 30 ppm of CO_2: a quarter of the warming power of the CO_2 increase.
- **Human sources:** about 60% agriculture and waste (livestock, landfills, rice paddies), 30% leaks from gas wells and coal mines, 10% biomass burning. Natural sources, mainly wetlands, are about as large.
- **Policy lever:** methane is destroyed in about 12 years, so cutting it cools the climate within decades. CO_2 cuts take centuries to show up as lower concentrations.

Source: Dessler (2021), Ch. 5

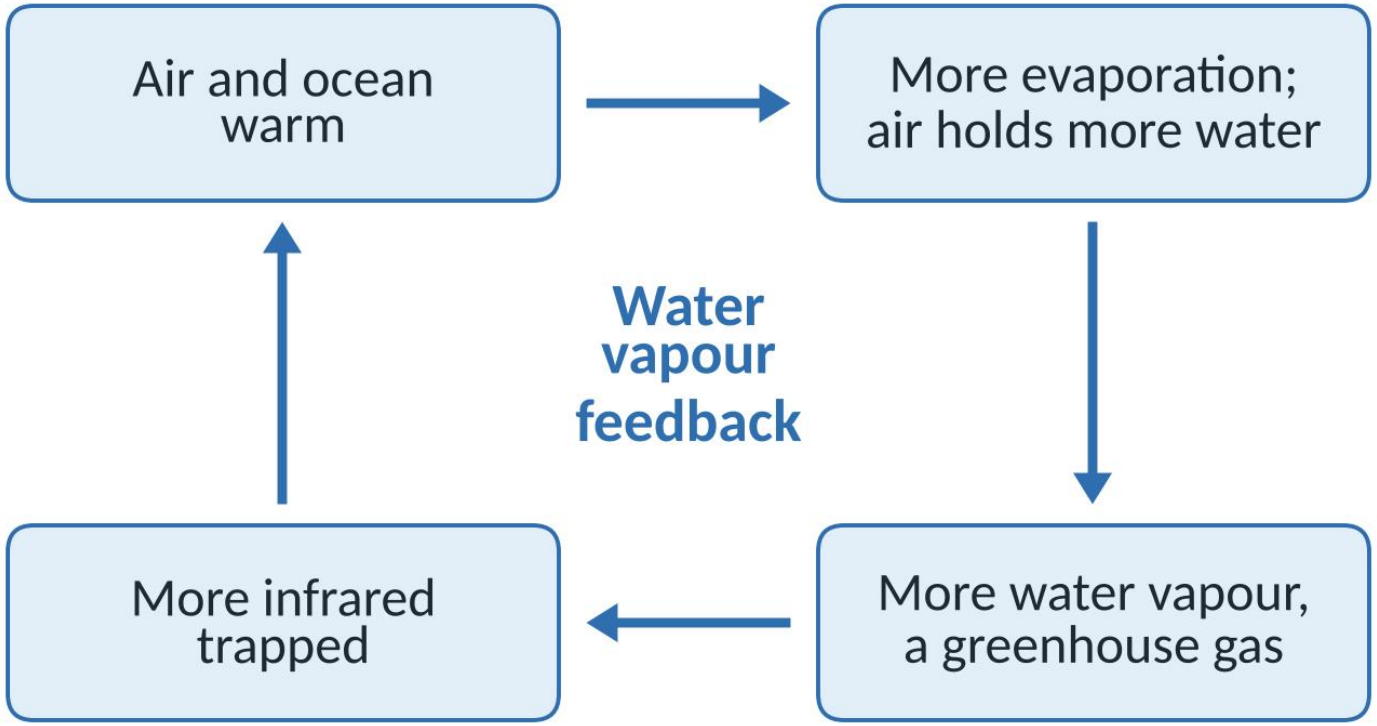
Radiative forcing: how much humans have pushed



- **Radiative forcing** is the change in (energy in − energy out) caused by some change, before the temperature responds. Positive values warm.
- **Greenhouse gases:** +3.5 W/m² since 1750, about 60% of it from CO_2.
- **Aerosols:** −1.1 W/m², offsetting about 30% of the greenhouse forcing.
- **Net:** +2.5 W/m², almost all from human activity.
- **Benchmark:** doubling CO_2 gives +4 W/m².

Source: Dessler (2021), Ch. 6 (values for 1750–2018)

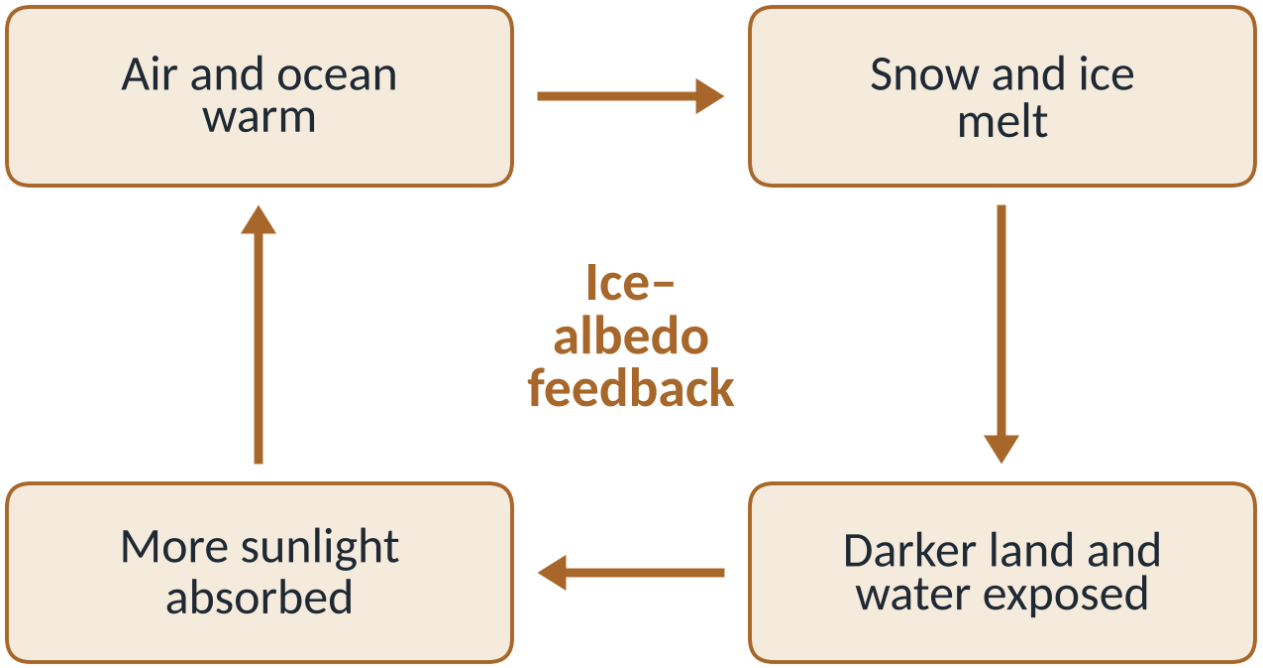
Feedbacks that amplify warming: water vapour



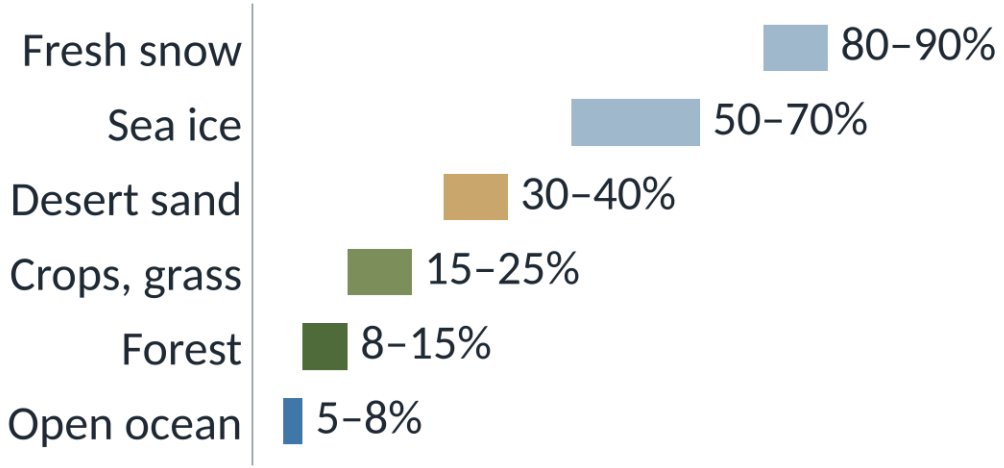
- Warmer air holds about 7% more water vapour per °C, and warmer oceans evaporate more.
- Water vapour is the most abundant greenhouse gas, so the extra moisture traps more infrared and adds to the warming.
- This roughly doubles the warming CO_2 would cause on its own. More moisture also feeds heavier downpours.
- Over India, specific and relative humidity are already rising.

Source: IPCC AR6 WG1 (2021), Ch. 7 and 8

Feedbacks that amplify warming: ice and albedo



Albedo: share of sunlight a surface reflects (Earth average 29%)



- As snow and ice retreat, bright surfaces give way to dark ones. This is one reason the Arctic and the high Himalaya warm faster than the global average.

Feedbacks and climate sensitivity

Fast feedback	Strength g	Mechanism
Water vapour	+0.6	Warmer air holds more water vapour, itself a greenhouse gas
Ice–albedo	+0.15	Melting ice exposes darker surface, which absorbs more sunlight
Lapse rate	−0.25	The upper atmosphere warms more than the surface and radiates more to space
Clouds	0 to +0.25	Clouds both reflect sunlight and trap infrared; the net change is the biggest uncertainty
Total	0.5 to 0.75	

Final warming = initial warming / (1 − g) so feedbacks multiply the initial warming by 2 to 4

- **Without feedbacks,** doubling CO_2 warms the planet by about 1.2 °C.
- **Climate sensitivity** (with fast feedbacks): 2.5–4.0 °C per doubling, best estimate 3 °C, or 0.75 °C per W/m².

Source: Dessler (2021), Ch. 6

Slow feedbacks, and forcing versus feedback

Ice sheets

Greenland and Antarctica take millennia to respond; losing them lowers albedo further.

Permafrost carbon

Thawing ground lets frozen organic matter decay, releasing CO_2 and methane.

Vegetation

Climate shifts vegetation, which changes albedo, water exchange and carbon uptake.

Weathering

More rain speeds rock weathering, removing CO_2. Stabilising, but over millions of years.

- **Slow feedbacks** are uncertain but mostly positive: more warming over centuries than the fast feedbacks alone predict.
- **Forcing** starts a climate change and is not itself set by the climate (human CO_2, the Sun, volcanoes).
- **Feedback** responds to temperature and amplifies or damps the change. CO_2 can be either: a forcing today, a feedback in the ice ages.

Source: Dessler (2021), Ch. 6

What scientists conclude

Established fact	Humans have influenced the climate system since the mid-20th century.
Extremely likely (95%)	Human influence is the main driver, responsible for more than half of the observed warming.
Likely (66%)	Humans are responsible for all, or nearly all, of the warming since the mid-19th century.

- **“CO_2 lagged temperature in the ice ages”:** true, because orbital changes started those warmings and CO_2 released from the ocean then amplified them. There CO_2 was a feedback; today humans add it first, so it is the forcing.

Source: Dessler (2021), Ch. 7

Part 2: Climate change over the Indian region

Part 1: The physics of climate change

What climate is, evidence of warming, energy balance, the greenhouse effect, the carbon cycle, forcing and feedbacks, attribution

Part 2: Climate change over the Indian region

Models, temperature, monsoon and rainfall, greenhouse gases and aerosols, droughts and floods, Himalayas, oceans, impacts

Part 3: Emissions and climate scenarios

What drives emissions, energy and carbon intensity, SSPs and RCPs, projected warming, impact models by sector

Part 4: Economics and policy

Exponential growth and discounting, response options, carbon pricing, history of science and negotiations, long-term goals

How climate models work

Weather forecast

An initial-value problem: it needs today's exact state, and small errors grow, so skill ends after about a week.

Climate projection

A boundary-value problem: it predicts the statistics of weather from the forcing, like knowing August is hotter than January.

- **Global climate models** split the atmosphere into about a million grid boxes, tens of km wide, and apply conservation of energy, mass and momentum.
- **Parameterisation:** clouds, rain, sea ice and vegetation are smaller than a grid box and are represented by tuned empirical rules. This is why rainfall is less certain than temperature.
- **Track record:** projections made decades ago matched the warming that followed.
- **Next slides:** RCP4.5 and RCP8.5 are moderate and high emission scenarios, explained in Part 3.

Source: Dessler (2021), Ch. 8 and 9

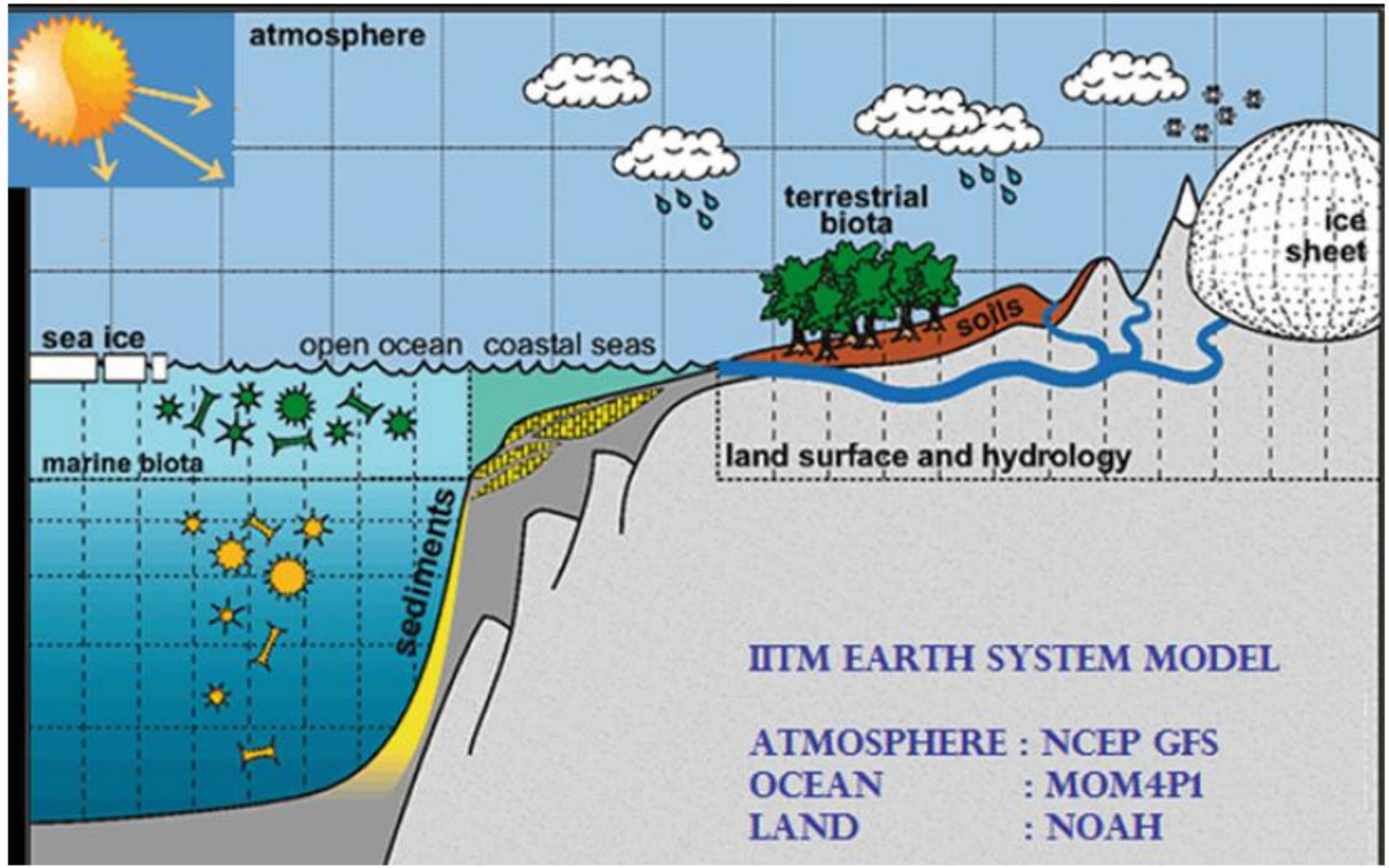


Fig. 1.1 Schematic of IITM Earth system model (IITM-ESM)

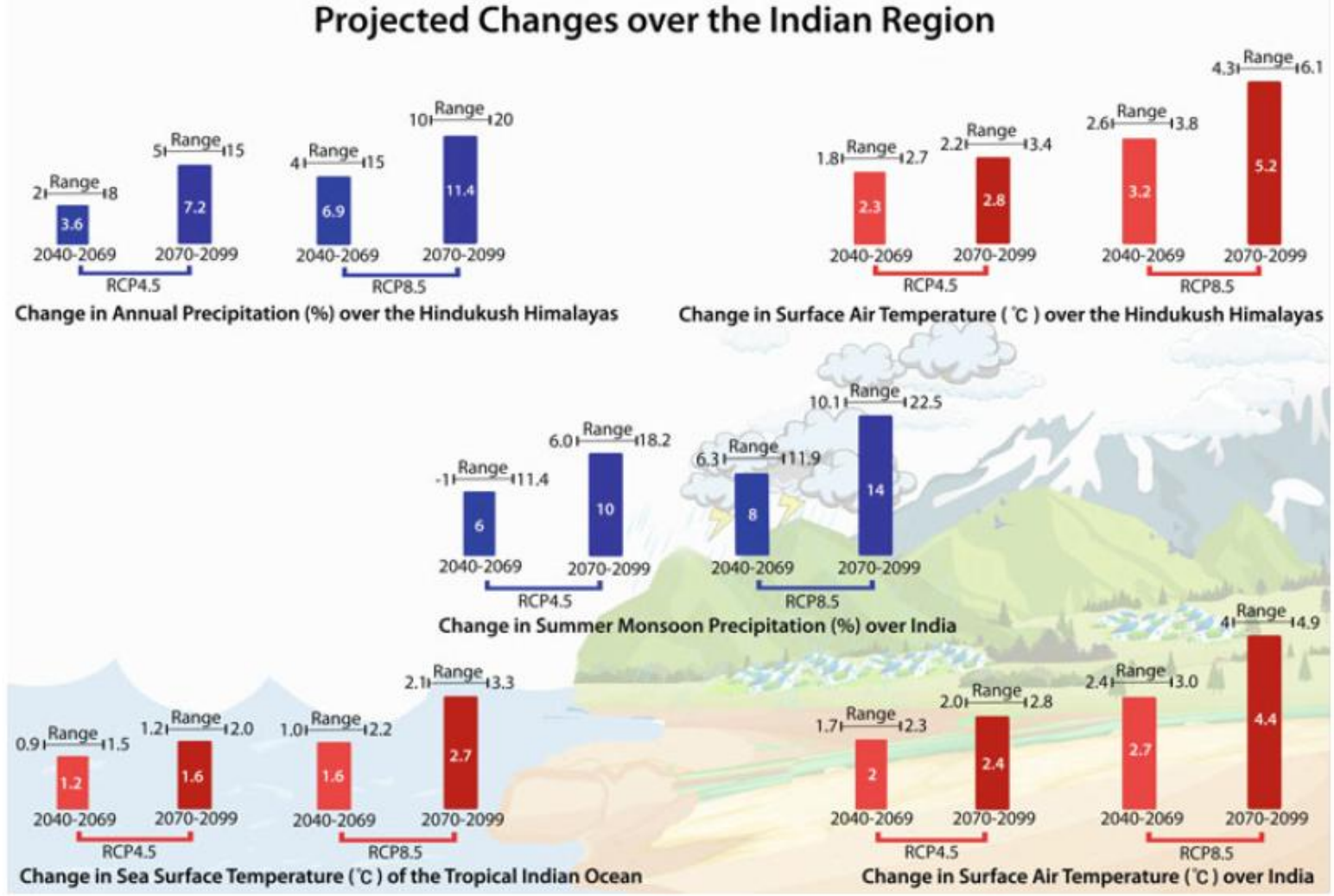


Fig. 1 Best estimate and range in climate model projections of future changes in 1. Surface air temperature over India (°C; bottom right panel), 2. Sea surface temperature of the tropical Indian Ocean (°C; bottom left panel), 3. Surface air temperature over the Hindu Kush Himalayas (°C; top right panel), 4. Summer monsoon precipitation over India (% change; centre panel), 5. Annual precipitation over the Hindu Kush Himalayas (% change; top left panel). All the changes are computed relative to their climatological average over the 30-year period 1976–2005. Projected changes are reported for the middle and end of the 21st century under the RCP4.5 and RCP8.5 scenarios (defined in Box 1). Details regarding the models and computations are discussed in the respective chapters

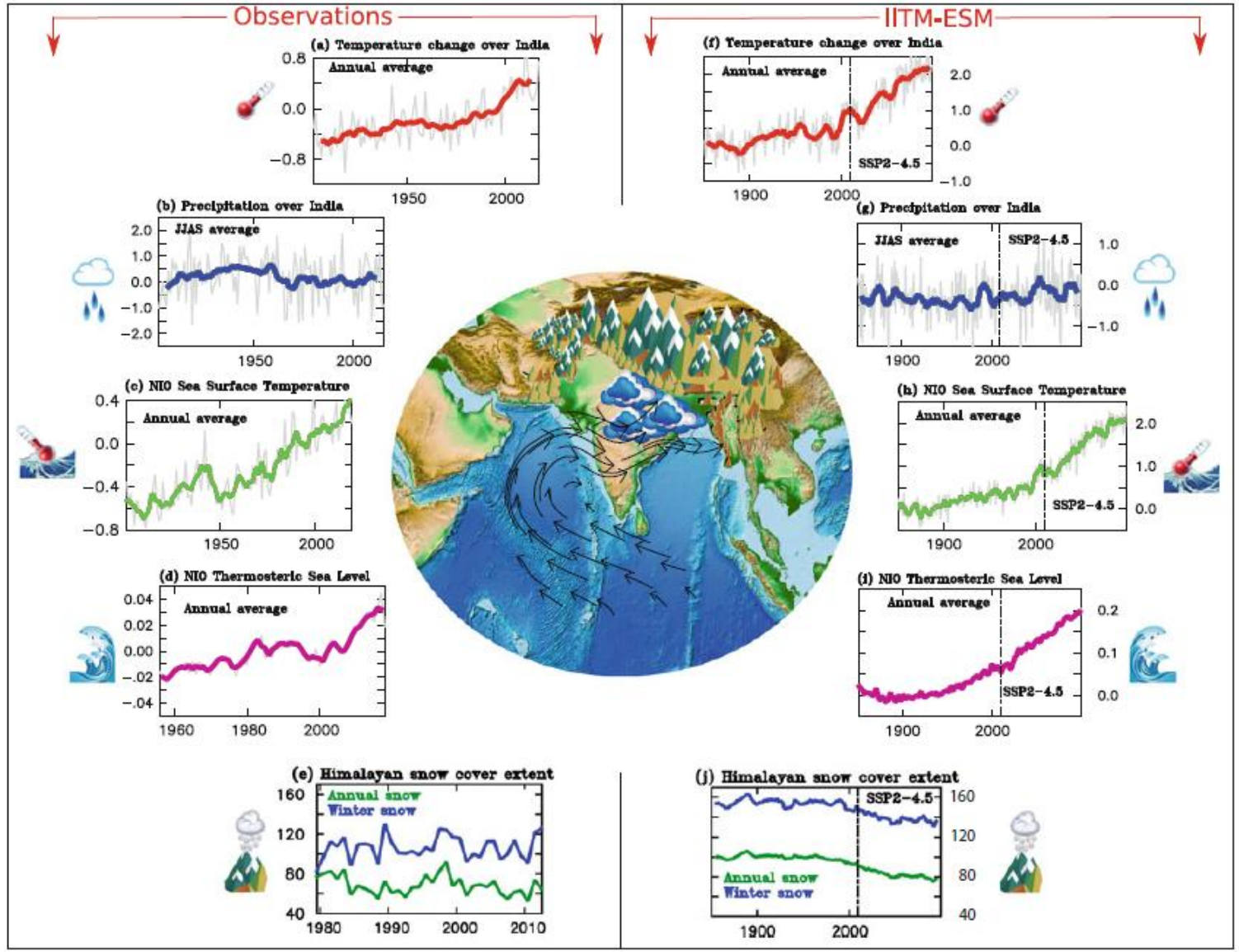


Fig. 1.6 Temporal variation of selected climate variables from observations (panels on the left) and IITM-ESM projections (panels on the right) : Near surface air temperature (°C) over India in (**a**) and (**f**); Summer monsoon precipitation (mm/day) over Indian land region in (**b**) and (**g**); Sea surface temperature of the north Indian Ocean (NIO; °C) in (**c**) and (**h**); Thermosteric sea level (m) in the north Indian Ocean in (**d**) and (**i**); and Himalayan snow cover extent (x 10000 km^2) in (**e**) and (**j**). Note that the time-series of all variables, except Himalayan snow cover extent, are shown as anomalies w.r.t. a specified baseline. Absolute values of the Himalayan snow cover extent are shown in (**e**) and (**j**). The IITM-ESM projections cover the period from 1850 to 2100, with the future projections based on the SSP2-4.5 scenario. All anomalies in the IITM-ESM projections are computed w.r.t. the pre-industrial baseline (1850–1900) and anomalies in observed changes (left panels) are w.r.t. the recent period (1976–2005)

Temperature changes in India

The climatological seasons in India are broadly classified as the winter (December–January–February), pre-monsoon (March–April–May), summer monsoon (June–July–August–September) and the post-monsoon (October–November) seasons.

'Near future' refers to the climatological average over the period 2040–2069 and is used interchangeably with 'middle of the century'. Likewise 'far future' refers to the climatological average over the period 2070–2099 and is used interchangeably with 'end of the century'.

- Surface air temperature, typically measured at 2 m above the ground.
- Annual mean, maximum and minimum temperatures averaged over India during 1986–2015 show significant warming trend of 0.15 °C, 0.15 °C and 0.13 °C per decade, respectively (high confidence).
- Pre-monsoon temperatures displayed the highest warming trend followed by post-monsoon and monsoon seasons.
- The **frequency** and **intensity** of warm days and warm nights are projected to increase over India in the next decades, while that of cold days and cold nights will decrease (high confidence). These extreme temperature indices are defined from daily temperatures as days when daily maximum (daytime) and minimum (nighttime) temperatures are above the 90th (warm) or below the 10th (cold) percentile.
- The pre-monsoon season heatwave frequency, duration, intensity and areal coverage over India are projected to substantially increase during the twenty-first century (high confidence).
- The all India averaged annual and seasonal mean specific humidity and relative humidity are found to be increasing significantly.

Temperature changes in India

Q. What is criterion for declaring heat wave?

Heat wave is considered if maximum temperature of a station reaches at least 40^0C or more for Plains and at least 30^0C or more for Hilly regions.

a) Based on Departure from Normal

Heat Wave: Departure from normal is 4.5^0C to 6.4^0C

Severe Heat Wave: Departure from normal is $>6.4^0$C

b) Based on Actual Maximum Temperature

Heat Wave: When actual maximum temperature $\geq 45^0$C

Severe Heat Wave: When actual maximum temperature $\geq 47^0$C

If above criteria met at least in 2 stations in a Meteorological sub-division for at least two consecutive days and it declared on the second day.

Q. What is a criterion for describing *Heat Wave* for coastal stations?

When maximum temperature departure is 4.5^0C or more from normal, *Heat Wave* may be described provided actual maximum temperature is 37^0C or more.

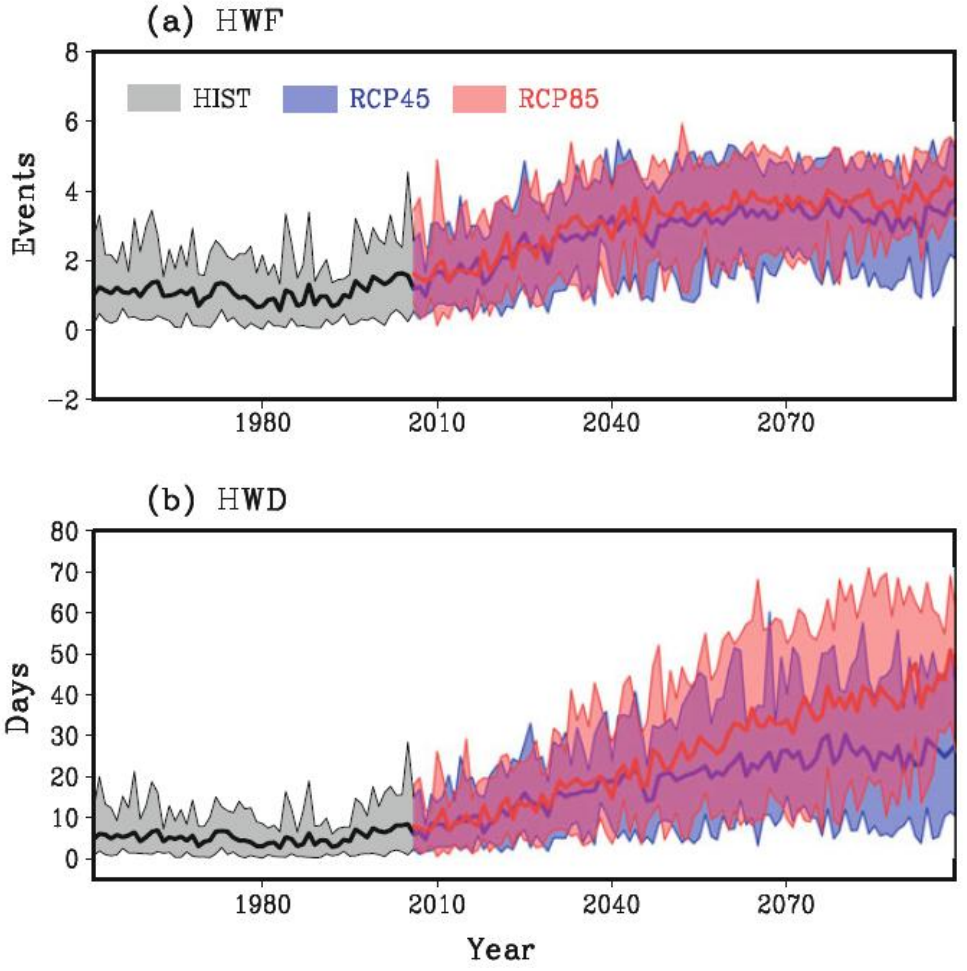


Fig. 2.10 Time series of all India averaged CORDEX South Asia multi-RCM projections of the summer (April–June) heatwave **a** frequency (HWF; events per season) and **b** total average duration (HWD; days per season) for the CORDEX South Asia RCM ensemble mean (solid lines) and the minimum to maximum range of the individual RCMs (shading) based on the historical simulations during 1951–2005 (grey), and based on the future projections during 2006–2099 under RCP4.5 scenario (blue) and RCP8.5 scenario (red)

Temperature changes in India

The annual probability of occurrence of heat waves with magnitude greater than the one in Russia in 2010 (the most severe of the present era) at 1.5 and 2 °C global warming above pre-industrial levels is assessed using CMIP5 ensemble to be different than zero in few parts of India when measured by means of a heatwave index, which takes into account both temperature and relative humidity. The yearly probability of occurrence of a heatwave at 4° global warming with magnitude greater than this most severe event in the present period will be greater than 10% in many parts of India, with the southern peninsula expected to experience such type of humid heatwaves with an annual probability greater than 50%, corresponding to an average return period of two years

Air temp, °C	Gap between dry-bulb and wet-bulb reading, °C									
	1	2	3	4	5	6	7	8	9	10
32	93	86	80	74	68	62	57	51	46	41
34	93	87	81	75	69	63	58	53	48	43
36	94	87	81	75	70	65	59	55	50	45
38	94	88	82	76	71	66	61	56	51	47
40	94	88	82	**77**	72	67	62	57	53	48
42	94	88	83	77	72	67	63	58	54	50
44	94	89	83	78	73	68	64	59	55	51

Cell = relative humidity (%). Colour = wet-bulb temperature:

below 31 °C	31 to 35 °C	35 °C or more

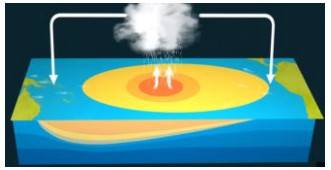
Row: air temperature on an ordinary (dry-bulb) thermometer.
Column: how much lower a thermometer wrapped in wet cloth reads. Drier air evaporates more and cools it more.
Wet-bulb = row – column. At 40 °C with a 4 °C gap: humidity 77%, wet-bulb 36 °C.
Sweat cools us the same way. At a wet-bulb of 35 °C it no longer can; lab studies suggest about 31 °C.

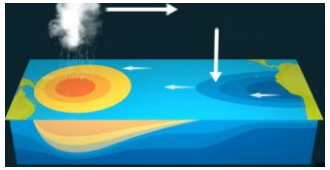
Precipitation changes in India

- There has been a decreasing trend in the all-India annual, as well as summer monsoon mean rainfall during 1951–2015, notably so over areas in the Indo-Gangetic Plains and the Western Ghats.
- The frequency of localized heavy rain occurrences over India has increased during 1951–2015 (high confidence).
- With continued global warming and expected reductions of aerosol concentrations in the future, climate models project an increase in the annual and summer monsoon mean rainfall, as well as frequency of heavy rain occurrences over most parts of India during the twenty-first century (medium confidence).
- The interannual variability of summer monsoon rainfall is projected to increase through the twenty-first century (high confidence).
- Variabilities in the Monsoon system
  - Onset and withdrawal
  - Intra-seasonal oscillations
  - Annual variations. Northeast Monsoons show greater inter-annual variability than Southwest Monsoons

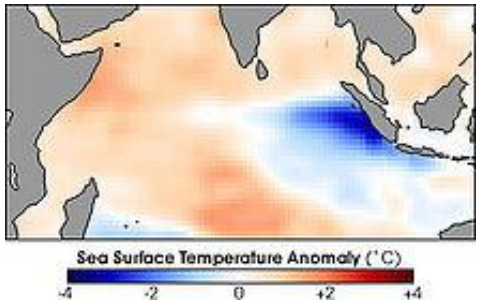
Processes influencing the Monsoon system

- Synoptic systems: Movement of low-pressure systems -> intra-seasonal variations. Influenced by land cover
- El Nino (Pacific) -> annual variations. Floods in Peru, droughts in India and Indonesia



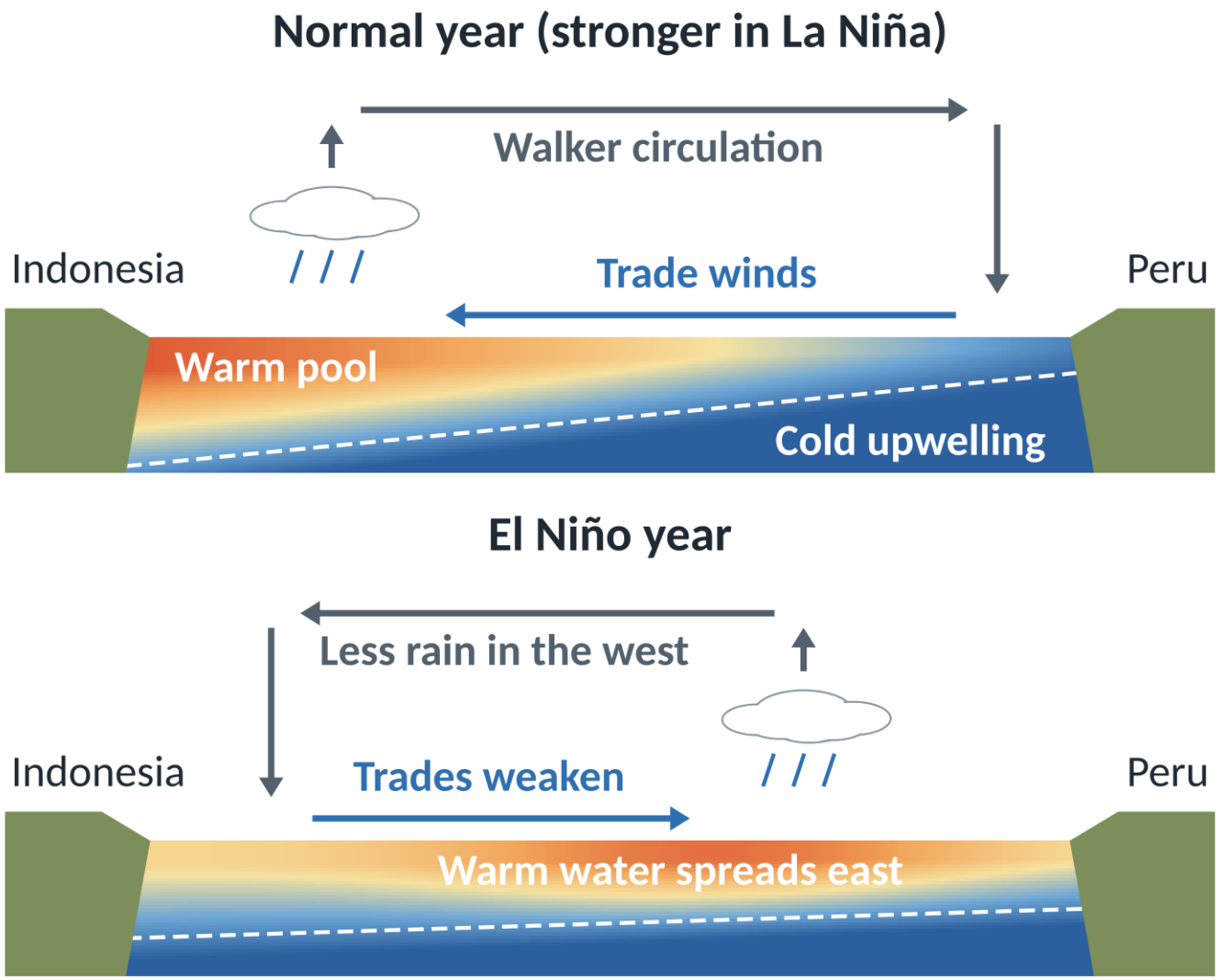


- Indian Ocean dipole: Positive IOD can dilute the effect of El Nino



- Eurasian snow cover: Positive snow cover during winter and spring tends to be followed by anomalous deficit summer monsoon
- Atlantic and Northern Pacific circulation changes also affect

El Niño and the monsoon

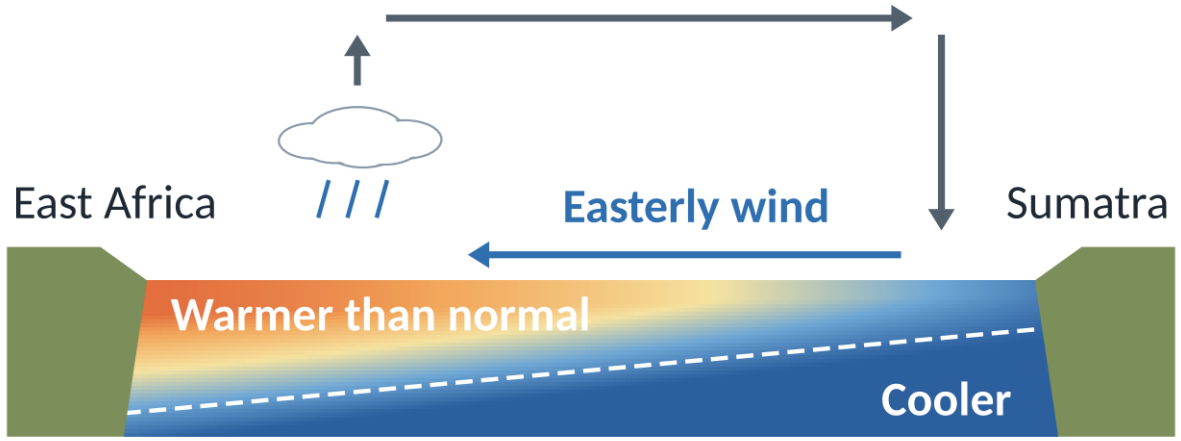


Sources: NOAA CPC; Kumar et al., Science, 2006

- **Normal:** trade winds push warm surface water west, so the warmest water (29–30 °C) and the heaviest rain sit near Indonesia. Cold water wells up off Peru.
- **El Niño:** every 2–7 years the trades slacken and warm water spreads east. Rain follows it, and air sinks over Indonesia and India, weakening the monsoon. La Niña is the reverse and usually brings good rains.
- **Index:** sea surface temperature anomaly in the Niño 3.4 box (5°N–5°S, 170°–120°W). El Niño is declared when its 3-month mean stays at or above +0.5 °C for five overlapping seasons.
- **India:** the droughts of 1965, 1972, 1982, 1987, 2002, 2009 and 2015 all fell in El Niño years, yet the very strong 1997 El Niño gave normal rain. Warming centred in the central Pacific does more damage than warming in the east.

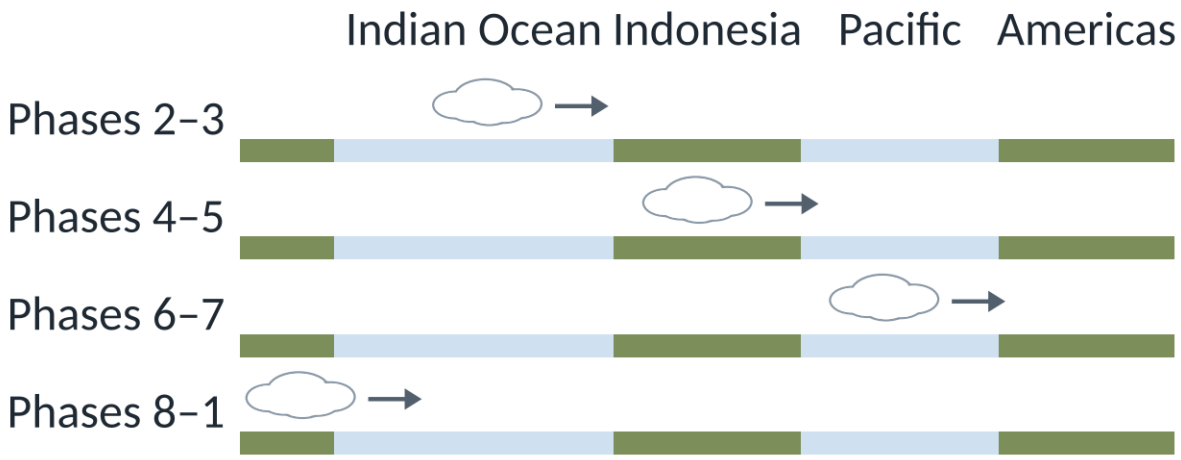
Indian Ocean Dipole and MJO

Positive Indian Ocean Dipole



- **What:** an east–west seesaw in Indian Ocean surface temperature. Index (DMI) = anomaly in the west (50–70°E, 10°S–10°N) minus the east off Sumatra (90–110°E, 10°S–0°).
- **Positive phase:** warm west, cool east. Rain shifts towards East Africa and India; Indonesia and Australia turn dry. Builds from May–June, peaks September–November.
- **Offsets El Niño:** 1997 had normal rain; in 2019 a strong positive IOD came with 110% of average monsoon rain.

Madden–Julian Oscillation



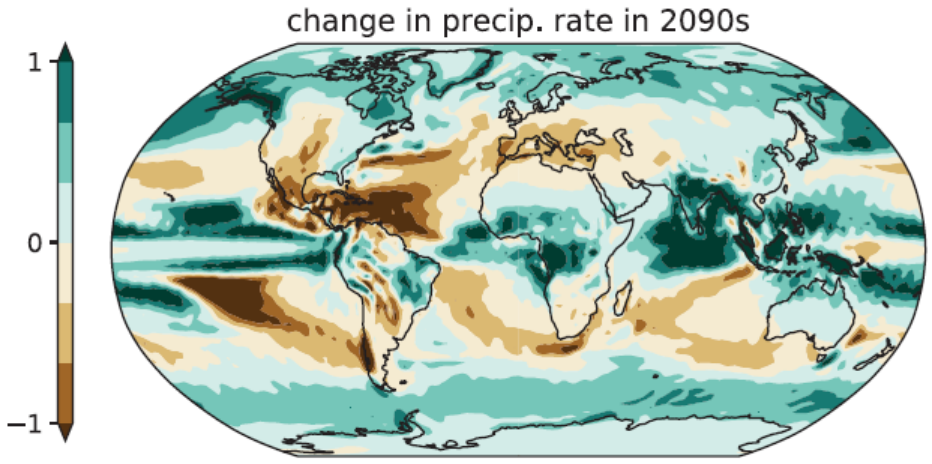
- **What:** a pulse of cloud and rain several thousand km wide that travels east along the equator at about 5 m/s, circling the globe in 30–60 days. Tracked in eight phases (RMM index).
- **Monsoon:** in summer the pulse also moves north over India, giving the active and break spells. It affects onset and cyclone formation.
- **Timescales:** ENSO and IOD shape the season's total; the MJO shapes its timing within the season.

Sources: Saji et al., Nature, 1999; Madden and Julian, 1971; Wheeler and Hendon, 2004; IMD monsoon reports

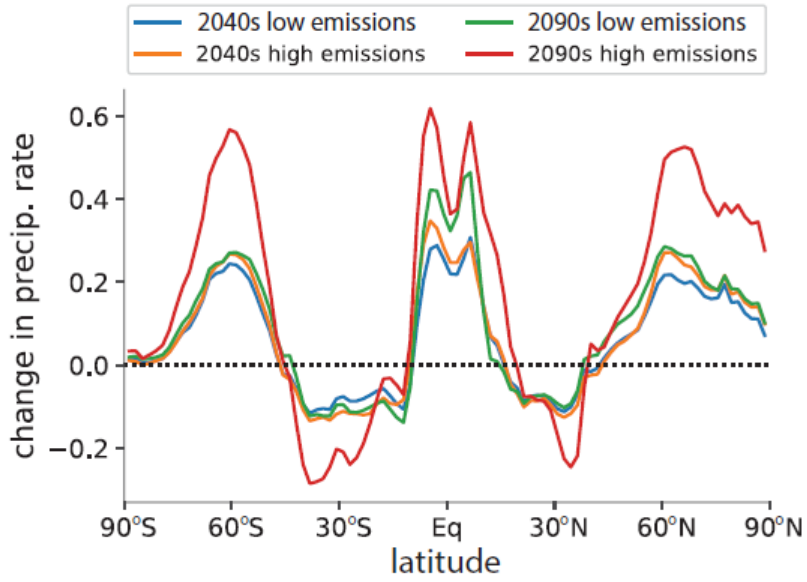
Why both floods and droughts increase

- **More evaporation:** a warmer surface evaporates more, so global precipitation rises a few percent per °C.
- **Wet gets wetter, dry gets drier:** most latitudes gain rain, but the dry belts near 30° lose it.
- **Heavier downpours:** more of the rain falls in intense events. Soil saturates, the rest runs off: floods, erosion, less water stored.
- **Longer gaps:** fewer rain days plus faster drying of soil means more drought.
- **Snow to rain:** winter rain runs off at once, leaving less meltwater for summer.

(a)

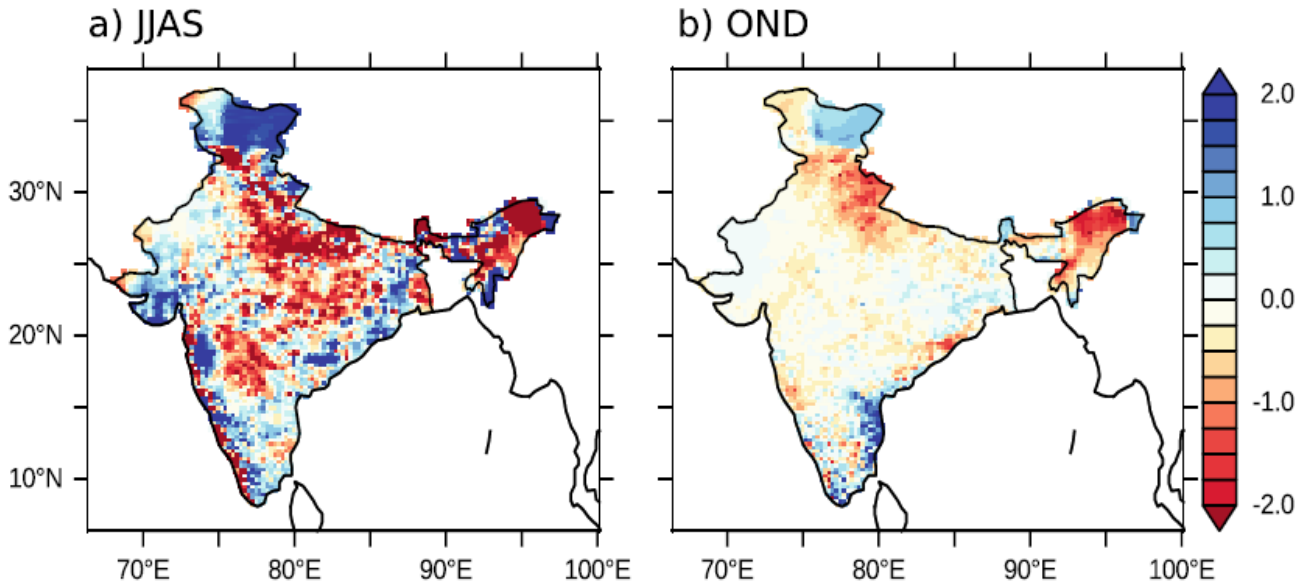


(b)



Precipitation changes in India

Fig. 3.4 Linear trends (mm/day over 64 years) in the southwest (left) and northeast (right) monsoon rainfall from 1951 to 2015 based on IMD data



What the colour scale shows

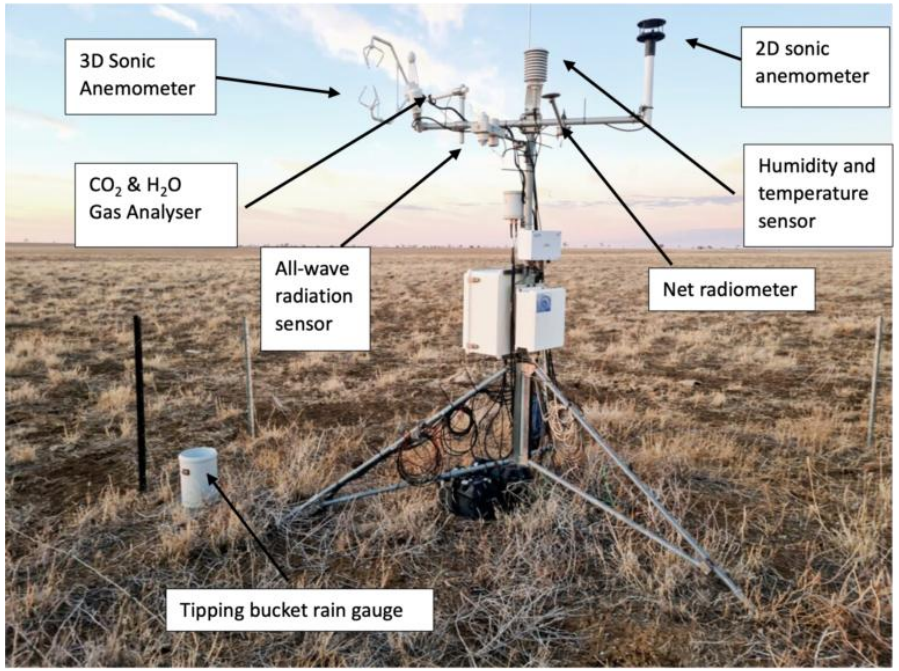
Change in the season's average daily rainfall between 1951 and 2015, in mm/day, from a straight-line fit at each grid cell.

Red is drier, blue wetter. −2 means about 240 mm less rain over the 122-day June–September season.

- A statistically significant increase in the frequency of dry spells (27% higher during 1981–2011 relative to 1951–1980) and intensity of wet spells and statistically significant decreases in the intensity of dry spells have been observed in recent six decades
- The present models fail to adequately simulate the intra-seasonal variability of monsoon. Increased extremes and increased spatial variability have been observed in recent decades and also projected to increase, and it is likely due to regional forcings such as aerosols and land-use/urbanization changes.

GHG concentrations and fluxes over India

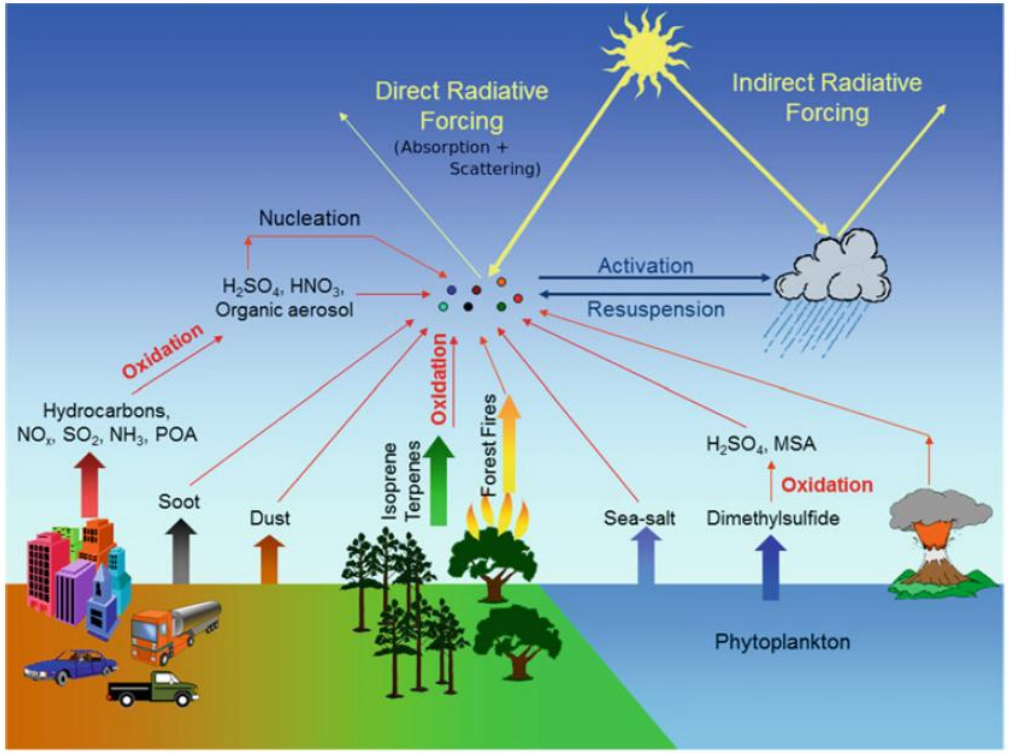
- CO_2 leaves the atmosphere slowly: half in about 50 years, but a quarter remains after 500 years and some beyond 10,000 years. CH_4 and N_2O last about 12 and 110 years. The increase in CH_4 concentration is mainly dominated by agricultural and animal husbandry operations, of N_2O is due to agricultural practices.
- The Kaziranga forest in Northeast India sequesters maximum carbon during pre-monsoon season, whereas the forests in Haldwani and Barkot in northern India sequester maximum carbon during the summer monsoon season. The forests in Betul in central India, mangroves in Sundarbans in east India and forests in Kosi-Katarmal in north India sequester maximum carbon during post-monsoon, whereas mangroves in Pichavaram in the east coast of south India sequester maximum carbon during the winter season.



- Modeling studies show that even though the Indian terrestrial ecosystem has not historically been a strong source or sink of carbon, it is behaving as a carbon sink since the 1980s. The terrestrial carbon sink is maintained primarily by the carbon fertilization effect aided to some extent by forest conservation, management, and reforestation policies in recent decades.
- Fertilizers such as urea, ammonium sulfate and potassium nitrate increased the CH_4 emission, while dicyandiamide helped reduce it. Intermittently irrigated rice paddy fields emit less CH_4 compared to the continuously flooded fields.
- Presently, agricultural activities alone account for more than 90% of the total anthropogenic N_2O emissions in India, with 65% of the total N_2O emission being attributable to the chemical fertilizers.

Aerosols and trace gases over India

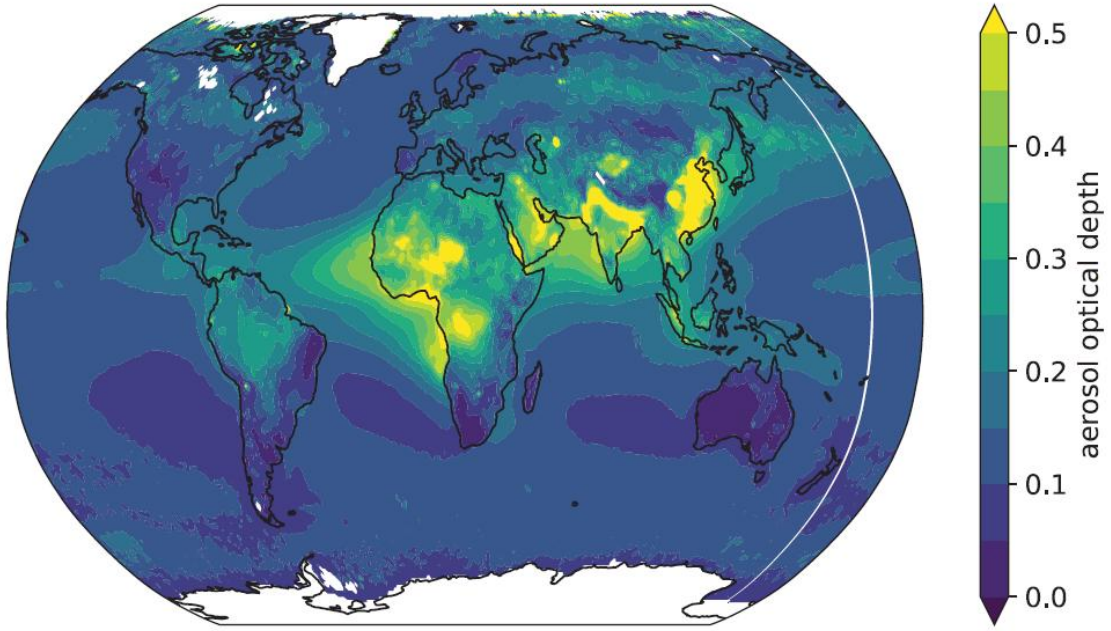
- Aerosol loading over India has substantially increased during the recent few decades. The annual mean 500 nm aerosol optical depth (AOD) from ground-based observations shows an overall increasing trend of about 2% / year during the last 30 years (high confidence). This trend in AOD is subject to seasonal variability. The rate of increase in AOD is significantly high during the dry months of December–March.



- Non-absorbing aerosols like sulfate, nitrate, and sea spray scatter shortwave radiation back to space leading to a net cooling of the climate system while absorbing aerosols produce the opposite effect. Carbonaceous aerosols (black carbon, organic carbon) and mineral dust can absorb and scatter sunlight producing either warming or cooling effects determined by aerosol properties and environmental conditions. Absorbing aerosols also affect climate when present in surface snow by lowering surface albedo, yielding a positive radiative forcing, directly changing the melting of snow and ice.
- Ozone acts as a greenhouse gas in the troposphere, while in the stratosphere, filter out the incoming ultraviolet radiation coming from the Sun. Thus, it helps in protecting life on the Earth. The human-made processes have injected new trace gases into the atmosphere, for example, chlorofluorocarbons (CFCs), which damage the ozone layer in the stratosphere.

Aerosols: cooling that masks warming

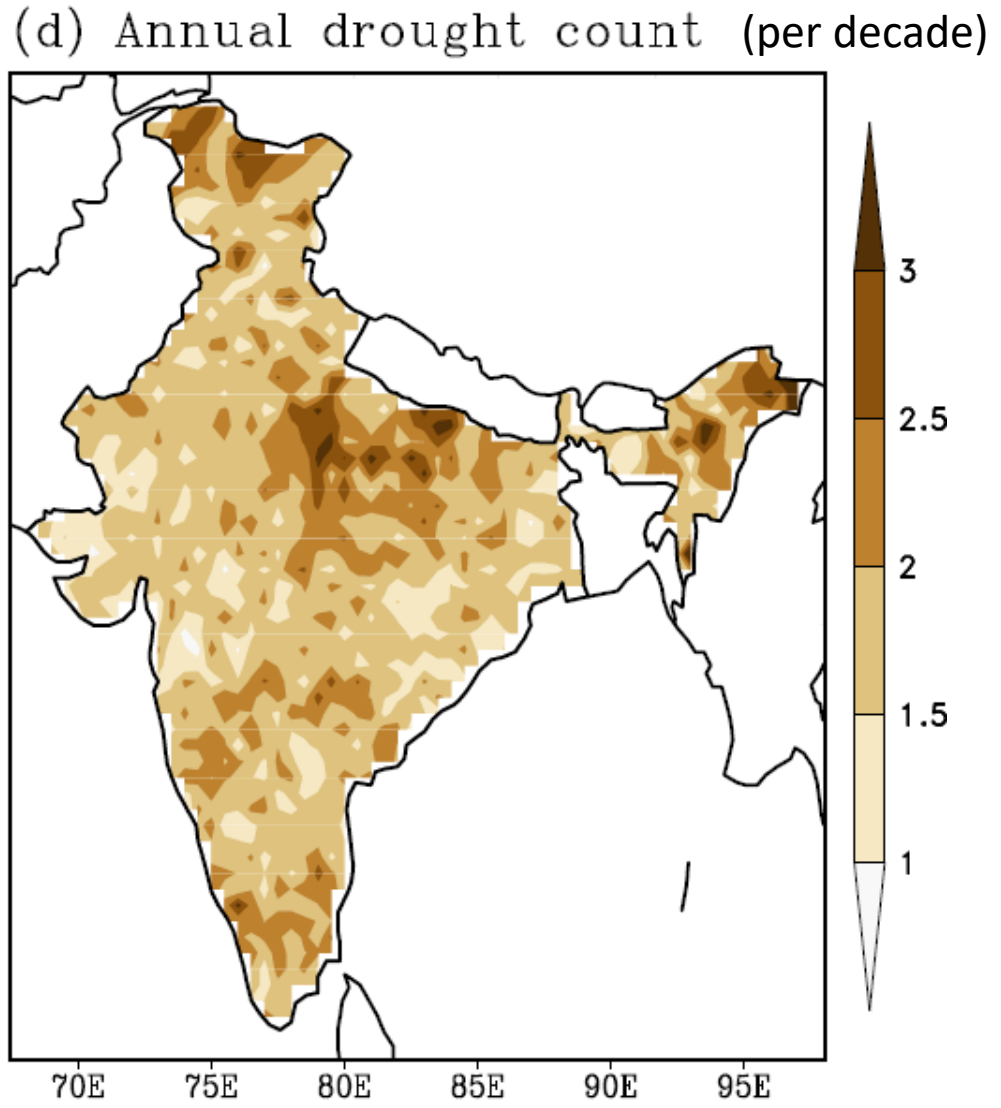
- **Types:** sulfate from burning sulfur-bearing fuel reflects sunlight (cools); black carbon absorbs it (warms); dust cools.
- **Direct effect:** about −0.3 W/m².
- **Cloud effect:** aerosols seed more, smaller droplets, making clouds brighter and longer-lived: about −0.9 W/m². Ship tracks show it.
- **Net:** −1.1 W/m², hiding about 30% of greenhouse warming.
- **Short-lived and local:** aerosols last weeks, so they pile up near sources, heavily over Asia.
- **Trade-off:** cleaner air saves millions of lives but unmasks warming.



Source: Dessler (2021), Ch. 6

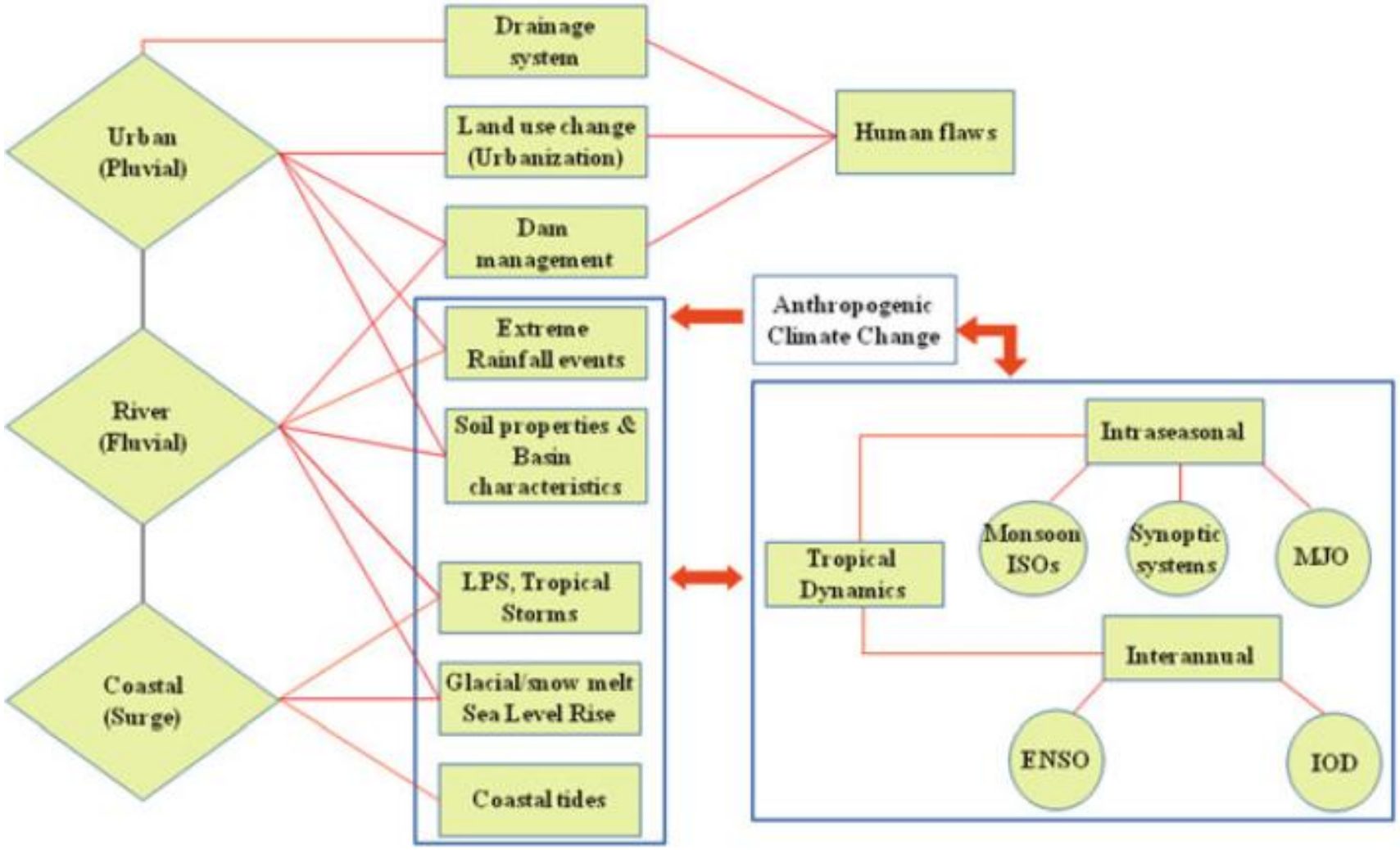
Droughts and floods

- The frequency and spatial extent of droughts over India have increased significantly during 1951–2015. An increase in drought severity is observed mainly over the central parts of India, including parts of Indo-Gangetic Plains (high confidence). These changes are consistent with the observed decline in the mean summer monsoon rainfall.
- Increased frequency of localized heavy rainfall on sub-daily and daily timescales has enhanced flood risk over India (high confidence). Increased frequency and impacts of floods are also on the rise in urban areas.
- Climate model projections indicate an increase in frequency, spatial extent and severity of droughts over India during the twenty-first century (medium confidence), while flood propensity is projected to increase over the major Himalayan river basins (e.g. Indus, Ganga and Brahmaputra) (high confidence).
- Droughts are broadly categorized into four major classes: (1) meteorological drought, as a deficit in precipitation; (2) hydrological drought, as a deficit in streamflow, groundwater level or water storage; (3) agricultural drought, as a deficit in soil moisture; and (4) socio-economic drought, incorporating water supply and demand.



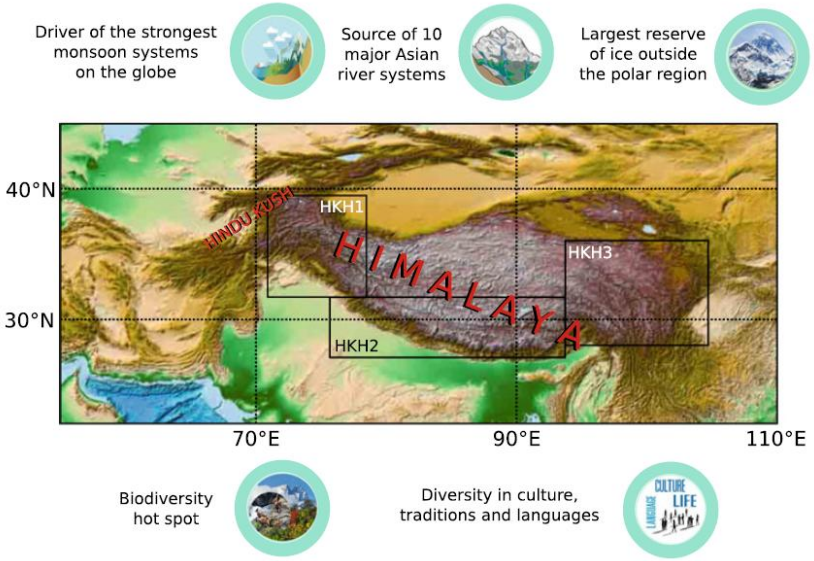
Droughts and floods

- Floods are classified into different types such as riverine (extreme rainfall for longer periods), flash (heavy rainfall in cities or steep slopes), urban (lack of drainage), coastal (storm surge) and pluvial (rainfall over a flat surface) flooding.
- Due to the proximity of Indus-Ganga-Brahmaputra river basins to the foothills of the Himalayas, the run-off is projected to increase primarily by an increase in precipitation and accelerated meltwater in a warming environment, at least until 2050.

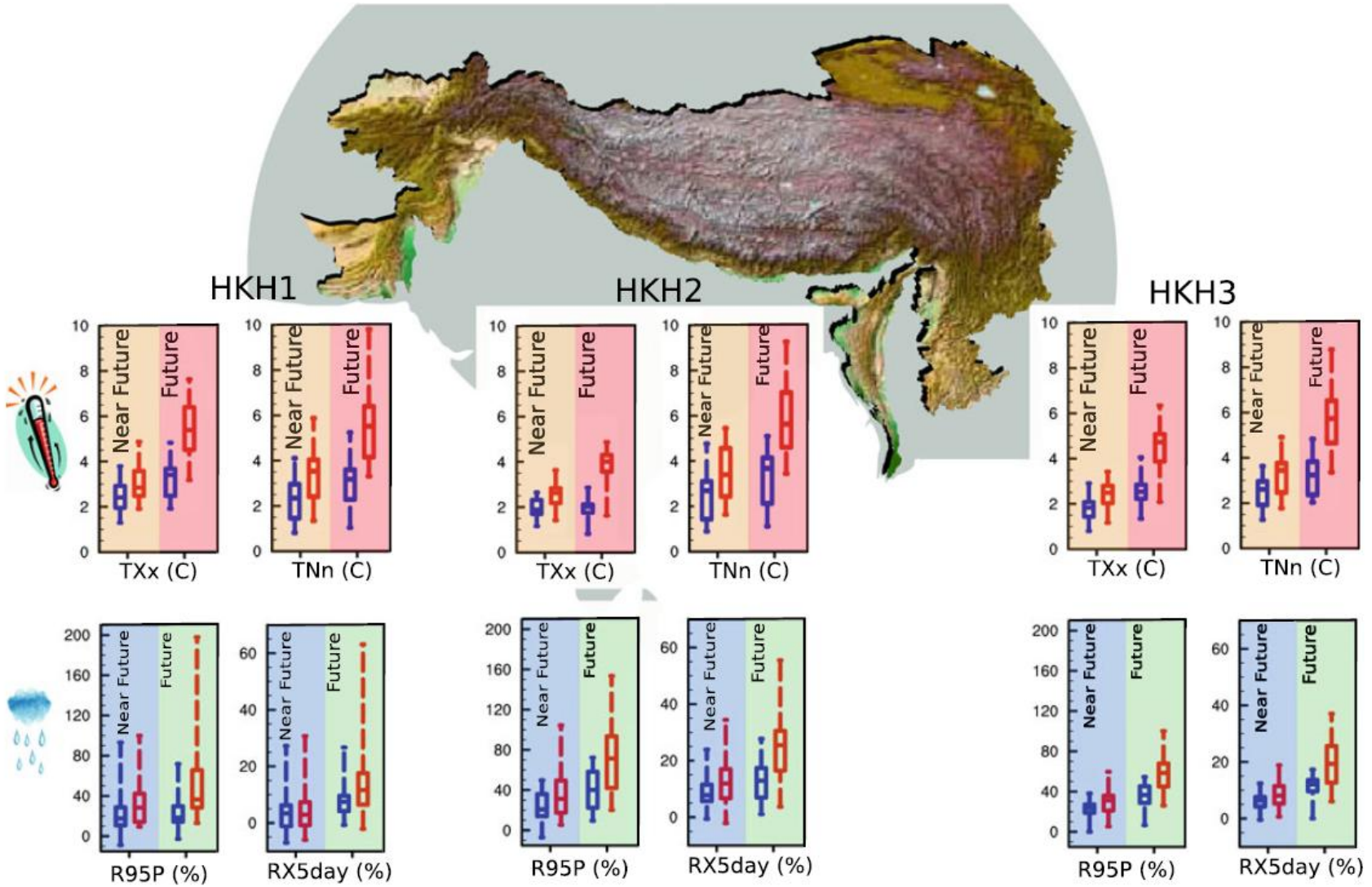


Climate change over the Himalayas

- The Himalayas and the Tibetan Plateau have experienced substantial warming during the twentieth century. The warming trend has been particularly pronounced over the Hindu Kush Himalaya (HKH) which is the largest area of permanent ice cover outside the North and South Poles.
- The annual mean surface-air-temperature in the HKH increased at a rate of about 0.1 °C per decade during 1901–2014, with a faster rate of warming of about 0.2 °C per decade during 1951–2014, which is attributable to anthropogenic climate change (High confidence). Additionally, high elevations (> 4000 m) of the Tibetan Plateau have experienced stronger warming, as high as 0.5 °C per decade, which is commonly referred to as elevation-dependent warming (EDW).
- Several areas in the HKH have exhibited declining trend in snowfall and retreating glaciers during the recent decades. Parts of the high-elevation Karakoram Himalayas have, in contrast, experienced increased wintertime precipitation in association with enhanced amplitude variations of synoptic western disturbances (Medium confidence).



- Future climate projections under various CMIP5 scenarios suggest warming of the HKH region in the range of 2.6–4.6 °C by the end of the twenty-first century. While future projections indicate significant decrease of snowfall in several regions of the HKH, high-elevation locations (> 4000 m) in the Karakoram Himalayas are projected to experience an increase in annual precipitation during the twenty-first century.



Reading the box plots

TXx (°C): hottest day of the year, the annual maximum of daily maximum temperature.

TNn (°C): coldest night of the year, the annual minimum of daily minimum temperature.

R95P (%): rain falling on very wet days, the annual total from days above the 95th percentile of daily rainfall.

RX5day (%): wettest five-day spell, the largest rainfall total over five consecutive days in the year.

Each panel shows the projected change from the 1976–2005 average in CMIP5 models, for the near future (2040–2069) and far future (2070–2099). Blue boxes are RCP4.5, red boxes RCP8.5; the box shows the spread across models. HKH1 = northwest Himalaya and Karakoram, HKH2 = central Himalaya, HKH3 = southeast Himalaya and Tibetan Plateau.

Other changes

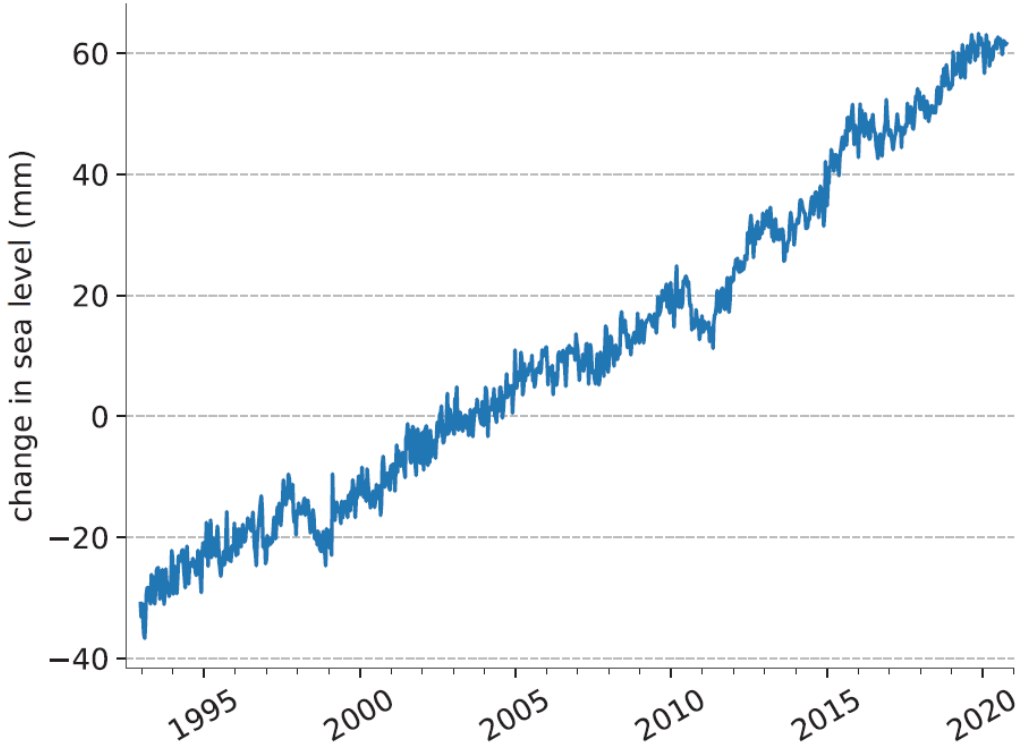
- Increased frequency and intensity of storms: cyclones, thunderstorms (localized convective storms), dust storms, cloudbursts
- Sea level rise. It is very likely that in the twenty-first century and beyond, the sea-level change will have a strong regional pattern, with some places experiencing significant deviations from the global mean sea-level rise. The underlying causes of regional sea-level changes are associated with dynamic variations in the ocean circulation as part of climate modes of variability, changes in the wind pattern and with an adjustment of Earth's crust to past and ongoing changes in polar ice masses and continental water storage.
- The tropical Indian Ocean has experienced rapid basin-wide sea surface temperature (SST) warming, and non-uniform SST warming trend in the tropical Indian Ocean is to continue in the future, under both medium and high emission scenarios (high confidence). Note that more than 90% of additional heat from global warming is stored in the ocean, increasing ocean heat content (OHC), while the residual heat is manifested in the form of melting of both land and sea ice, and in warming of the atmosphere and land surface.

Sea-level rise: causes and commitment

47–73 cm
projected rise by 2100, relative to 1995–2014

3.2 mm/yr
observed rate since 1993, and accelerating

- **Two causes:** melting ice on land adds water; warming water expands. Melting floating ice adds almost nothing.
- **Why it matters:** storm surges reach further inland and high-tide flooding becomes frequent.
- **Long commitment:** ice responds slowly. Past climates suggest a few metres per °C over millennia (ice age: 6 °C colder, 100 m lower).



Source: Dessler (2021), Ch. 2 and 9

Ocean acidification and tropical cyclones

Ocean acidification (certain)

CO_2 absorbed by the ocean forms carbonic acid, lowering pH. Surface pH is already the lowest in 2 million years.

Less carbonate is available, so corals and shell-builders struggle. About a billion people rely on the ocean for protein.

It happened before: the PETM dissolved sea-floor carbonate.

Tropical cyclones

Storm surge: worse as seas rise (certain).

Rainfall: heavier, because warmer oceans evaporate more (high confidence).

Number: about the same or fewer (moderate confidence).

Intensity: the storms that form will be stronger (moderate confidence).

- **Note:** reflecting sunlight to cool the planet would not stop acidification; only cutting CO_2 does.

Source: Dessler (2021), Ch. 9

Abrupt climate changes

- **Tipping points:** like a canoe that tilts and then suddenly flips, the climate may shift state within decades once a threshold is crossed.
- **Precedent:** the Younger Dryas, about 12,000 years ago, when meltwater disrupted Atlantic circulation and temperatures plunged.

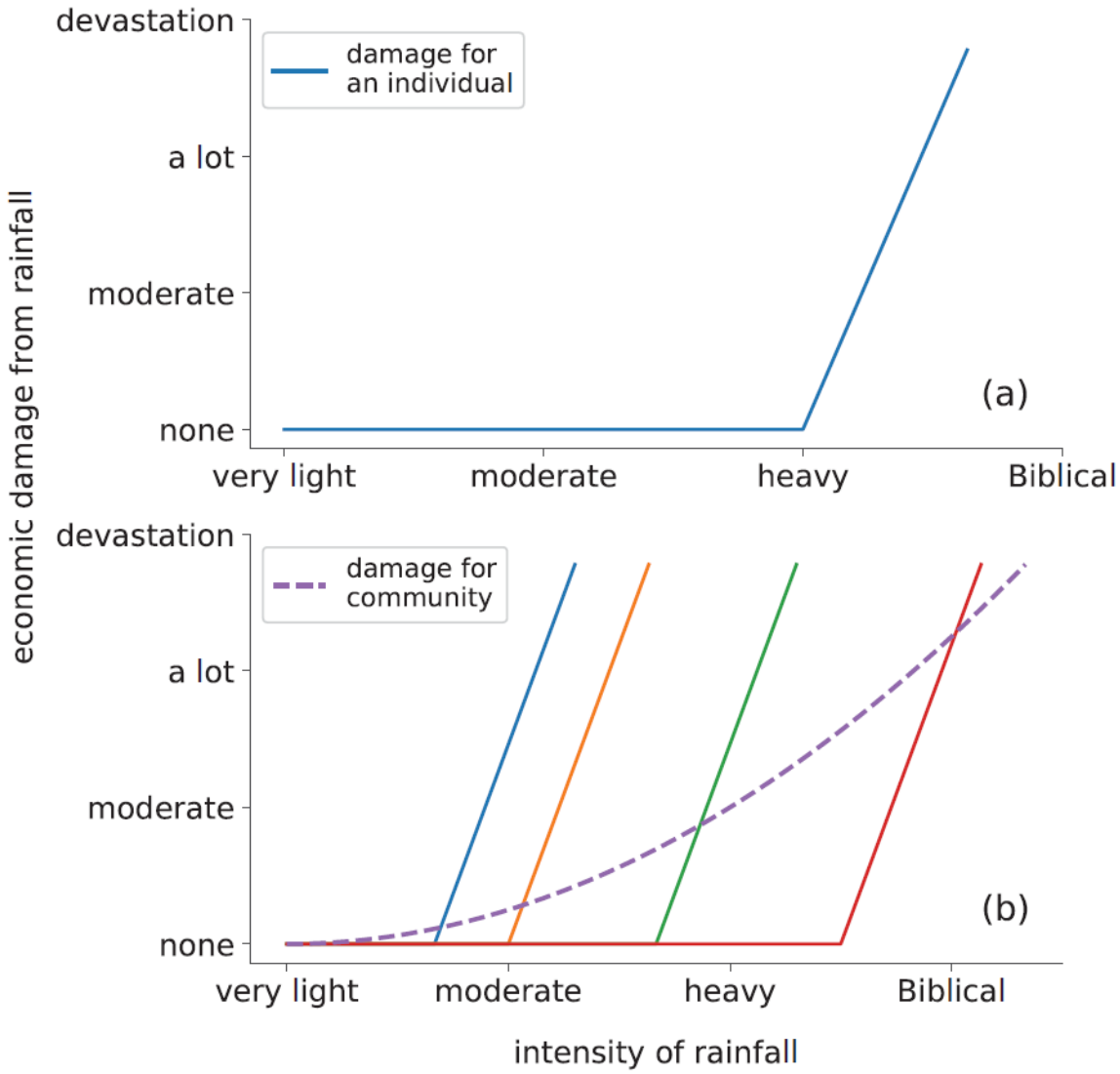
Possible abrupt change	Consequence
Shutdown of Atlantic currents (Gulf Stream)	Rapid, widespread shifts in regional climate
Collapse of West Antarctic or Greenland ice	Several metres of sea-level rise in a short time
Amazon dieback	Ecological and economic catastrophe for the region
Thawing permafrost and methane hydrates	Large greenhouse gas release that accelerates warming
Shift in the Indian monsoon	Changed timing and amount of rain that billions rely on

- **Low probability this century, very high consequence,** and hard to manage once under way.

Source: Dessler (2021), Ch. 9

Impacts are non-linear: fine until it isn't

- **Thresholds:** heavier rain does no harm until it exceeds what drains and defences can handle. Then damage is sudden and large.
- **Communities:** people have different thresholds, so each extra bit of change pushes more of them over. Total damage curves upward.



Source: Dessler (2021), Ch. 9

Adaptation is costly and has limits

Adaptation costs

Raising one house: about $175,000.

Sea barriers for one city: tens of billions of dollars.

Not adapting costs too: repairs after every storm.

Unmanaged systems

Acidifying oceans, species loss (40% of lizard populations by 2080).

Ecosystem services lost: mangroves that blunt storm surge, wild bees that pollinate crops.

Already happening

Hurricane Harvey: $125 billion.

California fires 2018: $148.5 billion.

Heat-related deaths: about 300,000 in 2018.

Work lost to heat: 300 billion hours in 2019.

- **A climate tax we already pay:** the choice is between the cost of acting and the cost of not acting, at only about 1 °C of warming so far.

Source: Dessler (2021), Ch. 9

Impacts and policy messages

- The impact of climate change on the availability of fresh water is a critical area of concern for India. The growing propensity for droughts and floods because of changing rainfall patterns caused by climate change would be detrimental to surface and groundwater recharge, posing threats to the country's water security.
- Likewise, the country's food security may be placed under progressively greater pressure due to rising temperatures, heat extremes, floods, droughts and increasing year-to-year rainfall variability that can disrupt rain-fed agricultural food production and adversely impact crop yield.
- Higher temperatures, extreme weather events, and higher climate variability have been associated with an elevated risk of heat strokes, cardiovascular and neurological diseases, and stress-related disorders. Heat stress in urban areas is often compounded by the heat island effect. Warmer, higher moisture conditions, on average, are also more favourable for the spread of vector-borne diseases such as malaria and dengue fever. In addition, a decrease in the availability or affordability of food and potable water caused by climate change may lead to reduced nutritional intake, particularly among economically weaker sections.
- India's long coastline, where some of its largest cities are located, is among the most densely populated regions of the planet, making it exceedingly vulnerable to the impacts of sea-level rise. Potential coastal risks include loss of land due to increased erosion, damage to coastal projects and infrastructure such as buildings, roads, monuments, and power plants, salinization of freshwater supplies and a heightened vulnerability to flooding.
- With the climate changing more rapidly than the usual pace of evolutionary adaptability of many species, they may face increasing threats on account of these changes.
- The risks posed by climate change can be considerably magnified when a cascade of climate-related hazards overlap or follow one another. For instance, a region may experience an abnormally long or intense summer heat wave followed by intense monsoon floods that alternate with lengthening dry spells.

Impacts and policy messages

- India’s climate change adaptation and mitigation response is achievable by a greater emphasis on widening observational networks, sustained monitoring, expanding research on regional changes in climate and their impacts, development of integrated, multiscale models that are capable of aiding predictions, scenario synthesis and providing information required for vulnerability assessment, and by continued investment in education and outreach programs including national and international collaborative partnerships.
- Outreach and communication of climate change risk to district and village-level communities would facilitate water-harvesting and farming decisions needed to adapt to a changing climate.
- Additionally, dedicated educational programmes from school to university would vastly improve awareness about climate change and its implications for humans and natural ecosystems. Such programs could encourage young minds to contribute through individual and collective efforts that are crucial for climate action.
- It is necessary to develop useful to usable (U2U) research and application agenda that can translate research to on-ground, effective decision tools for adapting to climatic change. Specific examples include passive reduction of indoor temperatures, water conservation and rainwater harvesting, groundwater regulation, reversing land degradation, reduction in food and water wastage, waste segregation and recycling, low impact urban development, expansion of urban green spaces and urban farming, pollution control, increasing the area under irrigation and improving the efficiency of agricultural water use, forest conservation and proactive afforestation, construction of coastal embankments and mangrove restoration, improvement in disaster response, phasing out fossil fuels and transition to renewables, electrification, expansion of walking, bicycling and public transport infrastructure, and carbon taxation, among others.
- Equity and social justice are critical to building climate resilience since the most vulnerable people such as the poor, the disabled, outdoor labourers and farmers will bear the brunt of climate change impacts.

Impacts and policy messages

- An increase in rainfall together with measures for water harvesting would aid the restoration of groundwater levels. Restoration of groundwater levels would not only improve water security and resilience to droughts but also help check land subsidence, consequently reducing the impacts of sea-level rise and storm surge.
- Ambitious afforestation efforts likewise offer myriad benefits. Aside from mitigating climate change through carbon sequestration, trees also enhance resilience to flash floods and landslides by improving soil retention, improve resilience to droughts by increasing percolation of surface water into the soil, improve resilience of coastal infrastructure and habitation by reducing coastline erosion due to storm surges and sea-level rise, reduce vulnerability to extreme heat by reducing ambient temperatures, and support native wildlife and biodiversity.

Part 3: Emissions and climate scenarios

Part 1: The physics of climate change

What climate is, evidence of warming, energy balance, the greenhouse effect, the carbon cycle, forcing and feedbacks, attribution

Part 2: Climate change over the Indian region

Models, temperature, monsoon and rainfall, greenhouse gases and aerosols, droughts and floods, Himalayas, oceans, impacts

Part 3: Emissions and climate scenarios

What drives emissions, energy and carbon intensity, SSPs and RCPs, projected warming, impact models by sector

Part 4: Economics and policy

Exponential growth and discounting, response options, carbon pricing, history of science and negotiations, long-term goals

What controls emissions

Emissions		**Population**		**Affluence**		**Energy intensity**		**Carbon intensity**
CO_2 per year	=	people	×	GDP per person	×	joules per $ of GDP	×	CO_2 per joule

Technology term: CO_2 per $ of GDP

- **Why consumption matters:** every good or service takes energy to produce, and most energy still comes from burning fossil fuels.
- **Population × affluence = GDP.** In recessions (2008, 2020) emissions fall with it.
- **Very unequal:** per person in 2017, the United States emitted 16.2 tonnes of CO_2, China 6.9 and Nigeria 0.6.
- **Known as** the IPAT relation (Impact = Population × Affluence × Technology) or the Kaya identity.
- **Units:** 1 GtC = 3.67 $GtCO_2$. Human emissions are about 11 GtC, or 40 $GtCO_2$, a year.

Source: Dessler (2021), Ch. 8

Energy intensity and carbon intensity

Energy intensity (J per $)

How much energy the economy needs to produce a dollar of output.

Depends on the mix of activity: a steel mill needs far more energy per dollar than a university.

Depends on efficiency: an LED gives the light of a 60 W bulb for 10 W. Efficient technology often costs more up front.

Carbon intensity (CO_2 per J)

How much CO_2 is released per unit of energy.

Depends on the energy sources: coal emits about twice as much per joule as natural gas, with oil in between.

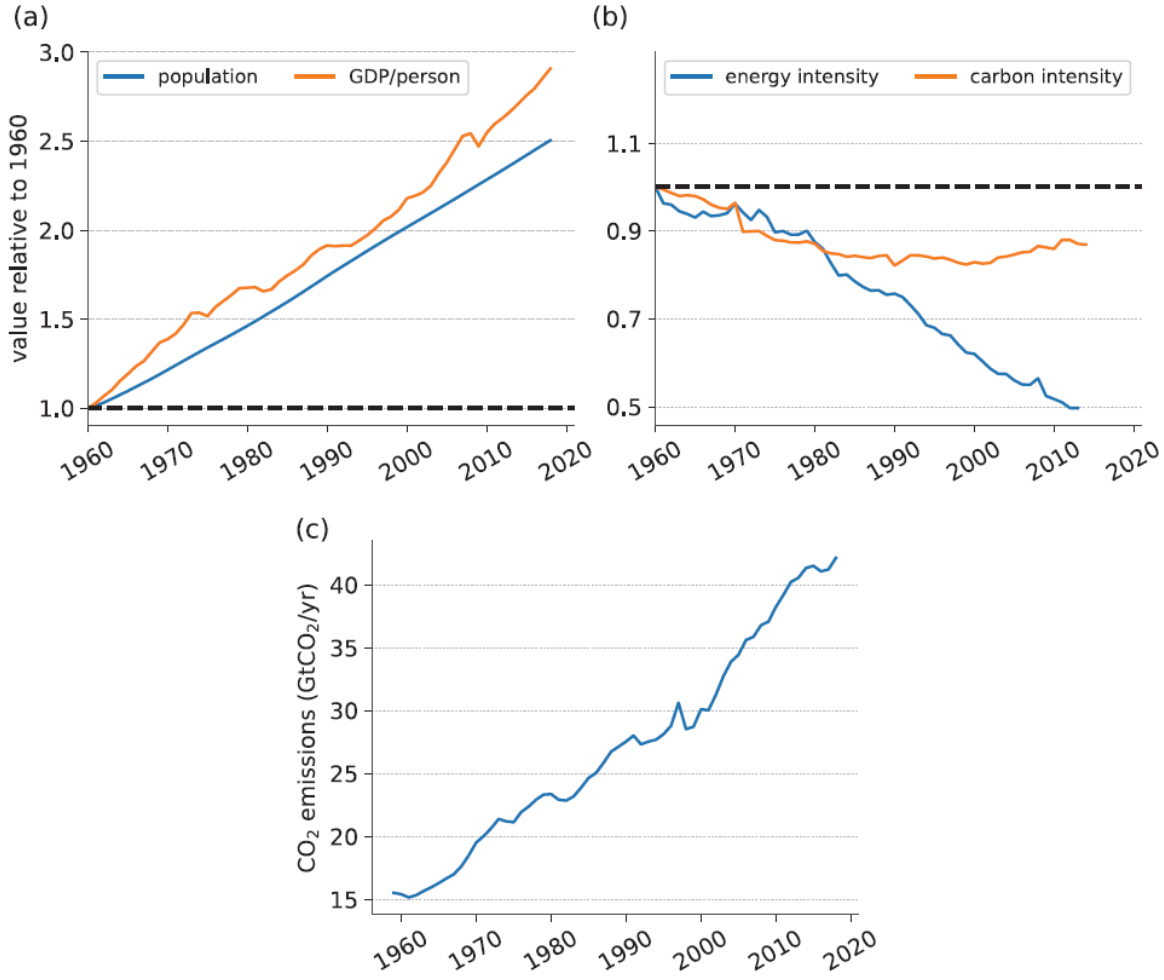
Hydro, nuclear, wind and solar emit none. France (nuclear) has a far lower carbon intensity than China or the United States.

- **Greenhouse-gas intensity = energy intensity × carbon intensity.** These two are the levers that technology and policy can pull.

Source: Dessler (2021), Ch. 8

How the four factors have changed since 1960

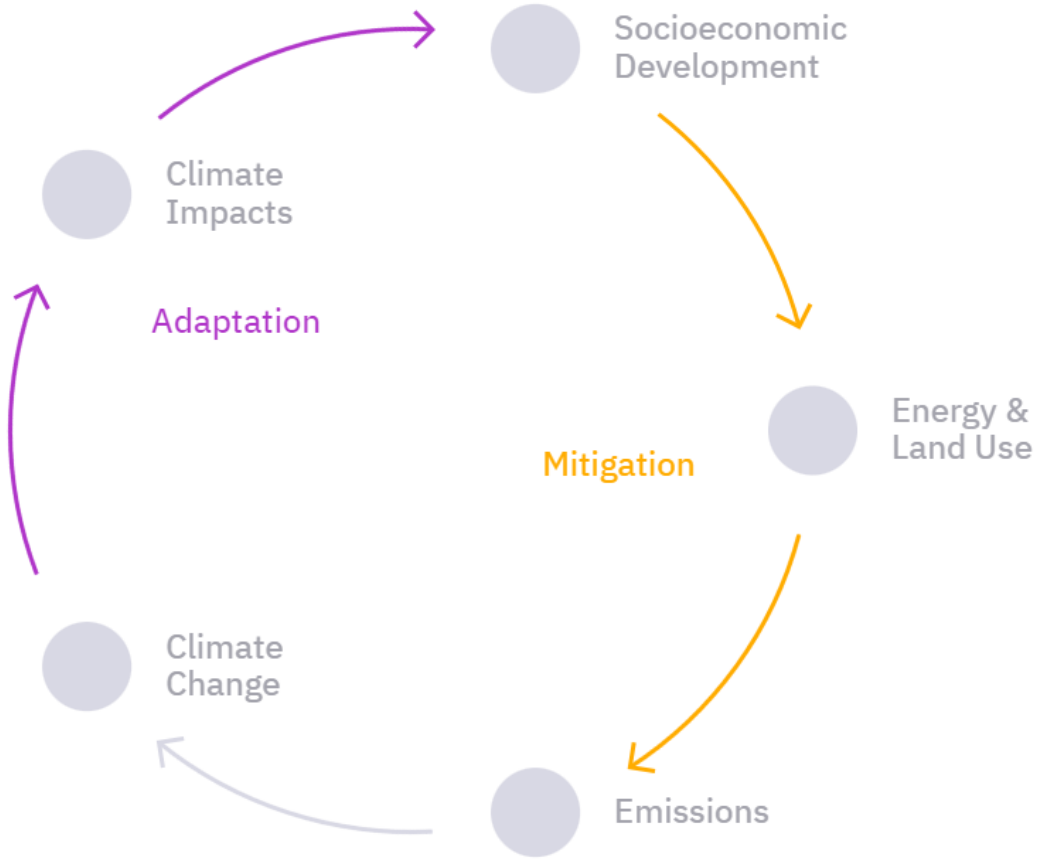
- **Population:** 2.5 times larger; 7.8 billion in 2020, growing about 1% a year.
- **Affluence:** GDP per person more than doubled.
- **Energy intensity:** fell steadily, through efficiency and a shift from heavy industry to services.
- **Carbon intensity:** fell to 2000 (coal to gas), rose in the 2000s with China's coal build-out, and is falling again.
- **Net result:** growth in population and affluence outran the gains, so emissions nearly tripled between 1960 and 2018.



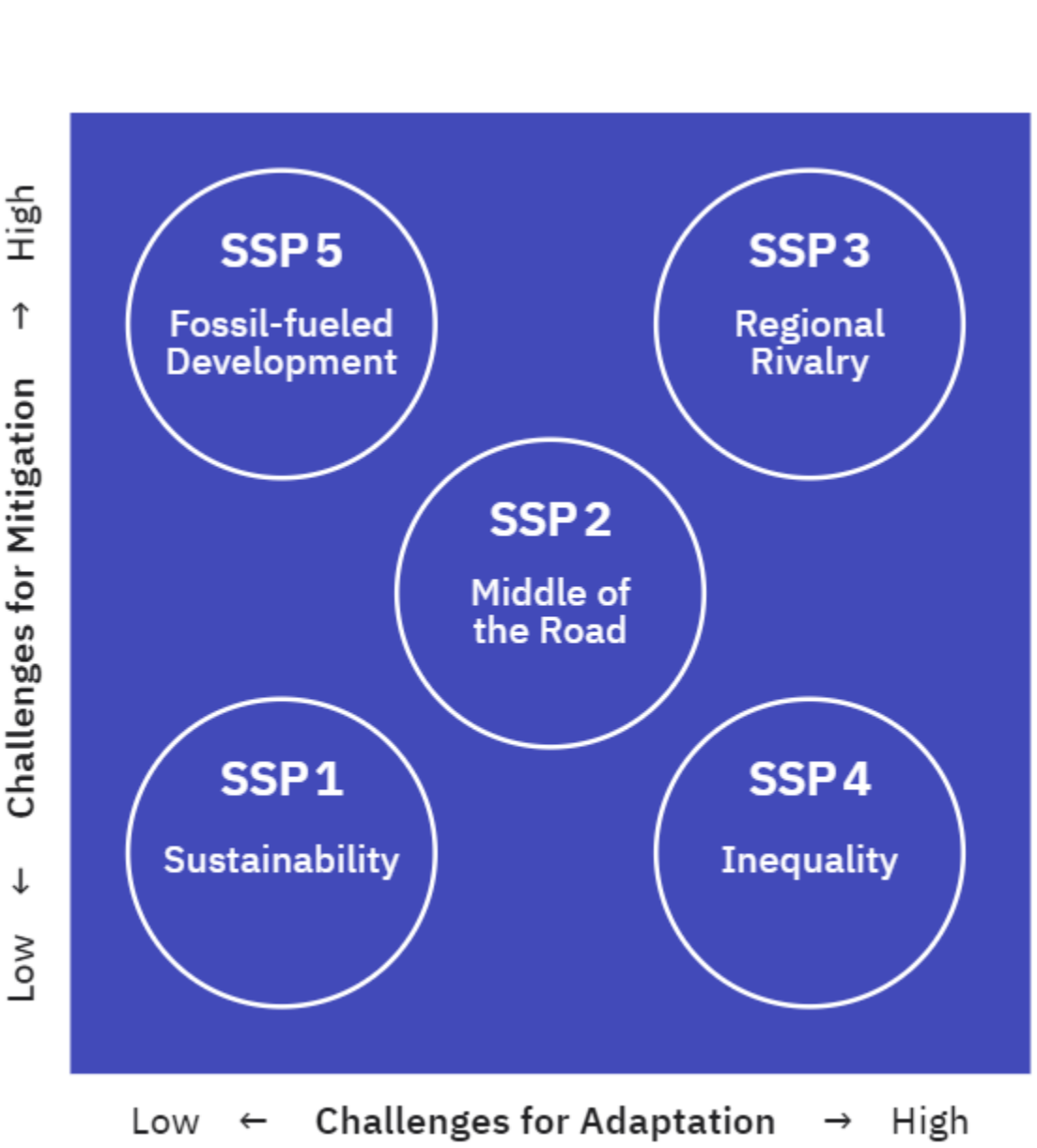
Source: Dessler (2021), Ch. 8

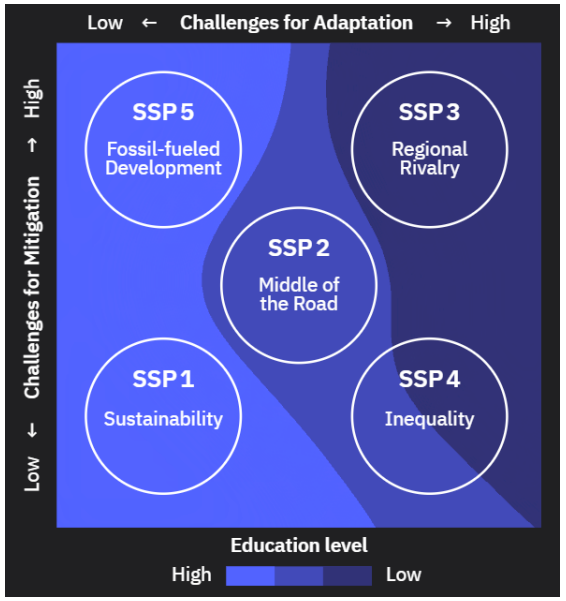
Climate change scenarios

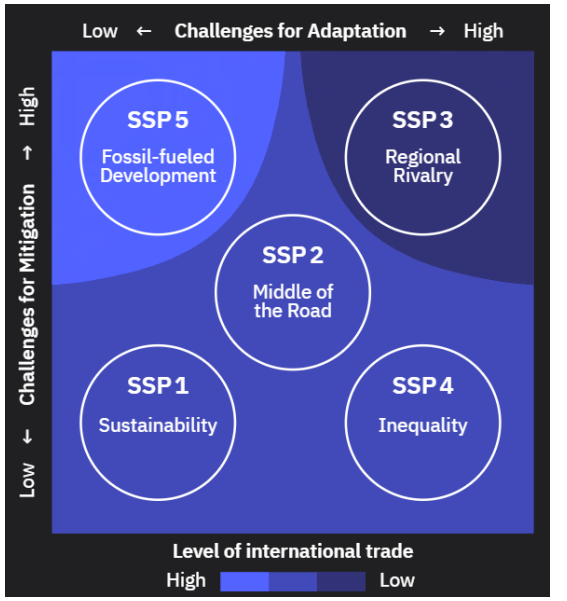
- Differences between scenarios are based on underpinning assumptions about future changes in fossil fuel consumption, land use change, etc., and were developed using integrated assessment models that combined economic, demographic and policy modelling, with simplified physical climate models to simulate the global economic impacts of climate change under different mitigation scenarios.
- https://climatescenarios.org/primer/

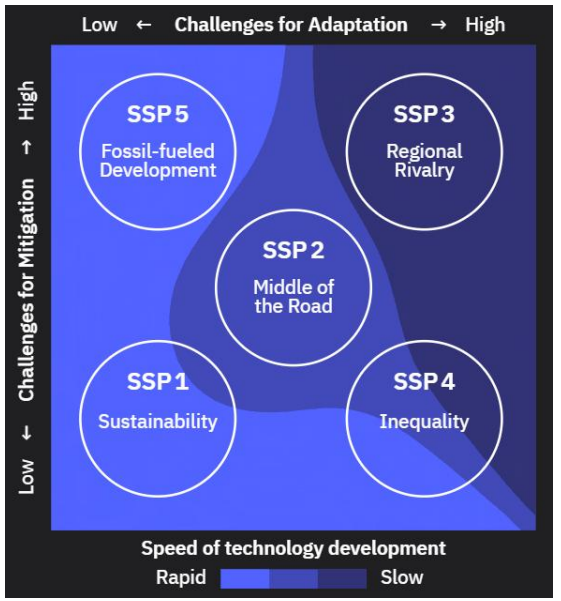


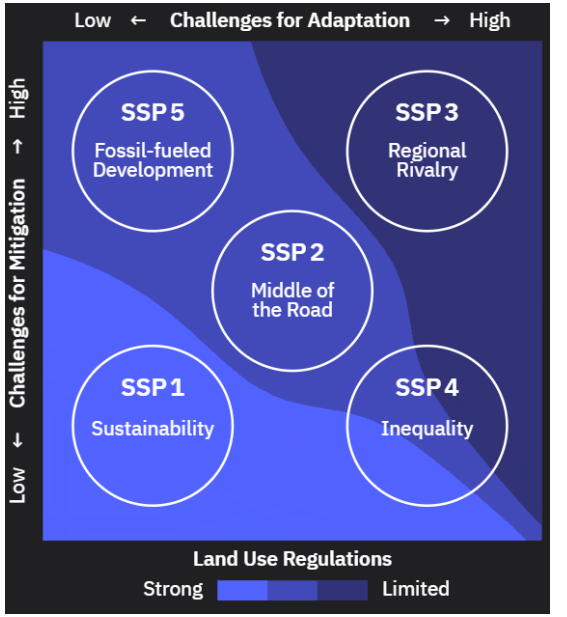
Climate change scenarios











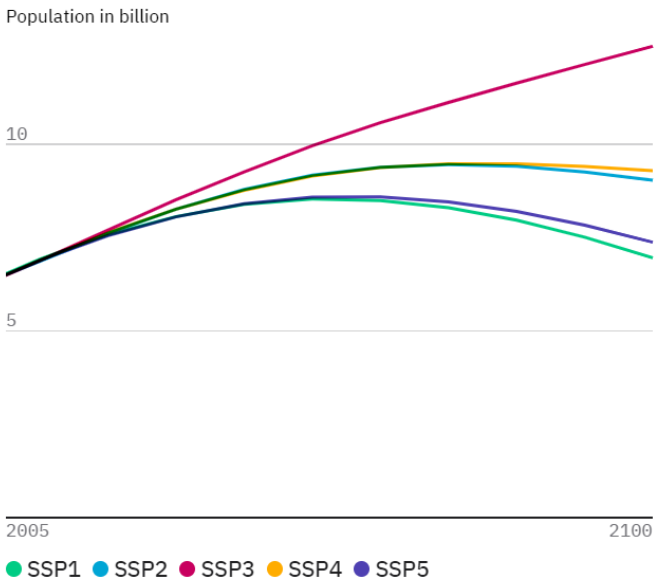
The five shared socioeconomic pathways

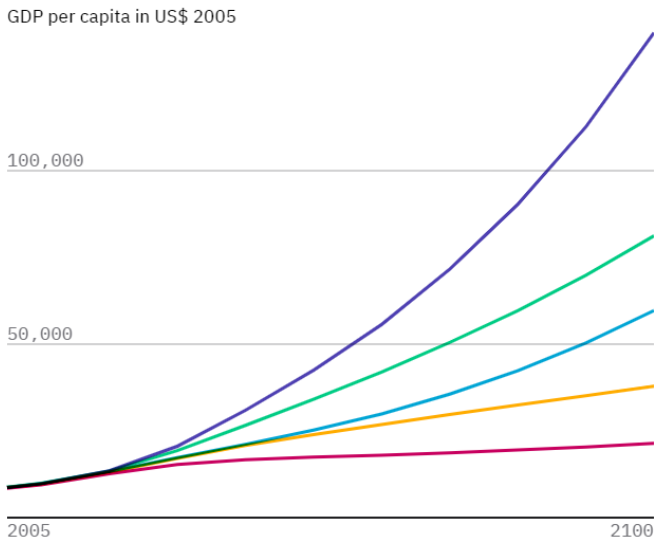
Pathway	Storyline
SSP1 Sustainability	A gradual, pervasive shift to an environmentally friendly path; poor countries catch up; population peaks mid-century; renewables
SSP2 Middle of the road	Social, economic and technological trends continue much as in the past
SSP3 Regional rivalry	Nationalism and conflict; the poor stay poor; high population growth in poor countries; slow technology and reliance on coal
SSP4 Inequality	A divided world like SSP3, but with faster development of low-carbon energy
SSP5 Fossil-fuelled development	Rich, high-growth and more equal like SSP1, but powered by fossil fuels

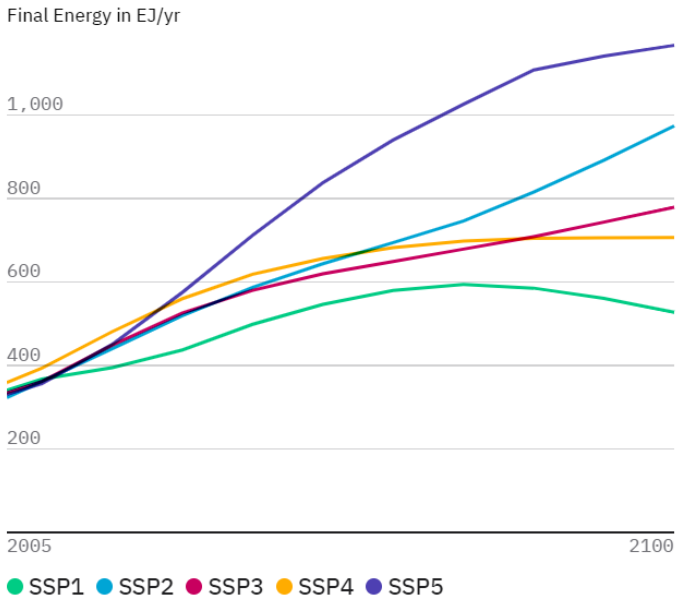
- **Internally consistent:** richer, better-educated societies have fewer children, so SSP1 and SSP5 have slower population growth than SSP3.

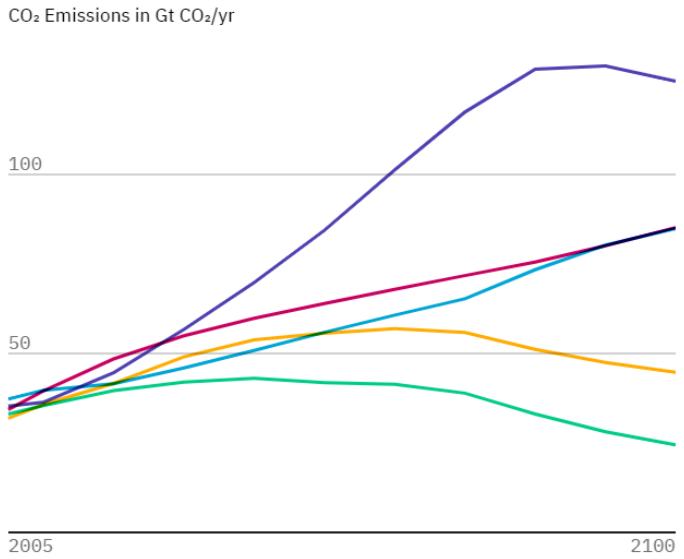
Source: Dessler (2021), Ch. 8

Climate change scenarios









Climate change scenarios

- The scenarios used by models participating in the Coupled Model Intercomparison Project Phase 5 (CMIP5) that contributed to the IPCC AR5 were termed the Representative Concentration Pathways (RCPs) and covered the period from 2006 to 2100.

RCP	Forcing	Temperature	Emission Trend
1.9	1.9 W/m2	~1.5 °C	Very Strongly Declining Emissions
2.6	2.6 W/m2	~2.0 °C	Strongly Declining Emissions
4.5	4.5 W/m2	~2.4 °C	Slowly Declining Emissions
6.0	6.0 W/m2	~2.8 °C	Stabilising Emissions
8.5	8.5 W/m2	~4.3 °C	Rising Emissions

Linking SSPs and RCPs

Scenario	Forcing in 2100	Comparable RCP	Emissions path
SSP1-2.6	2.6 W/m²	RCP2.6	Peaking about now; net negative from about 2075
SSP2-4.5	4.5 W/m²	RCP4.5	Peak around 2050, then decline
SSP3-7.0	7.0 W/m²	none (RCP6.0 was lower)	More than double by 2100; no new climate policy
SSP5-8.5	8.5 W/m²	RCP8.5	About three times today's by 2100 (126 $GtCO_2$ a year)

- **RCPs** (used for IPCC AR5) are concentration pathways built to reach a set radiative forcing in 2100. They carry no social or economic story.
- **SSPs** add the story. Each has a no-policy baseline and several policy variants; SSP1 variants span 1.9 to 5.0 W/m².
- **The name** joins the two: SSP1-2.6 = storyline SSP1 reaching 2.6 W/m² in 2100.
- **In this deck,** the Indian assessment uses RCP4.5 and RCP8.5: close to SSP2-4.5 and SSP5-8.5.

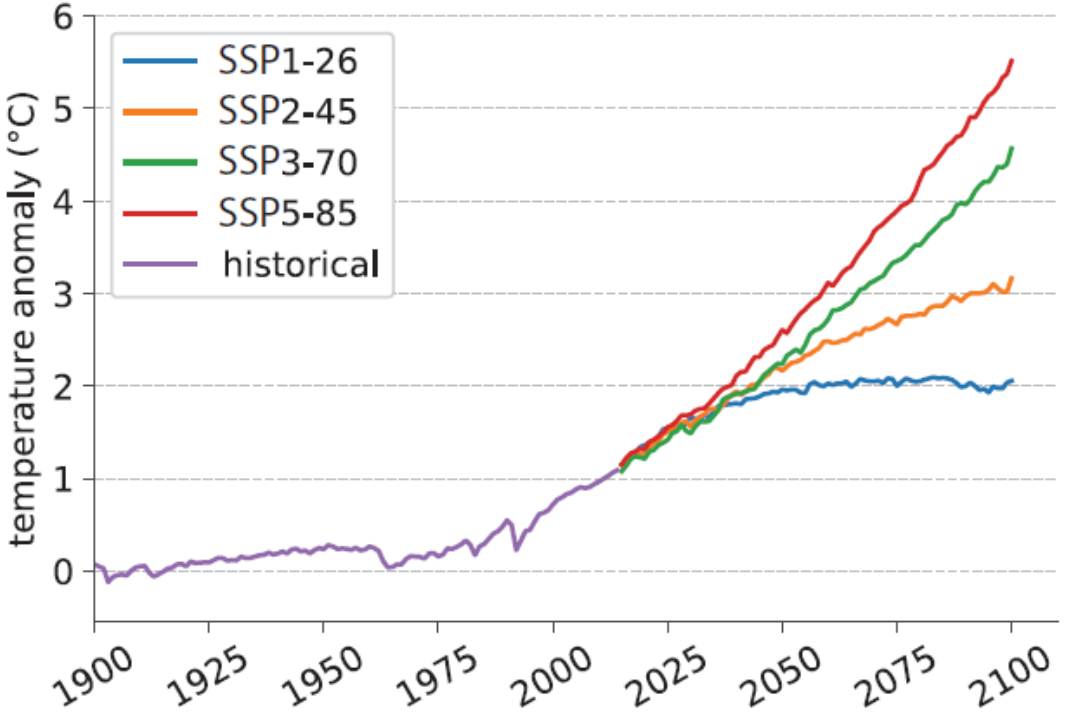
Source: Dessler (2021), Ch. 8

What the scenarios mean for warming

2.0 °C
by 2100 under SSP1-2.6, above pre-industrial

5.5 °C
by 2100 under SSP5-8.5, above pre-industrial

- **Slow to diverge:** in 2045 emissions differ threefold between scenarios, but CO_2 in the air by only 16%. The next few decades are largely set.
- **Our choice:** nearly 4 °C separates the scenarios by 2100.
- **CO_2 dominates** future forcing because it accumulates.
- **Beyond 2100:** temperatures stay high for 1,000 years after emissions stop.



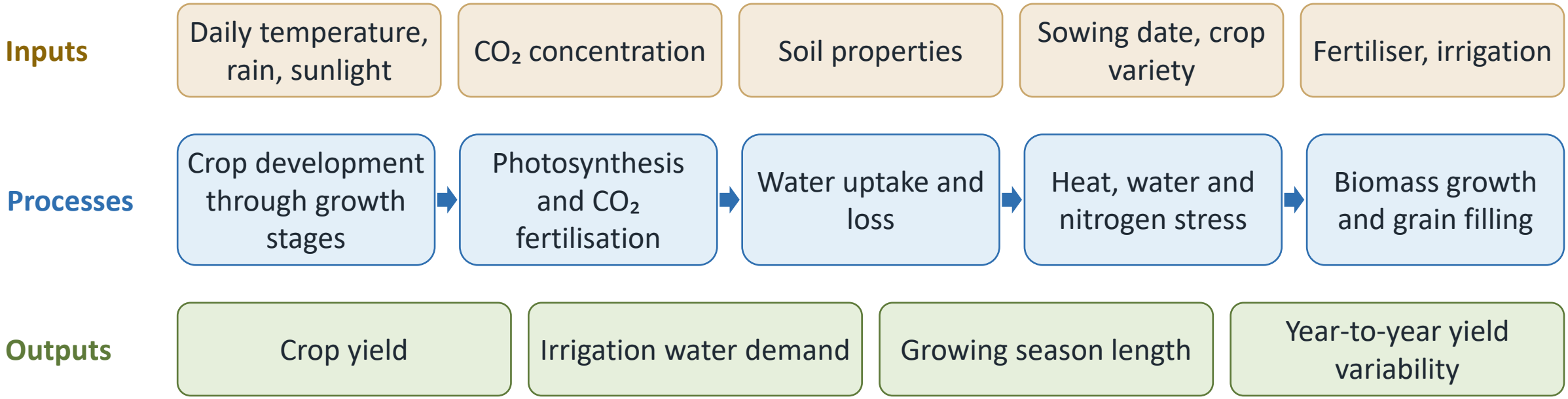
Source: Dessler (2021), Ch. 8

Climate change scenarios

Agriculture

The agriculture sector in ISIMIP seeks to understand how crops will be affected by changing atmospheric composition and climate. Increasing carbon dioxide concentrations may help some crops perform better, and in some regions increased temperatures could extend the growing season. But in other regions, more frequent and more intense heat waves and droughts could harm crop yields. This sector also examines how changing management practices (crop varieties, fertiliser, and irrigation) could be used to counteract negative effects of climate change.

In the models: global gridded crop models simulate each crop day by day in every grid cell.

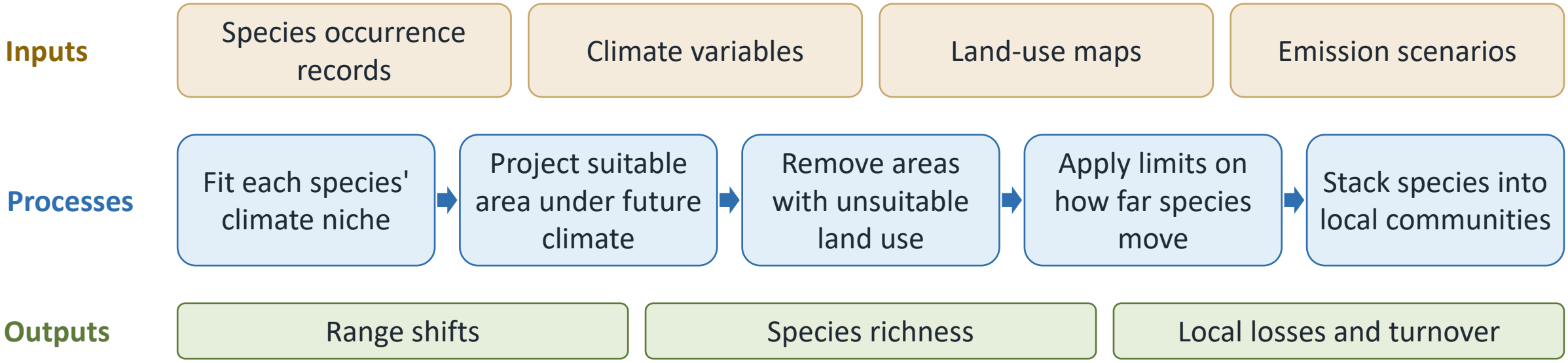


Climate change scenarios

Biodiversity

The ISIMIP Terrestrial Biodiversity Sector investigates the potential future impacts of climate and land-use change on global terrestrial biodiversity. We compare projected impacts under different emission scenarios as well as for different modelling approaches to understand and predict how the diversity of animals and plants in the terrestrial biosphere is influenced by multiple anthropogenic threats.

In the models: species distribution models link where each species lives today to climate and land use, then project that niche forward.

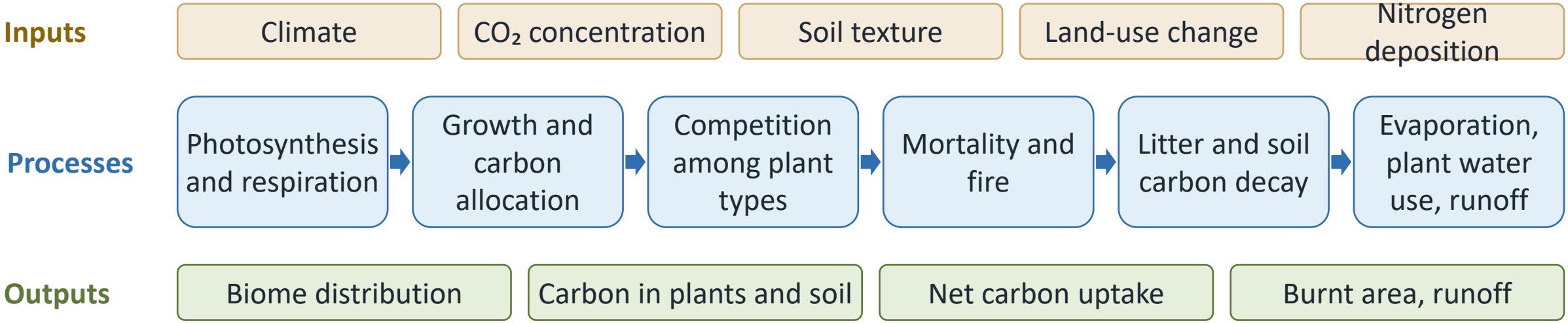


Climate change scenarios

Biomes

The Global Biomes Sector studies how climate and land-use change will affect the distribution and functioning of natural vegetation around the globe. Especially carbon and water cycling as well as the biogeographic distribution of plants under climate change are particularly relevant for this sector. The simulations contribute to better understand the capacity of natural vegetation to sequester and store carbon under increasing pressure from climate change but also from land-use changes.

In the models: dynamic global vegetation models grow competing plant types and track carbon and water through plants and soil.

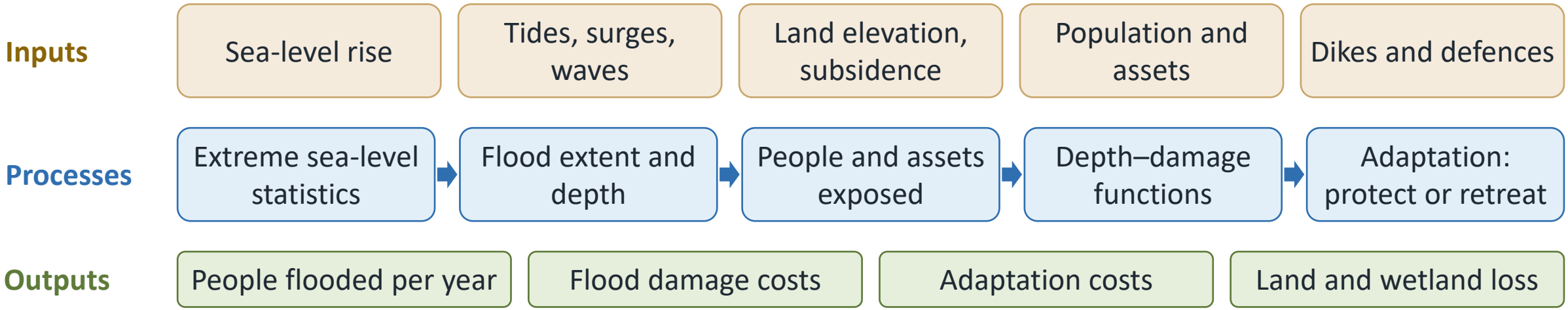


Climate change scenarios

Coastal Infrastructure

The increase of damage due to coastal extreme sea-level events (from tides, mean sea level anomalies, tropical cyclones, extratropical storm surges and waves) may be one of the costliest aspects of climate change and sea-level rise. The coastal sector of ISIMIP assesses these impacts at global and continental scales in order to inform a range of policy decisions including setting global mitigation targets, designing global financial mechanisms for adaptation and loss & damage, and strategic long-term coastal adaptation planning. In order to adequately inform such decisions, we systematically consider all major uncertainties in assessments of current and future coastal risk, including climate and socioeconomic scenario uncertainty, climate model and impact uncertainty, natural internal variability, as well uncertainties methods applied for extreme value analysis.

In the models: extreme sea levels are overlaid on elevation, population and asset maps, with and without coastal protection.

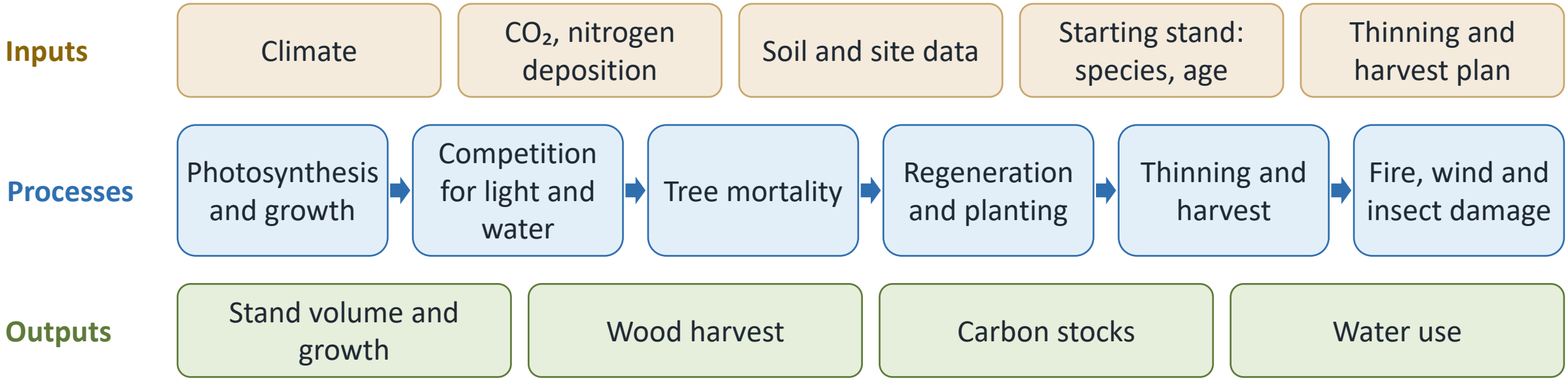


Climate change scenarios

Forestry

The Regional Forest Sector strives to understand how climate change will affect managed forests. Currently, simulations are carried out at the forest stand level for a wide range forest types covering the main forest tree species in Europe. The simulations are designed to disentangle the effects of different management schemes, of changing CO_2-concentrations and of different climate scenario on key forest services and functions such as plant productivity and growth, wood harvests and carbon and water cycling. Climate change is impacting forests in manifold ways potentially increasing productivity unless water is limited or disturbances such as fire or insect outbreaks lead to catastrophic damages.

In the models: stand-scale forest models grow individual stands under a prescribed management schedule.

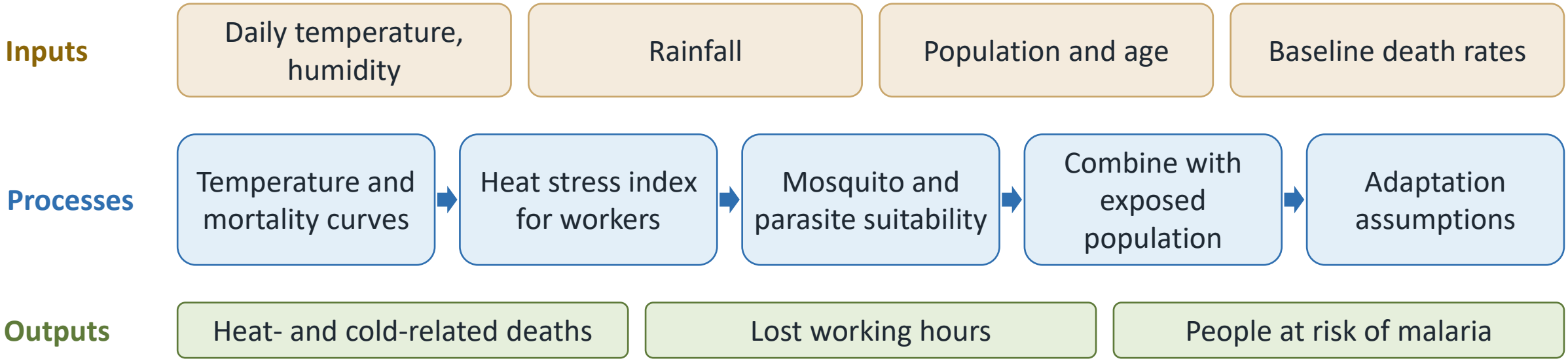


Climate change scenarios

Health

The ISIMIP health sector strives to provide improved knowledge on how human health will be affected by climate change. So far, from the myriad of potentially climate sensitive health outcomes only temperature related mortality, heat effects on labour productivity, and geographic changes in malaria suitability, have been considered in past modelling activities. For more comprehensive assessments of the health impacts of climate change extensions are planned to include (water-borne) infectious diseases and nutritional defects arising from food shortages. Modelling covers a wide range of spatial scales from local approaches (e.g., specific cities) to global assessments.

In the models: exposure–response relationships from epidemiology are applied to projected climate and population.

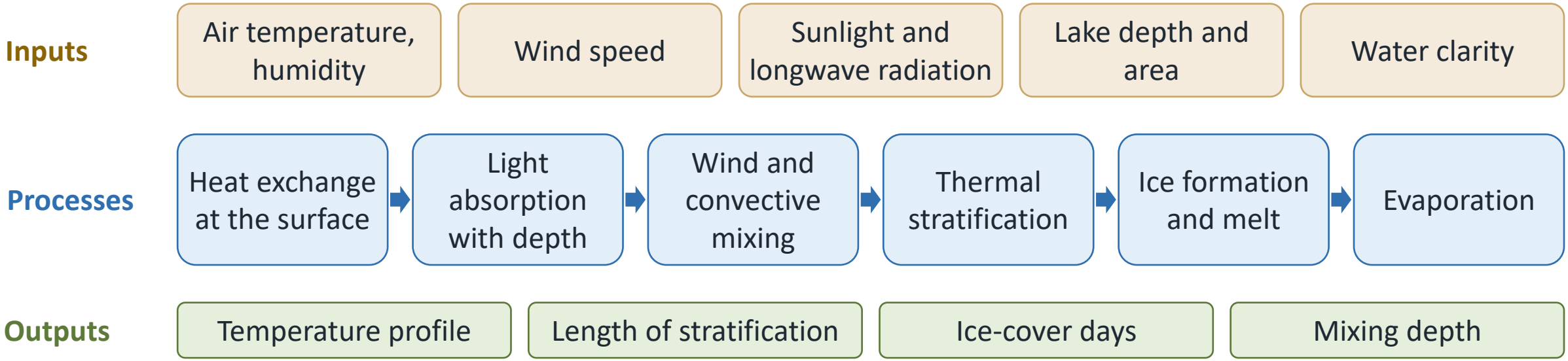


Climate change scenarios

Lakes

Analyses in the ISIMIP lake sector currently focus on how climate change is affecting the physical properties of lake functioning. Lake sector simulations follow two synergetic tracks, whereby one-dimensional lake models are either (i) tailored to reproduce observed processes in a selection of around 50 well-studied lakes, or (ii) run across all lake regions of the globe with one consistent set-up. The resulting data set provides an unprecedented opportunity to assess lake model performance and uncover global warming impacts on e.g., thermal stratification or lake ice dynamics, thereby forming the basis for understanding how lacustrine systems will respond to anthropogenic climate change.

In the models: one-dimensional lake models solve the heat balance layer by layer down the water column.

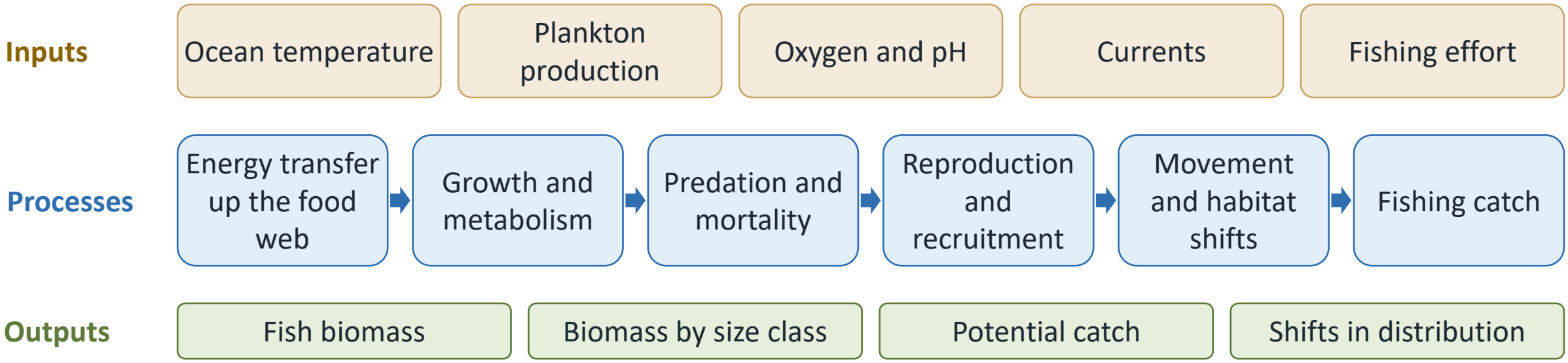


Climate change scenarios

Marine Fishery (FishMIP)

FishMIP looks at the future of marine ecosystems, and consequences for fisheries, biodiversity, and food supply. Climate change impacts the oceans in many ways, such as warming, ocean acidification, and deoxygenation, and we have a range of models to explore these issues with. The oceans provide many services to human beings, and the health and productivity of marine ecosystems remains of critical importance. Marine ecosystem models are extremely varied, so FishMIP allows us to look at these questions and make projections from many different angles.

In the models: marine ecosystem models pass energy from plankton up the food web; many are organised by body size.

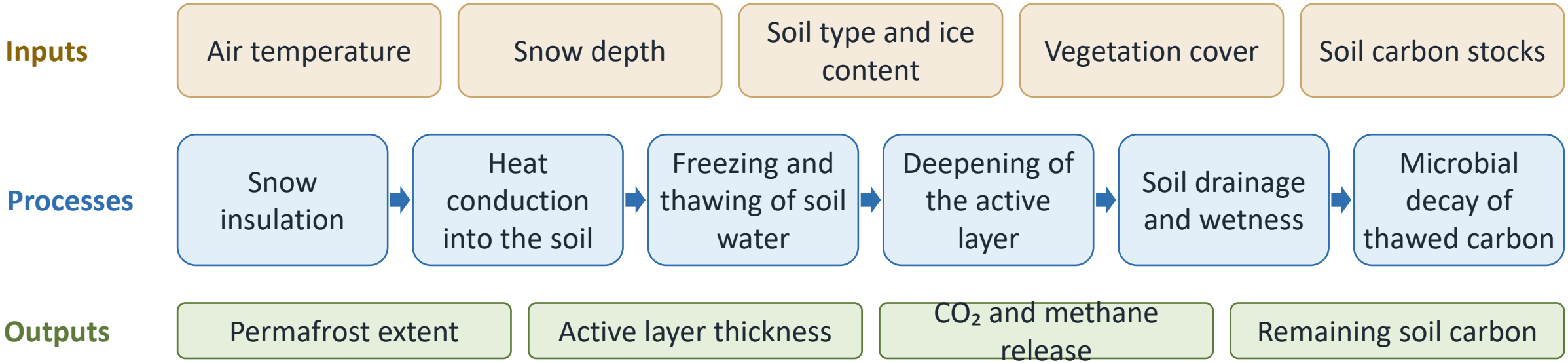


Climate change scenarios

Permafrost

Permafrost is defined as material that remains frozen for two consecutive years. Climate warming, which is amplified in the polar regions, can induce environmental changes that accelerate microbial breakdown of organic carbon that has previously been locked in the permafrost. The magnitude and timing of greenhouse gas emissions from thawing permafrost remain, however, highly uncertain. Thawing of permafrost affects ecosystems and climate on various spatial and temporal scales ranging from local to regional topography, hydrology, and biology as well as global scale biogeochemical cycling. Thawing of permafrost is also of high social (indigenous culture) and economical relevance (infrastructure, particularly oil and gas development).

In the models: land surface models with frozen-soil physics track heat, water and carbon through deep soil layers.

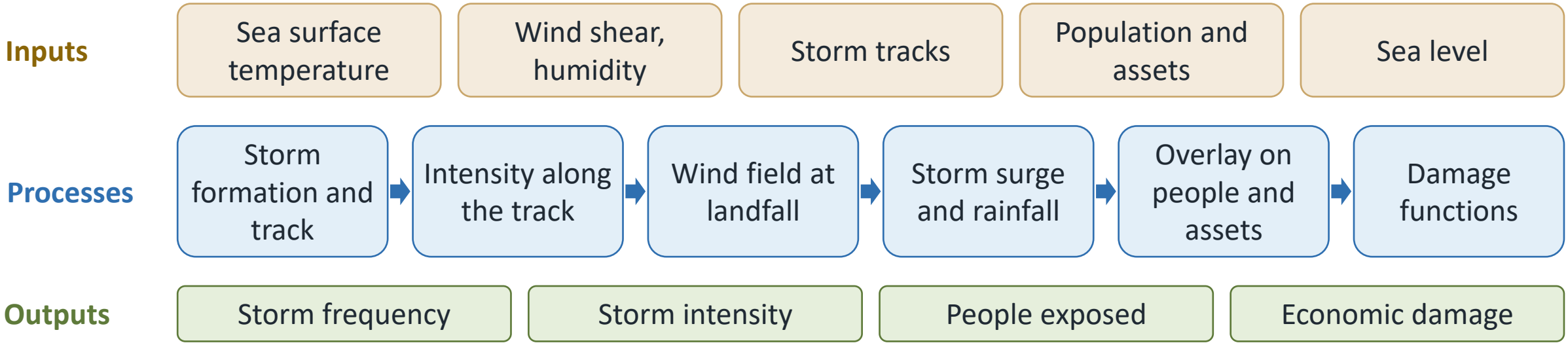


Climate change scenarios

Tropical Cyclones

The tropical cyclone sector in ISIMIP studies the change in global tropical cyclone activity and the corresponding impacts on society. This sector analyses how spatio-temporal patterns of tropical cyclone occurrence may change with climate change and, in particular, looks at changes in tropical cyclone intensity as well as frequency. Impacted societies do not only suffer from destructive winds but also from many other co-hazards like extreme precipitation, storm surges, and sea-level rise. These co-hazards provide a direct link to other sectors within ISIMIP, e.g., the coastal infrastructure sector. The simulations in this sector can provide a climate-proof risk assessment for coastal communities and provide guidance on potential adaptation strategies.

In the models: observed or synthetic storm tracks are turned into wind, surge and rain fields and overlaid on exposure maps.

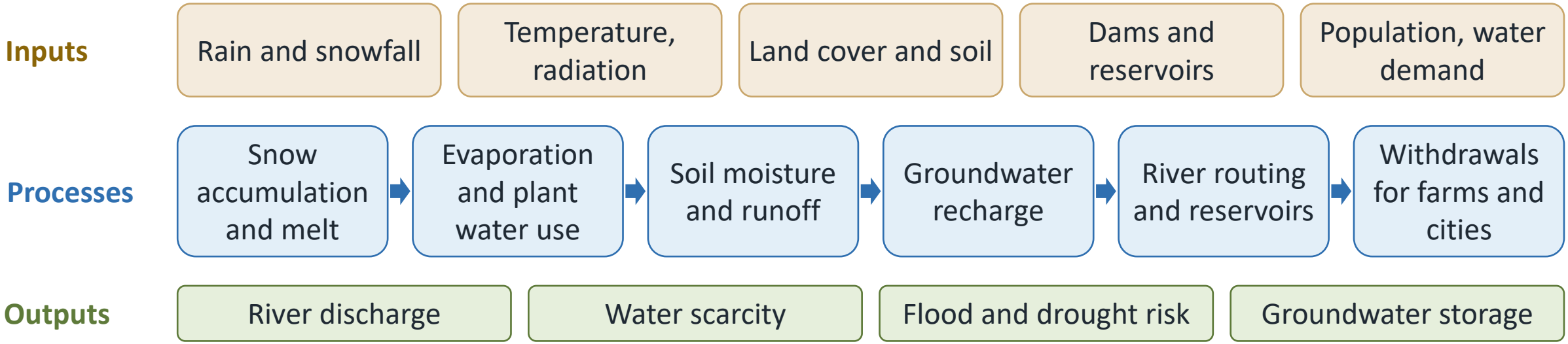


Climate change scenarios

Water (global)

The global water sector runs models to understand what water resources look like across the globe nowadays and how they might change in the future. We assess risks such as water scarcity, droughts and flooding. The models are able to account for factors such as human management of the water cycle, including dams, irrigation and water abstractions from rivers. A lot of work in this sector focuses on understanding how water-related risks might be affected by changes in future population and a changing climate. Our results show that climate change is likely to exacerbate regional and global water scarcity considerably, whilst the magnitude of flooding could increase across around a third of the global land surface.

In the models: global hydrological models compute a daily water balance in every grid cell and route the runoff downstream.

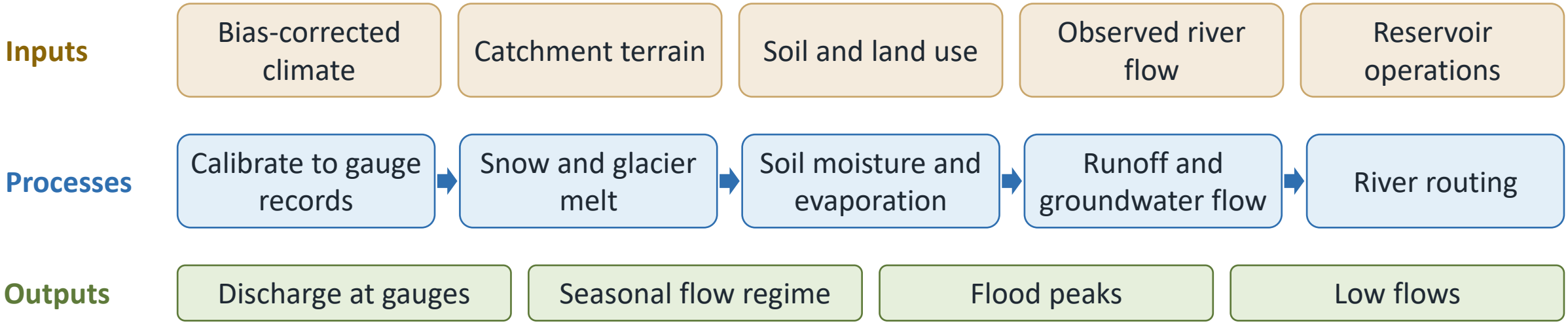


Climate change scenarios

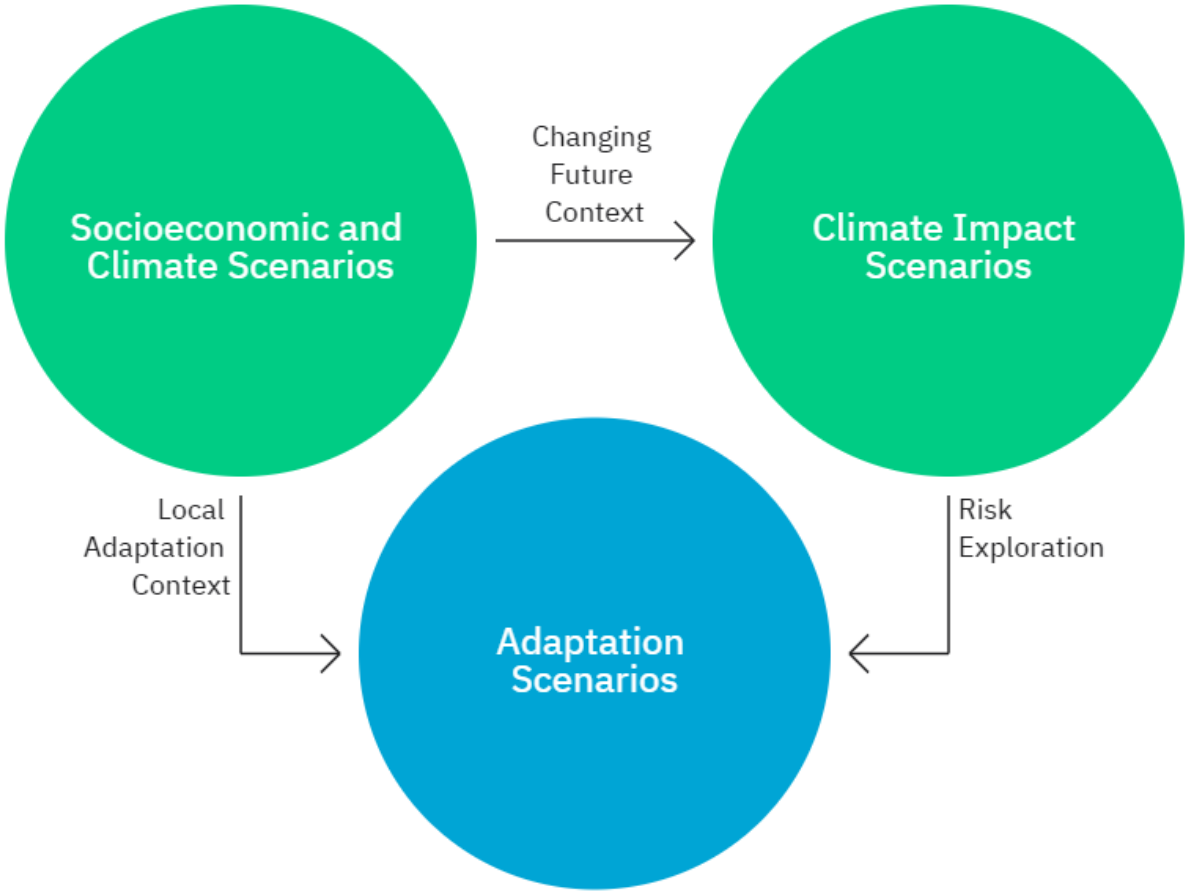
Water (regional)

Climate change is a global phenomenon, but the impacts manifest at the regional scale, and the regional scale is also where most adaptation measures in the water sector are planned and implemented. The aim of the regional water group in ISIMIP is therefore to underpin global scale assessments with more detailed regional and cross-scale studies, where both sides learn from each other and benefit from cross-pollination of ideas and methods. Eleven large scale river basins have been selected to act as case studies to investigate climate change impacts on water resources and extremes and across sectors.

In the models: catchment models are calibrated against measured river flow in each basin before being driven by climate projections.



Climate change scenarios



How is local adaptation contextualised from socioeconomic and climate scenarios?

- Multiscale scenarios, i.e. outcomes nested within a wider regional or a global context
- Downscaling climate scenarios to produce Regional Climate Models (RCMs)
- Linking Socioeconomic and Climate scenarios. Depends heavily on the availability of reliable data of sufficient spatial resolution.

Part 4: Economics and policy

Part 1: The physics of climate change

What climate is, evidence of warming, energy balance, the greenhouse effect, the carbon cycle, forcing and feedbacks, attribution

Part 2: Climate change over the Indian region

Models, temperature, monsoon and rainfall, greenhouse gases and aerosols, droughts and floods, Himalayas, oceans, impacts

Part 3: Emissions and climate scenarios

What drives emissions, energy and carbon intensity, SSPs and RCPs, projected warming, impact models by sector

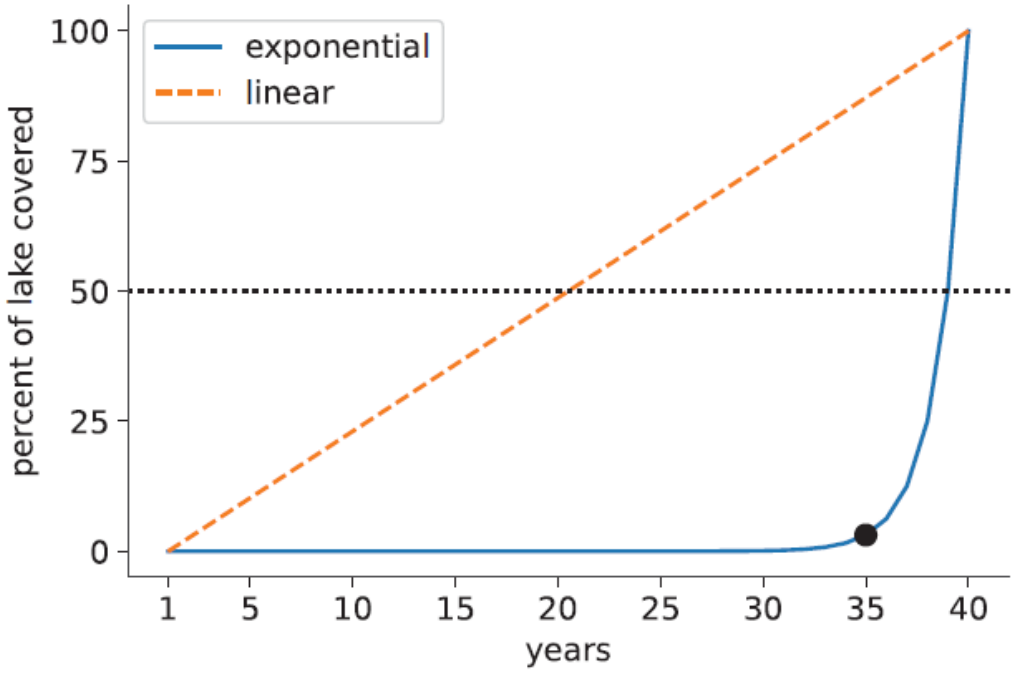
Part 4: Economics and policy

Exponential growth and discounting, response options, carbon pricing, history of science and negotiations, long-term goals

Exponential growth: catastrophe is closer than you think

$$F = P\,(1 + r/100)^n \qquad \text{doubling time} \approx 72 / r$$

- **Definition:** growth proportional to present size. Anything growing at “r percent a year” is exponential.
- **Lily pads** double every year and cover a lake in 40 years. At year 39 they cover half; at year 35, only 3%.
- **End-loaded:** half the growth comes in the last doubling. Clearing half the lake buys one year.
- **Lesson:** act on exponential threats early, before the danger is obvious. A US lockdown one week earlier in March 2020 would have cut cases by about 80%.



Source: Dessler (2021), Ch. 10

Four ways to respond

Adaptation

Respond to the impacts: sea walls, drought-resistant crops, relocation. Needed in any case, because some change is unavoidable.

Mitigation

Cut greenhouse gas emissions so that the change does not happen. The centrepiece of any serious policy.

Solar radiation management

Reflect a little more sunlight, for example with stratospheric aerosols. Fast and cheap, but risky and incomplete.

Carbon dioxide removal

Pull CO_2 out of the air faster than nature does. Treats the cause, but the scale needed is unproven.

Source: Dessler (2021), Ch. 11

Adaptation and equity

- **Needs government:** no individual can build a sea wall. Public money, rules (water conservation, flood insurance) and reliable information are required.
- **Needs resources:** so the poorest are least able to adapt. In Hurricane Katrina (2005) those with cars and credit left; those without made up most of the dead.
- **The fairness problem:** rich countries grew rich by emitting and can afford to adapt. Poor countries emitted little and cannot.
- **Aid falls short:** in 2019 the US and Australia declined to contribute to the Green Climate Fund.
- **Bottom line:** relying on adaptation alone is a gamble for the rich and certain hardship for the poor.

Haiti, 2010

Magnitude 7.0 earthquake. More than 200,000 dead; a million homeless.

Chile, 2010

Magnitude 8.8, about 500 times stronger. A few hundred dead, because a richer country could build to stricter codes.

Source: Dessler (2021), Ch. 11

Mitigation: which lever can we pull?

Kaya term	Can it deliver deep cuts?
Population	No. It would need a falling population, against deeply held social and religious norms. Not seriously proposed.
Affluence	No. People equate consumption with well-being, and about 2 billion people need growth to escape poverty.
Energy intensity	Partly. It falls 1–2% a year and efficiency saves money, but that is not fast enough.
Carbon intensity	Yes. Replace fossil fuels with climate-safe energy. This term must do most of the work.

- **Decades of lag:** because CO_2 accumulates, cuts made now show up in temperature only after 2050. People are asked to pay today for benefits they may not live to see.

Source: Dessler (2021), Ch. 11

Climate-safe energy options

Source	Potential	Limits
Solar	Over 100,000 TW falls on Earth vs 15 TW used; needs about 0.2% of the surface	Intermittent
Wind	Turbines up to 10 MW; land can still be farmed; offshore winds are steadier	Intermittent; millions of turbines
Biomass	Carbon-neutral in principle	About 15% of land; competes with food
Hydro	16% of world electricity	Most big rivers already dammed
Nuclear	About 16% of electricity; mature	Safety, waste, proliferation, cost
Carbon capture	Captures 85–95% of a plant's CO_2	Uses 10–40% of the energy; unproven

- **Dealing with intermittency:** storage (batteries, hydrogen), long-distance transmission, and dispatchable sources such as hydro, geothermal and nuclear.

Source: Dessler (2021), Ch. 11

Geoengineering: two very different ideas

Solar radiation management

How: inject sulfur into the stratosphere, as volcanoes do, or brighten low clouds.

For: works at once; about $10 billion a year.

Against: does nothing for ocean acidification; shifts rainfall; who decides?

Termination: stop, and warming rebounds within a decade or two.

Role: a stopgap, not a solution.

Carbon dioxide removal

How: trees (not permanent), ocean iron fertilisation (risky), direct air capture, bioenergy with carbon capture.

For: addresses every effect of CO_2, not only temperature.

Against: about 40 billion tonnes a year would need infrastructure on the scale of today's fossil industry.

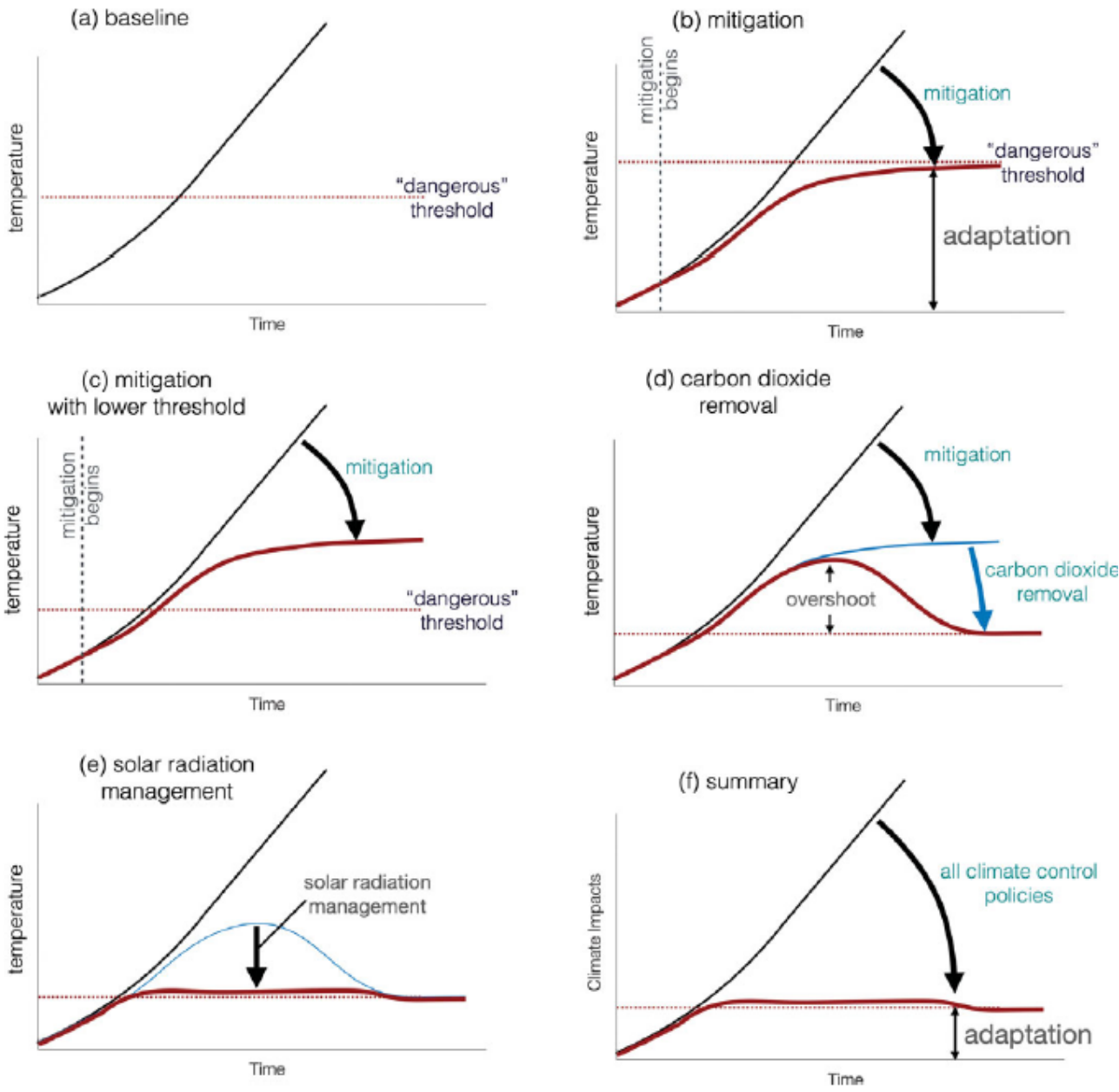
Role: probably needed, but the ability to do it affordably is speculative.

Source: Dessler (2021), Ch. 11

Combining the responses

- **Start early:** mitigation alone keeps warming under a chosen “dangerous” limit, with adaptation for the change that remains.
- **Start late:** even the fastest mitigation overshoots the limit. This is likely the world we are in.
- **Then add carbon removal** to bring temperatures back down after a temporary overshoot.
- **And perhaps solar radiation management** to shave the peak if even a temporary overshoot is unacceptable.
- **Always:** adapt to the warming that is not avoided.

Source: Dessler (2021), Ch. 11



Why markets alone will not fix it

Today's process

A widget costs the maker $10 and causes $2 of climate damage, paid by everyone. True cost: $12.

A clean process

Costs the maker $11 and causes no damage. True cost: $11, so society is better off.

- **But the maker will not switch:** to the maker, $10 beats $11. The $2 is an externality, a cost imposed on people outside the transaction.
- **Market failure:** when dumping CO_2 is free, the rational choice for each emitter harms everyone. A tragedy of the commons, as with overfishing.
- **So:** the choice is between government action and unchecked emissions. The fix is to make emitters pay for the damage (internalise the externality).

Source: Dessler (2021), Ch. 12

A carbon tax: worked example

- **A tax of $4 per tonne:** each plant cuts as long as cutting is cheaper than paying.
- **Plant A** cuts 4 tonnes (cost $10).
- **Plant B** cuts 2 tonnes (cost $6).
- **Total:** 6 tonnes cut for $16.
- **A uniform rule** of 3 tonnes each also cuts 6 tonnes, but costs $6 + $12 = $18.
- **Why the tax is cheaper:** it moves the cuts to wherever they cost least.

Marginal cost: what it costs each plant to cut one more tonne

		Plant A's cost ($)		Plant B's cost ($)	
Emissions reduced by (tonnes)	Units emitted (tonnes)	Marginal	Total	Marginal	Total
0	10	–	–	–	–
1	9	$1	$1	$2	$2
2	8	$2	$3	$4	$6
3	7	$3	$6	$6	$12
4	6	$4	$10	$8	$20
5	5	$5	$15	$10	$30
6	4	$6	$21	$12	$42
7	3	$7	$28	$14	$56

Source: Dessler (2021), Ch. 12

Carbon tax versus cap and trade

	Carbon tax	Cap and trade
Government sets	The price per tonne	The total quantity (number of permits)
Left uncertain	How much emissions fall	What it will cost
How it works	Pay for every tonne emitted	Hold a permit for every tonne; permits can be traded
Over time	The tax rate rises	The number of permits falls
Political issue	“Tax” is unpopular	Auction permits, or give them away free?
Experience	Australia introduced, then repealed	EU trading system; US acid-rain programme cut sulfur cheaply

- **Same economics:** emitters cut until the marginal cost equals the price, so cuts happen where they are cheapest. Both raise revenue that can be returned to citizens.

Source: Dessler (2021), Ch. 12 and 13

Offsets, standards and information

Offsets

Pay for removal elsewhere, such as planting trees.

Hard to verify, and they must be additional: would it have happened anyway?

Standards

Rules such as fuel-economy limits or bans on new coal plants.

Costlier than a price, but useful where a price acts slowly.

Information

Labels, flood-risk maps, emission registries.

Helpful, but too weak to drive deep cuts.

- **The centrepiece:** a price on greenhouse gas emissions. Nordhaus: whether a proposal raises the price of carbon is the test of whether it is serious.

Source: Dessler (2021), Ch. 12

Two centuries of climate science

When	What
1820s	Fourier describes how an atmosphere warms a planet
1856–59	Foote, then Tyndall, show that CO_2 and water vapour absorb infrared
1896	Arrhenius estimates 5–6 °C of warming for doubled CO_2 (today's estimate: 2.5–4 °C)
1938	Callendar argues that human CO_2 is already warming the planet
1957–58	Keeling begins measuring atmospheric CO_2
1979	US National Academy: “no reason to doubt that climate changes will result”
1988	Hansen testifies to the US Congress; the IPCC is formed

- **Old, tested science:** the core ideas are over a century old.

Source: Dessler (2021), Ch. 13

Manufacturing doubt: the tobacco strategy

“Doubt is our product.” (tobacco company memo, 1969)

- **The method:** recruit a few credentialed contrarians, cherry-pick anomalies, raise questions already answered, and demand “equal time”.
- **Reused** against action on CFCs and ozone, acid rain, and then climate.
- **The outcomes:** CFCs were phased out under the Montreal Protocol (1987) at a cost so small that almost no one noticed. Acid rain was cut cheaply by cap and trade.

The four arguments

1. The harm is not happening.
2. If it is, our product is not the cause.
3. If it is, fixing it will cost too much.
4. The scientists are biased or corrupt.

Uncertainty always exists; by itself it says nothing about how serious a problem is.

Source: Dessler (2021), Ch. 13

How confidence in attribution grew

Report	Statement on the human role
1990	The observed warming “could be largely due to natural variability”
1995	“The balance of evidence suggests a discernible human influence”
2001	Most of the warming of the last 50 years is likely (66%) due to human activities
2007	Most of the warming since the mid-20th century is very likely (90%) due to greenhouse gases
2013	Extremely likely (95%) that humans caused more than half of the warming since 1951

- **The trend:** each assessment is more confident than the last, as evidence accumulates.
- **The 1970s “cooling scare”:** scientists debated whether aerosol cooling or greenhouse warming would win. There was never a consensus for cooling.

Source: Dessler (2021), Ch. 13

Climate negotiations

Agreement	What it did
1992 Framework Convention	Goal: avoid “dangerous” interference with the climate. Non-binding. Near-universal membership.
1997 Kyoto Protocol	Binding: industrialised countries to cut 5% below 1990 levels by 2008–2012. None for developing countries. The US withdrew in 2001.
2009 Copenhagen Accord	Defined dangerous as 2 °C. Countries set their own targets.
2015 Paris Agreement	Well below 2 °C, pursuing 1.5 °C. Nationally determined contributions, reviewed every 5 years; no penalties.
2020 onwards	Net-zero pledges: EU, UK, Japan, Korea by 2050; China by 2060

- **The gap:** the 2015 Paris pledges implied 3–3.5 °C by 2100: better than over 4 °C without them, but far from the goal.

Source: Dessler (2021), Ch. 13

Principles in the Framework Convention

Avoid dangerous interference

Stabilise greenhouse gases low enough to prevent “dangerous” change.

What counts as dangerous is a value judgment, not a scientific one.

Common but differentiated responsibilities

Everyone acts, but rich countries go first.

They have the resources, they caused most of the emissions, and early action lowers costs for others.

Precautionary principle

Lack of full scientific certainty is not a reason to postpone action against serious or irreversible damage.

- **The fault line:** rich countries wanted poorer ones to commit to cuts; poorer ones would not give up growth. China's 2015 pledge to peak by 2030 shifted this.

Source: Dessler (2021), Ch. 13

Deciding under uncertainty

	If climate change turns out mild	If it turns out severe
We cut emissions	Some cost, but cleaner air and an energy shift that had to come anyway	Catastrophe avoided
We do not cut	Nothing lost	Millennia of warming; catastrophic and irreversible

- **Waiting is also a choice:** every year of delay commits the planet to more warming.
- **Which error is worse?** Juries accept letting the guilty go free to avoid convicting the innocent. Defence planners act on a 1% chance. For climate, doing too little is the worse mistake.
- **Irreversibility:** cutting emissions can be undone if it proves unnecessary. Emitted CO_2, lost ice sheets and extinct species cannot.
- **Result:** almost every government has concluded that emissions should be reduced.

Source: Dessler (2021), Ch. 14

Discounting: how much is the future worth?

$$\text{Present value} = F / (1 + r/100)^n$$

- **Money later is worth less than money now;** the discount rate r says how much less per year.
- **Time discounting:** we prefer to consume now. Across generations this means valuing ourselves above our descendants; many argue it should be near zero.
- **Growth discounting:** future people will be richer, so a dollar matters less to them.
- **A value judgment:** economists argue for rates from 1% to 4%, and the choice can reverse the policy conclusion.

Discount rate	Present value
1%	$370 billion
2%	$138 billion
3%	$52 billion
4%	$19 billion
7%	$1 billion

Value today of $1 trillion of climate damage 100 years from now

Source: Dessler (2021), Ch. 10 and 14

The limits of cost–benefit analysis

- **The idea:** compare the cost of a policy with the damage it avoids, and pick the policy with the largest net benefit.

Regional extremes

Damage comes from local extremes, which are the hardest thing to project.

Adaptation

How well will people adapt? Assumptions range from none to perfect foresight.

Valuation

What is a species or an ecosystem worth? Leaving it out prices it at zero.

Discount rate

$1 trillion in 100 years is worth $370 billion today at 1%, and $1 billion at 7%.

Distribution

Totals hide who gains and who loses. Efficient is not the same as fair.

Catastrophe

Small chances of terrible outcomes are usually ignored, so valued at zero.

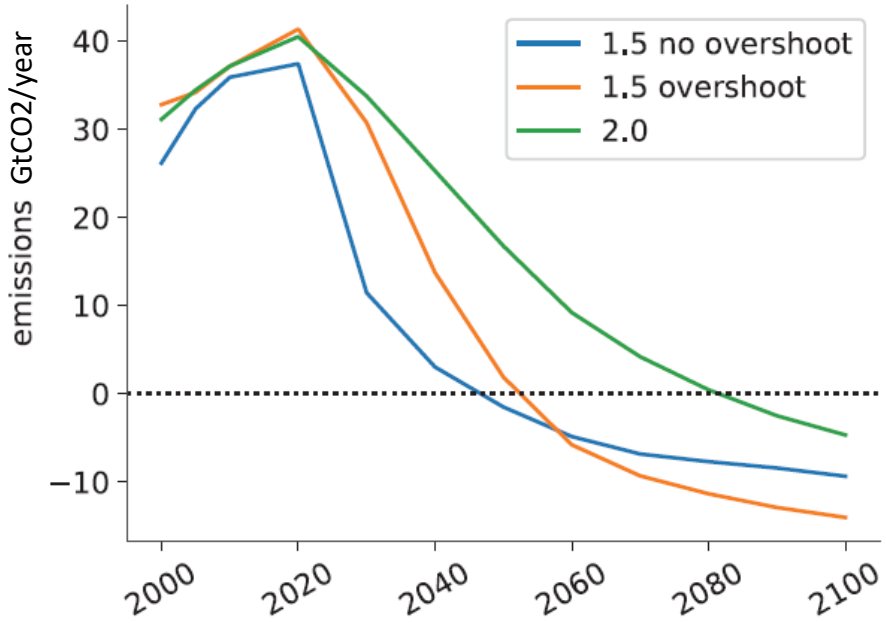
Source: Dessler (2021), Ch. 14

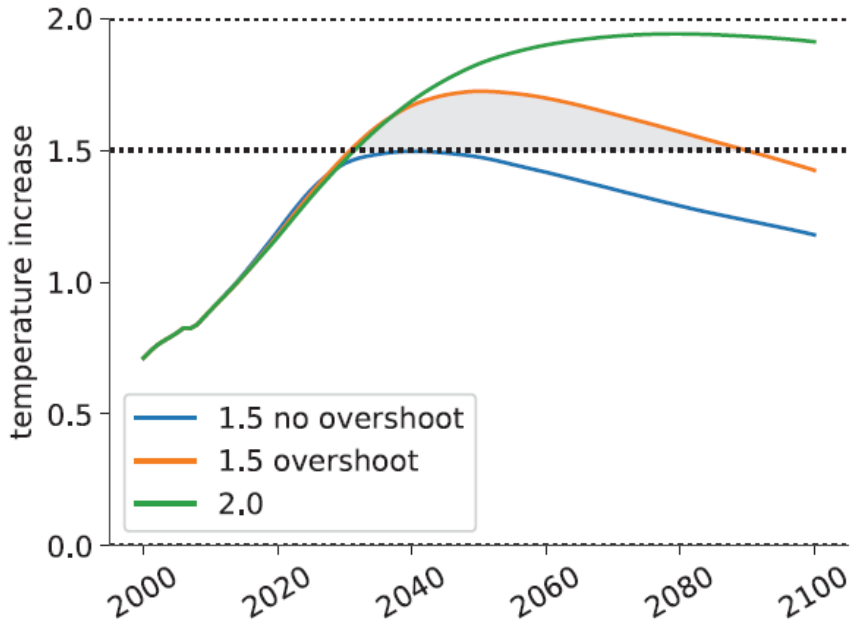
Targets and the carbon budget

580 $GtCO_2$
left from 2018 for 1.5 °C:
gone by the early 2030s

1,500 $GtCO_2$
left from 2018 for 2 °C:
gone by the 2050s

- **Peak warming follows cumulative CO_2:** 0.3–0.6 °C per 1,000 $GtCO_2$. About 2,200 Gt was emitted by 2017; we add over 40 Gt a year.
- **1.5 °C path:** halve emissions by 2030, net zero by the mid-2040s, negative after.
- **2 °C path:** halve by about 2050, zero around 2080.
- **Political targets,** not physical thresholds.





Source: Dessler (2021), Ch. 14

What the pathways require

Annual change needed in the 2020s	1.5 °C (with overshoot)	2 °C
Emissions	−8%	−3%
Population × affluence (expected)	+2% to +4%	+2% to +4%
So greenhouse-gas intensity must fall	−11%	−6%
Energy intensity (expected)	−1% to −2%	−1% to −2%
So carbon intensity must fall	up to −10%	about −4%

- **In hardware:** retire about 1 TW of fossil power a year and build about 2 GW of climate-safe power every day for decades. A large power station is 1 GW.
- **Possible?** The world already builds the equivalent of a city of a million people every five days.
- **Delay raises the rate:** every year of waiting makes the required cuts steeper, and both paths need negative emissions later.

Source: Dessler (2021), Ch. 14

Elements of a coherent climate policy

1 Price emissions, everywhere

A global price on greenhouse gases; if big emitters stay out, the rest must cut faster.

2 Regulate where prices are slow

Fuel standards, limits on coal.

3 Fund research

Storage, new energy sources, carbon removal.

4 Prepare to adapt

Build long-lived infrastructure for the future climate; aid poorer countries.

5 Research solar radiation management

A fallback if mitigation comes too late.

6 Review and revise

Update policy as science and technology change.

- **The real cost comparison:** fossil fuels already carry hidden costs (climate damage, air pollution deaths, oil-price shocks). Most economic analyses find the switch a net benefit.

Source: Dessler (2021), Ch. 14